STATE OF CLIMATE TECH 24

A *Net Zero Insights'* yearly analysis on global funding and deal activity in the private market venture space

PRODUCED BY

Net Zero Insights

SPONSORED BY

EDB SINGAPORE

STRATEGIC PARTNER

CLIMATE FIRST

Table of Contents

A word from Net Zero Insights

This marks the third year of our annual State of Climate Tech report.

As we've meticulously tracked investment activity over the years, the Climate Tech ecosystem has shown rapid development, growth, and transformation—from the investment boom of 2021 to the resilience of 2022 and the downturn of 2023. Beyond the rise and fall of many remarkable companies, we've observed significant evolution in the investor landscape.

Non-dilutive funding has emerged as a defining factor, particularly for financing solutions involving significant physical assets. This includes both lower-TRL technologies striving for demonstration and early commercialization, as well as market-ready solutions poised for scale and full adoption.

These varied and complex capital needs require diverse forms of investment and a broader range of investors beyond venture capital. In particular, we've been struck by the steadily growing role of corporates and banks. Corporates are critical for industrial validation, scaling, and driving market demand, while banks play a pivotal role in financing these efforts.

While the ecosystem has grown substantially, achieving the necessary progress to enable the transition demands an even greater scale of effort—and participation from all market players is essential.

We look forward to closely tracking the progress and evolution toward a Net Zero Economy.

Federico Cristoforoni
Founder and Director
Net Zero Insights

A word from our sponsor

On behalf of the Singapore Economic Development Board, I am pleased to contribute to the foreword of this year's State of Climate Tech Annual Report. This report offers valuable insights into the progress and challenges in Climate Technology, a critical enabler of global climate goals.

Singapore is committed to achieve net zero emissions by 2050, and is taking bold steps to build a resilient and sustainable economy that thrives on innovation and collaboration. Beyond our own shores, Asia also holds immense potential for decarbonization with its rich natural resources and renewable energy opportunities. We believe that Climate Tech and carbon services companies play a vital role in accelerating the region's low-carbon transition. These firms offer innovative solutions that can reduce emissions and unlock low-carbon alternatives in Asia. In addition, renewable energy services firms looking to tap into Asia's growth opportunities can develop, finance and execute their projects from Singapore.

With our supportive business-friendly environment, robust ecosystem of industry partners and connectivity to Southeast Asia, Singapore is an ideal launchpad and hub for companies looking to scale their innovations and tackle regional and global challenges. We therefore invite Climate Tech, renewable energy and carbon services companies to join us in co-creating solutions that advance Singapore's net zero goals and contribute to a more sustainable future for all.

Sincerely,

Dino Tan
Senior Vice President
Head of Region, Europe & UK
Singapore EDB

Executive Summary

The State of Climate Tech 2024 report explores key shifts in the Climate Tech landscape, providing insights to guide decisions. Based on a specialized taxonomy, extensive investment data, and diverse ecosystem perspectives, It aims to be the market's most comprehensive analysis.

In 2024, deal activity declined by 23%, driven largely by a significant drop in equity deals, particularly at early stages. In contrast, non-dilutive funding surged, supporting sectors like batteries, hydrogen, and EV. Debt financing made up 41% of total funding, reflecting a shift toward mature, scalable technologies with lower risk profiles.

Regional dynamics and trends

- In 2024, Europe led in funding, driven by large debt rounds, while North America remained stable. In contrast, emerging markets saw a decline in investments
- Comparing the two most funded markets, Europe and the US:
  - Both markets are shifting toward fewer, higher-value investments, driven by a smaller investor pool compared to 2023
  - Both markets reversed the 2023's total funding decline trend, while Europe experienced a slight drop in equity funding
  - Despite the strong focus on physical solutions, the US invests more toward breakthrough innovations than Europe

Corporate commitment to Climate Tech

- Over the past five years, corporations have shown strong dedication to the Climate Tech space, but investors surveyed for the SOCT 2024 stress that greater engagement is vital for driving innovation
- An increasing number of corporations are entering the Climate Tech space, with 27.6% of corporations participating in 2024 deals as first-time investors
- Corporations ramped up their M&A activity in 2024, with 60% of respondents in the State of Climate Tech 2024 survey indicating plans to further intensify their M&A strategies over the next 2 to 3 years

Sectoral insights and focus areas

- Since 2020, investments have increasingly focused on energy solutions, with 2024 seeing significant funding for battery technologies, solar energy and hydrogen
- The transport sector, including electric vehicles (EVs) and charging infrastructure, follows closely in 2024 investment choices
- In 2024, the GHG capture, removal, and storage challenge area, saw an expressive YoY increase, driven by the growing need for carbon capture, utilization and storage (CCUS) technologies
- Despite a clear technological investment focus, industry experts recommend a holistic approach to Climate Tech deployment—one that prioritizes market needs instead of overly focusing on any single solution category

Research methodology

This report combines quantitative and qualitative analyses to offer a deep dive into the evolving Climate Tech investment landscape. It draws on data from over 43,000 deals, 16,000 organizations, and 20,000 investors spanning 01/01/2020 to 30/11/2024

Market intelligence for the Net Zero Economy

This report, produced by Net Zero Insights, leverages a proprietary database of millions of data points on business, technology, and financial information related to Climate Tech companies, deals, and investors worldwide.

Trusted by top investors, corporates, and advisors, the Net0 Platform helps identify new technologies and monitor emerging trends and opportunities.

Get in Touch

Key Takeaways

Global equity funding into Climate Tech declines for second consecutive year

Global equity investment fell 14% YoY since 2023, though the US and Europe remained stable. 60% of surveyed founders cited challenges in fundraising, including limited follow-on capital and a shortage of industry-specific investors.

$1B+ debt rounds in Europe reversed 2023's downward funding trend

Total funding is projected to reach $92B in 2024, slightly surpassing 2023, thanks to a few giga debt rounds. Debt financing plays an increasingly important role in scaling Climate Tech, funding almost 50% of physical solutions.

70% of investors expect more capital to flow into Climate Tech in coming years

Despite the deal activity slowdown and fundraising challenges, 70% of surveyed investors remain optimistic, anticipating increased capital flows into Climate Tech and highlighting confidence in the sector’s long-term potential.

Key Takeaways

Europe overtakes the US in total funding with surge in debt giga-rounds

A surge in large debt financing rounds in early 2024 made Europe the global leader in total funding, surpassing the US for the first time. Approximately half of Europe's 2024 funding came from eight of the year's ten giga-rounds.

Equity funding in the US remained stable despite the election year

Despite a global equity decline, the US is set for a 4% growth in Climate Tech equity funding and a 12% rise in total funding, marking its first positive YoY growth since 2022 despite election-year uncertainty and broader market challenges.

The US dominates in funding breakthrough innovations

While Europe and the US provide comparable funding for market-ready solutions, in the last five years US companies developing R&D-intensive technologies raised more than twice the amount secured in Europe.

Key Takeaways

Corporations and Governments play crucial roles to scale Climate Tech

Amid a funding slowdown and long ROI timelines in Climate Tech, government support through funding and regulatory easing, along with corporate collaboration, are crucial for the success of capital-intensive hardware solutions.

1 out of 2 corporations plan to increase their involvement in Climate Tech

Half of surveyed corporations plan to expand partnerships and deepen collaborations, driven primarily by goals of creating new revenue streams or business models and gaining a competitive edge in the market.

60% of corporates expect M&A activity to increase in coming years

M&A dominated exits, accounting for 95% of activity, with strategics driving 84% of deals. Notably, 60% of corporates expect M&A activity to grow further in the coming years, underscoring momentum in Climate Tech consolidation.

01

A year in review

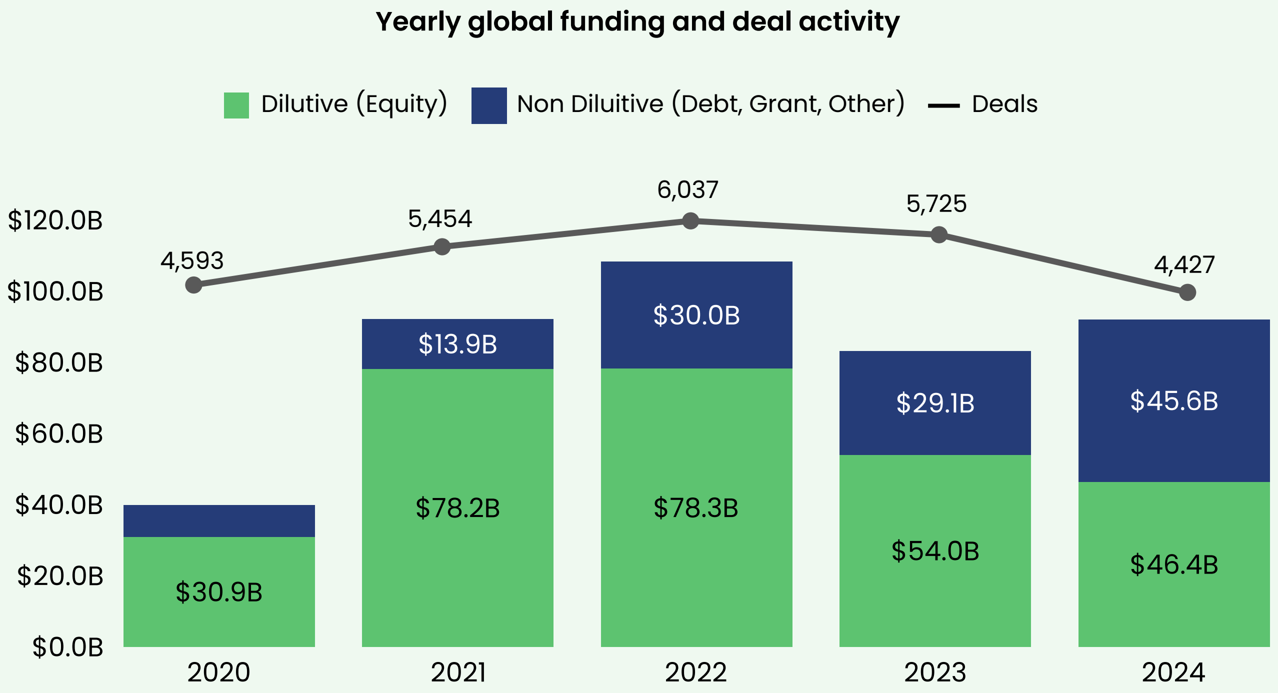
01 Funding and Deal Activity

Analyzing Climate Tech funding and deal activity reveals trends, highlights dynamics, and provides foresight into future performance, helping investors and innovators adapt strategies to seize emerging opportunities.

Record-high non-dilutive funding reversed the 2023 downward trend

Climate Tech funding remains robust, with $92.2B projected in 2024, despite a decline from the 2022 peak of $108.3B and a sharp drop in deal activity from 6,037 deals in 2022 to 4,427 in 2024.

Investors are increasingly concentrating capital in fewer, larger deals, prioritizing high-impact, market-ready solutions.



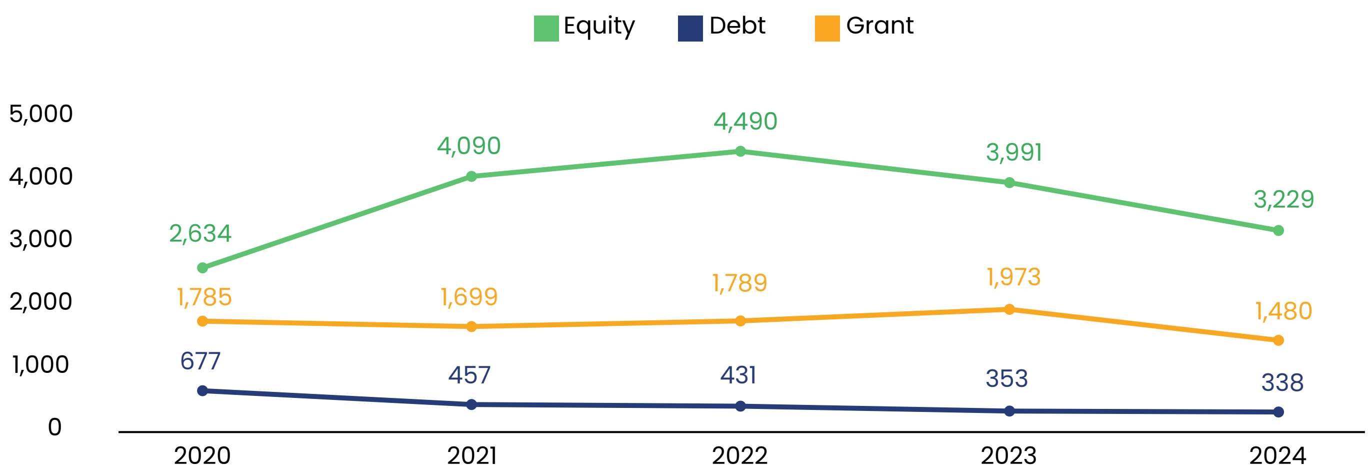
Source: Net Zero Insights

The figures show funding and deal activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, but exclude exits and post-exit financing. Only deals with disclosed round amounts are included. 2024 data incorporate end-of-year projections.

Non-dilutive funding accounts for $45.7B in 2024, but nearly half stems from just 10 giga-rounds ($1B+), highlighting a focus on scalable, later-stage projects. Global equity funding fell 14% year-over-year, driven by a decline in emerging markets. However, the US showed resilience with a 4% funding increase, and Europe saw only a modest 3% decline, reflecting stability in the two most funded regions.

The trends reveal a clear shift toward large-scale investments while smaller, early-stage ventures face growing challenges. Despite fewer deals and reliance on giga-rounds, funding levels remain strong, underscoring investor commitment to tackling climate challenges through high-value, scalable innovations.

Yearly global deal activity by financing type



Source: Net Zero Insights

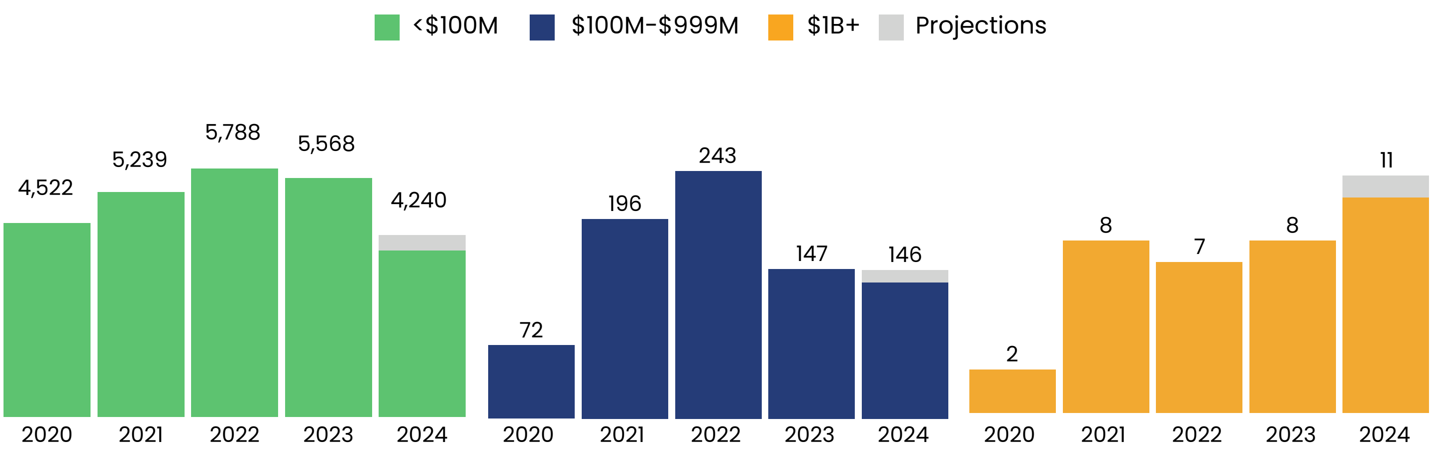
The figures show funding and deal activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, but exclude exits and post-exit financing.

While both non-dilutive and equity deals declined in 2024, the number of giga-rounds deals reached a record-high

The equity deal count in 2024 is projected to decline by 19% year-over-year, with debt deals expected to decrease by 4%. This trend reflects a shift in funding priorities, highlighted by the stabilization of mega-rounds ($100M–$999M) and a record-setting rise in giga-rounds ($1B+), signaling a focus on more mature, market-ready solutions.

Year-to-date, Climate Tech has marked a five-year high in giga-rounds. While mega-round funding remained steady year-over-year, smaller rounds experienced a significant drop. This polarization indicates a growing concentration of capital in large-scale projects, leaving early-stage ventures facing heightened challenges in securing investment.

Deal count by deal size



Source: Net Zero Insights

$1B+ debt rounds dominated 2024 funding in Europe

In 2024, giga-round deals—primarily fueled by debt financing—saw notable growth in both volume and value, highlighting the substantial capital being deployed to accelerate the scaling and adoption of proven, market-ready technologies.

Banks and corporations played a pivotal role in these rounds, with banks financing 4 out of 10 and corporates participating in 3 out of 10 deals, underscoring their critical involvement in advancing large-scale projects.

Company	Tags	Deal type	Amount
northvolt	#Battery #LithiumIon	Debt Deal	$5B
acc	#Battery #LithiumIon	Debt Deal	$4.75B
Stegra	#Steel #Hydrogen	Debt Deal	$4.2B
solørbioenergi	#DistrictHeating #Fuels	Debt Deal	$2.11B
HOLTEC INTERNATIONAL	#NuclearEnergy	Debt Deal	$1.52B
VERKOR	#Battery #LithiumIon	Debt Deal	$1.41B
Enpal.	#Solar #HeatPumps	Debt Deal	$1.2B
INFINIUM	#Electrofuel	Equity Deal	$1.1B
IM MOTORS	#ElectricVehicle	Equity Deal	$1.12B
WAYVE	#AutonomousVehicle #AI	Equity Deal	$1.05B

Key Highlights

$23.5B
Funding in 2024

+300%
YoY funding variation

Q1 '24	7 deals	$19.9B
Q2 '24	2 deals	$2.5B
Q3 '24	1 deal	$1.1B

Share of funding by financing instrument

9% Equity

91% Debt

Share of funding by region

84% Europe

11% North America

5% Asia

Investor participation

4/10 Banks

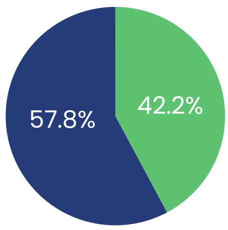
1/10 Governments

3/10 Corporations

Capital utilization

Capacity expansion

Facility construction



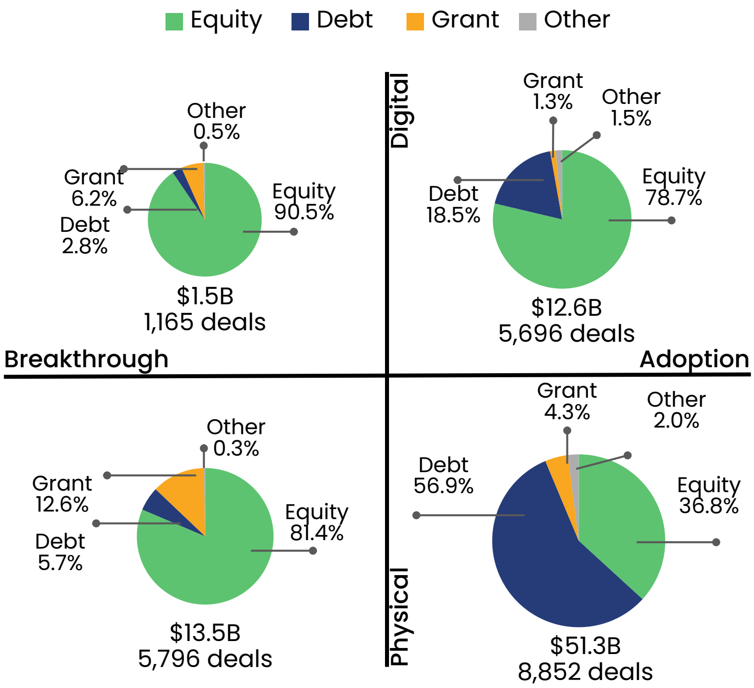
Capacity expansion involves allocating capital to enhance existing manufacturing and distribution capabilities, while facility construction refers to deals where most of the capital is used to build new facilities.

Debt fuels the financing of market-ready, physical solutions

Non-dilutive capital is playing a pivotal role in scaling market-ready Climate Tech solutions, with debt financing taking center stage. In 2024, non-dilutive funding reached record levels, showcasing its critical importance in driving the adoption of physical solutions. Over 56% of physical solutions in the adoption stage were funded by debt, reflecting the dominance of debt in this category. In contrast, equity remained the primary funding source for digital solutions,

Grants, on the other hand, served as a cornerstone for breakthrough innovation, representing 12.6% of funding in physical solutions and 6.2% in digital solutions,

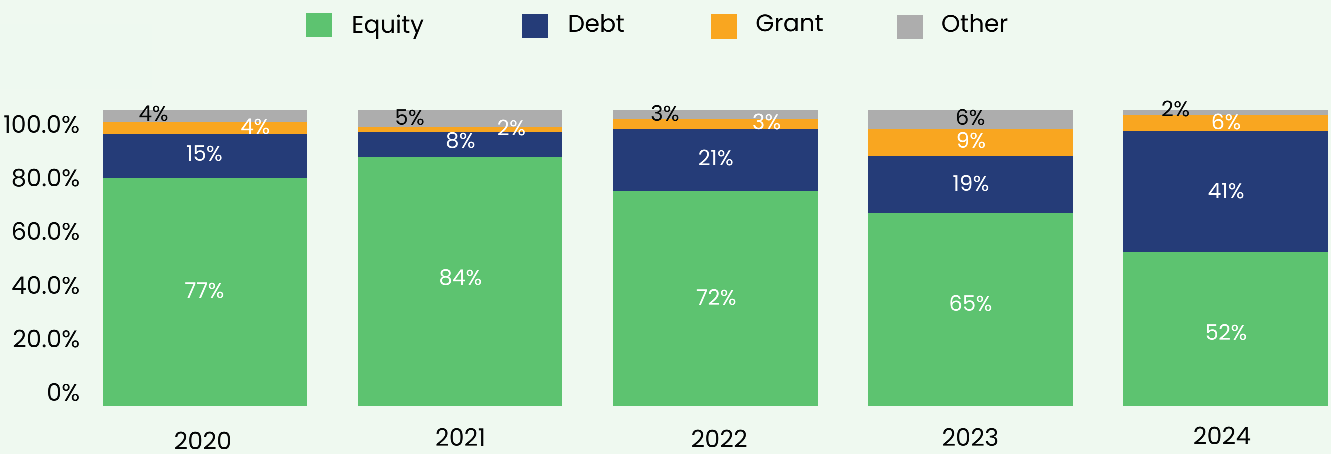
2024 funding by financing type across PI framework



Source: Net Zero Insights

The figures represent funding activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, while excluding exits and post-exit financing. For details on the product-innovation framework, see the research methodology section.

Yearly global share of funding by financing type



Source: Net Zero Insights

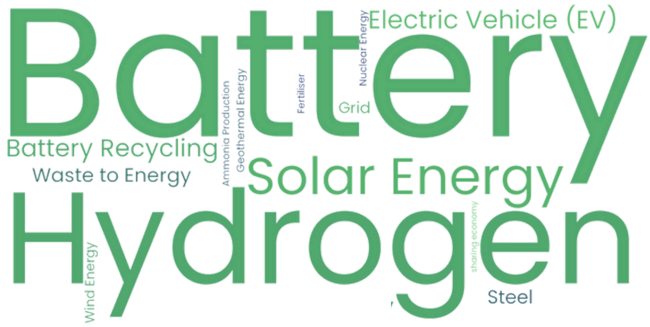
The figures illustrate funding activity in private Climate Tech ventures, covering equity, debt, grants, and other financial instruments, while excluding exits and post-exit financing.

Batteries and hydrogen lead 2024 debt deals as Europe takes charge

Battery and hydrogen-based solutions dominated debt financing in 2024, with notable investments also flowing into solar, and electric vehicle (EV) technologies, highlighting investor confidence in their scalability and readiness for widespread adoption.

While major deals included US startups, Europe led Climate Tech debt financing, securing eight of the top 10 deals across five countries.

2024 most funded solutions in debt deals



Top 2024 debt deals

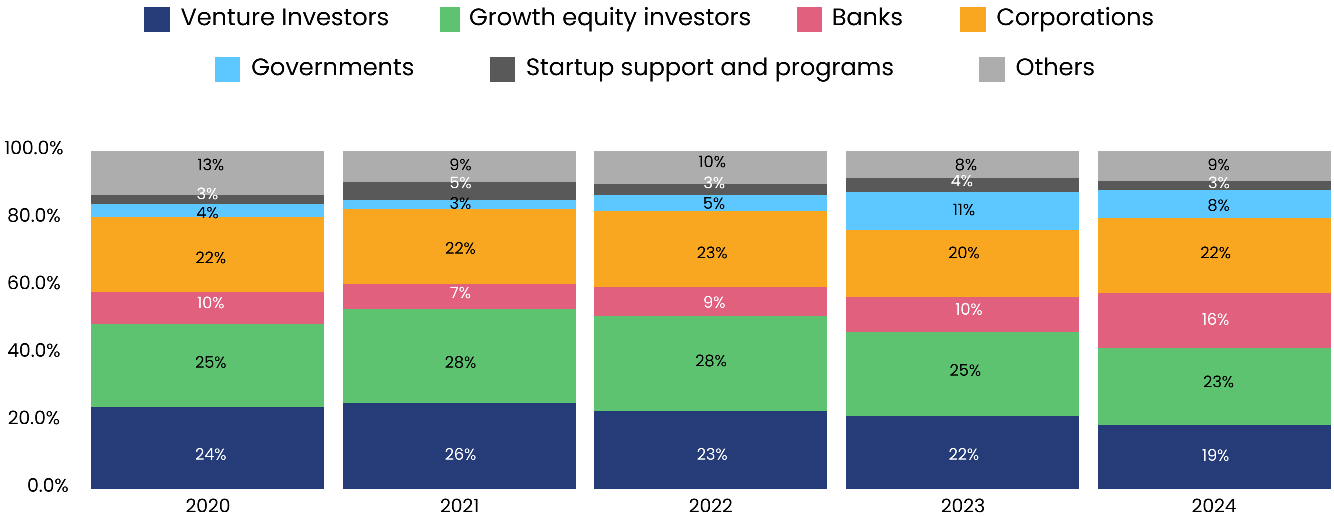
Company	Amount	HQ Country	Tags
northvolt	$5.0B	Sweden	#Battery #LithiumIon
acc	$4.75B	France	#Battery #LithiumIon
solørbioenergi	$2.11B	Sweden	#DistrictHeating #Fuel
HOLTEC INTERNATIONAL	$1.52B	United States	#NuclearEnergy
VERKOR	$1.41B	France	#Battery #LithiumIon
Enpal.	$1.2B	Germany	#Solar #Heatpumps
VoltaGrid	$650M*	United States	#Fuels
Equipment Share	$600M	United States	#Rental #Equipment
Hometree	$318M	United Kingdom	#Heating #Maintenance
Bolt	$238M	Estonia	#SharedMobility

Source: Net Zero Insights

*$500M term loan facility + $50M additional accordion feature + $100M revolving credit facility.

Banks and corporations further ramp up their activity in 2024

Yearly share of global funding by investor type participation



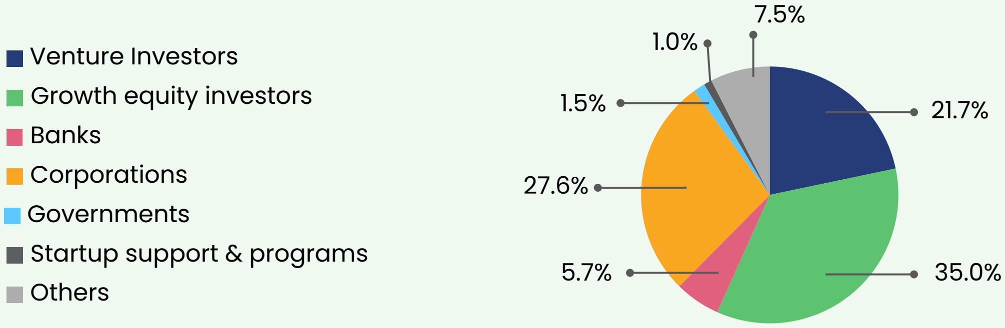
Source: Net Zero Insights

The figures illustrate funding activity in private Climate Tech ventures, covering equity, debt, grants, and other financial instruments, while excluding exits and post-exit financing. For more information on the investor categories, please refer to the research methodology section of the report.

Venture investors, growth equity investors, governments and startup support programs saw their share of global Climate Tech funding decline in 2024. Conversely, banks and corporations increased their participation, with shares rising by 6% and 2%, respectively. This highlights a growing reliance on corporations and banks, particularly through debt financing, as equity venture investment slows. The drop in government-linked funding, from 11% in 2023 to 8% in 2024, underscores reduced support for early-stage, breakthrough solutions.

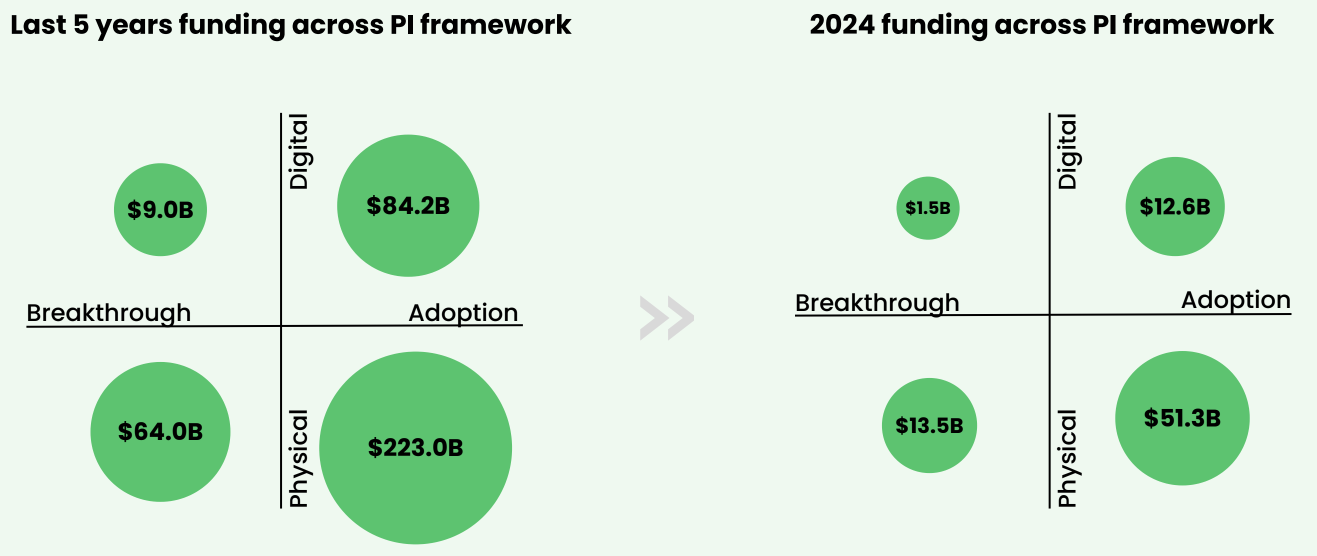
Banks and corporations together made up one-third of first-time investors in 2024.

2024 share of first-time investors by type



Source: Net Zero Insights

A focus on market-ready solutions across several challenge areas

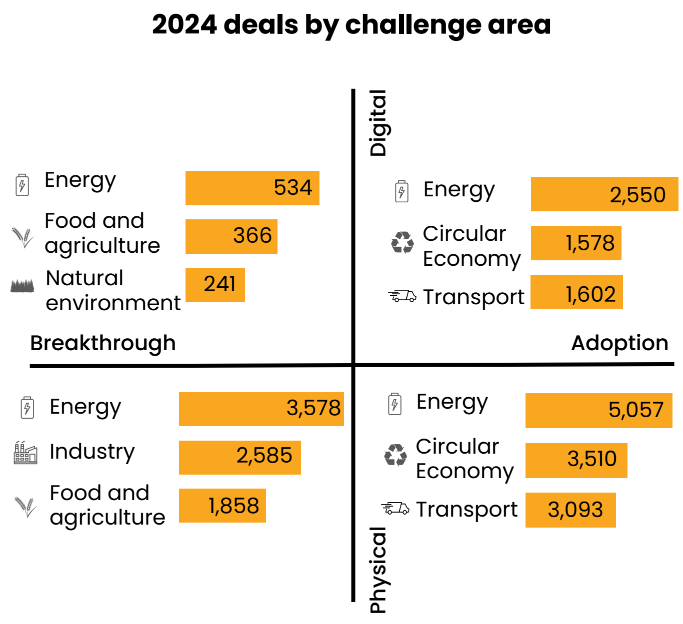


Source: Net Zero Insights

The figures show funding in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. For details on the product-innovation framework, see the research methodology section.

The 2024 funding trend aligns with the PI framework's past five years, with a slight increased focus on physical solutions compared to digital solutions. Adoption-stage innovations continue to attract the most funding, particularly in physical solutions for energy, circular economy, and transport.

While energy remains a focus across the PI framework, market-ready solutions have similar focuses, wheter they are digital of physical solutions. In contrast, food and agriculture are being emphasized as areas for breakthrough solutions.



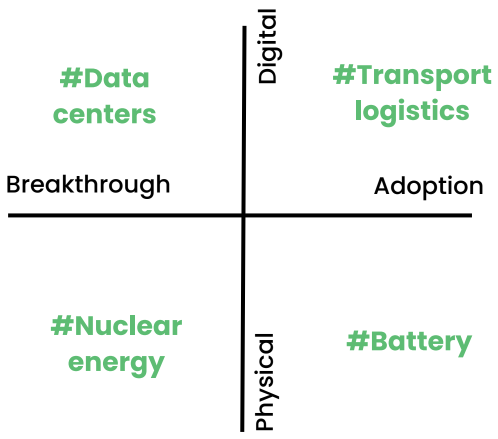
Source: Net Zero Insights

The figures show funding in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. Refer to the research methodology section for details on the product-innovation framework.

Data centers took the lead in digital breakthroughs, driven by the growing need for low-carbon infrastructure. Meanwhile, nuclear energy emerged as a frontrunner in physical breakthroughs, reflecting a renewed push for clean energy solutions.

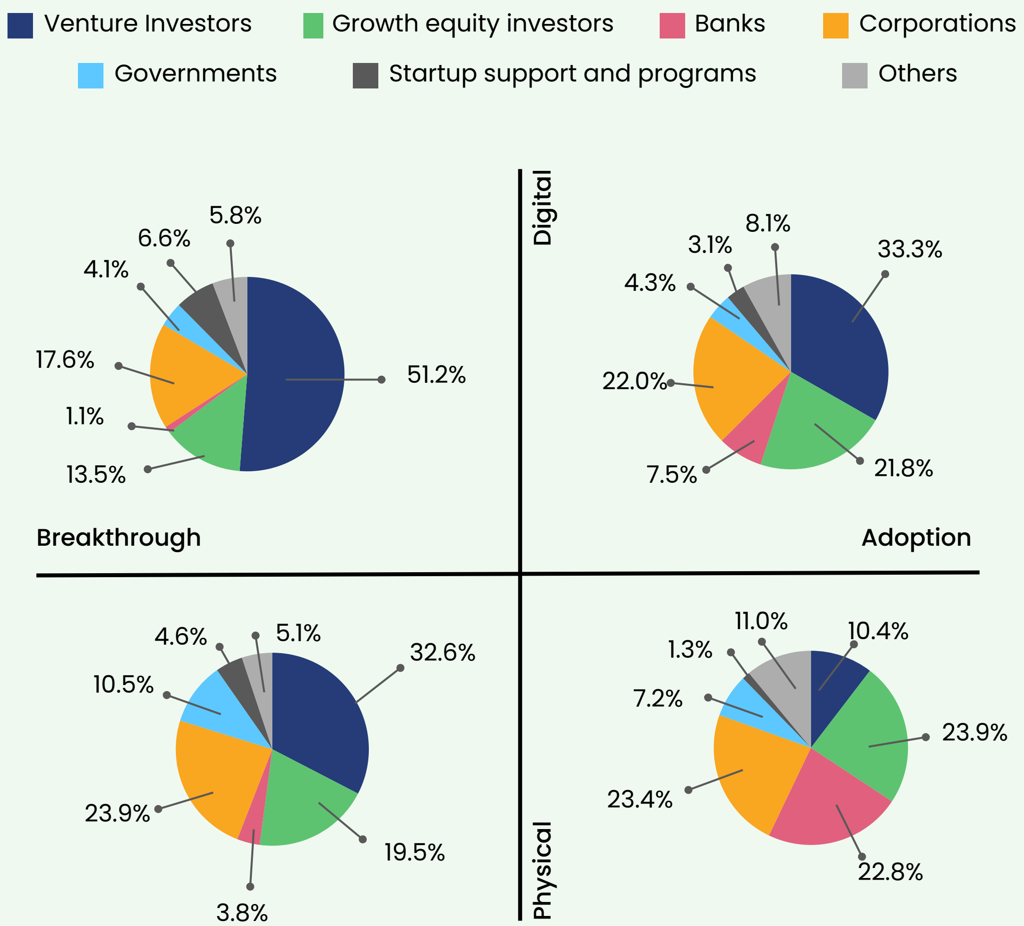
In the adoption type of innovations, transport logistics became a focal point, highlighting its role in decarbonizing supply chains, while battery technologies dominated market-ready physical solutions funded in 2024, reinforcing their importance in critical areas such as energy storage and electrifying transport.

2024 most funded solutions across the PI framework



Source: Net Zero Insights

2024 funding by investor type



Source: Net Zero Insights

The figures show private Climate Tech funding, including equity, debt, and grants, excluding exits. See the report's methodology for investor categories and product-innovation framework details.

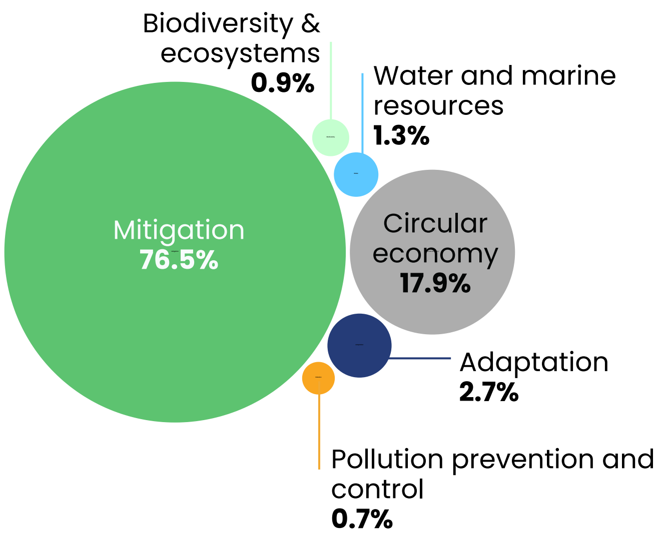
Climate change mitigation keeps securing vast majority of funding

In 2024, Climate Tech funding remains concentrated on climate mitigation (76.5%) and circular economy (17.9%), emphasizing global priorities of decarbonization and additionally towards solutions that also promote resource efficiency.

Though adaptation receives less funding, the rising investment in the circular economy underscores the growing focus on sustainable practices alongside mitigation.

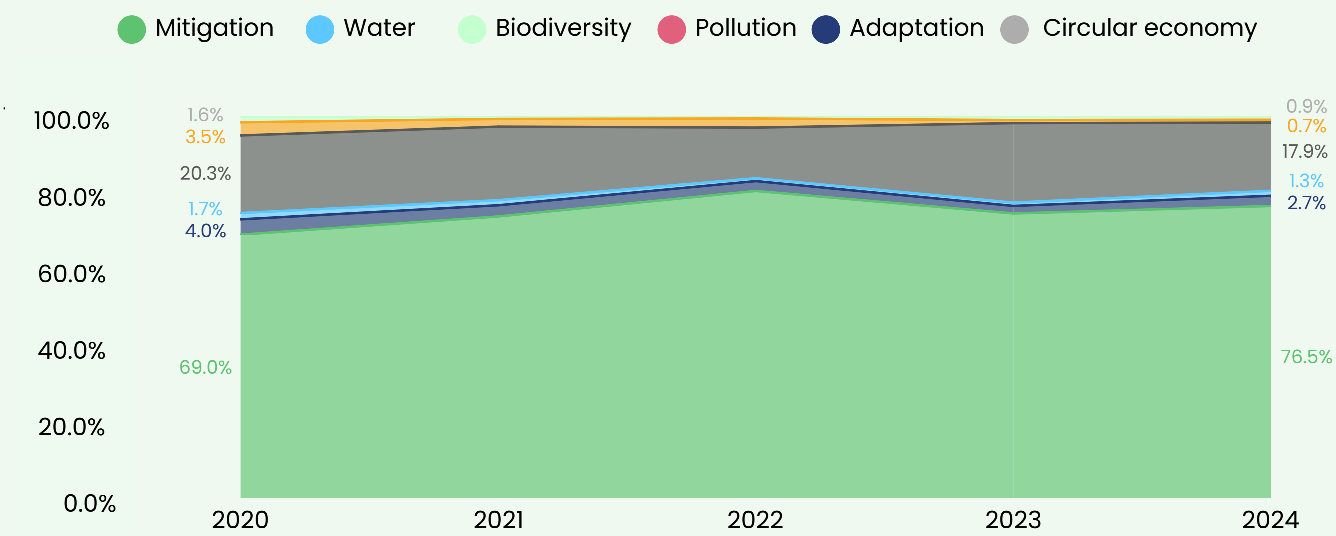
Over the past five years, funding shares across environmental objectives have remained stable, reflecting a consistent emphasis on cutting GHG emissions, fostering sustainability, and advancing innovative recycling and reuse strategies.

2024 share of global funding by environmental objective



Source: Net Zero Insights

Yearly share of global funding by environmental objectives



Source: Net Zero Insights

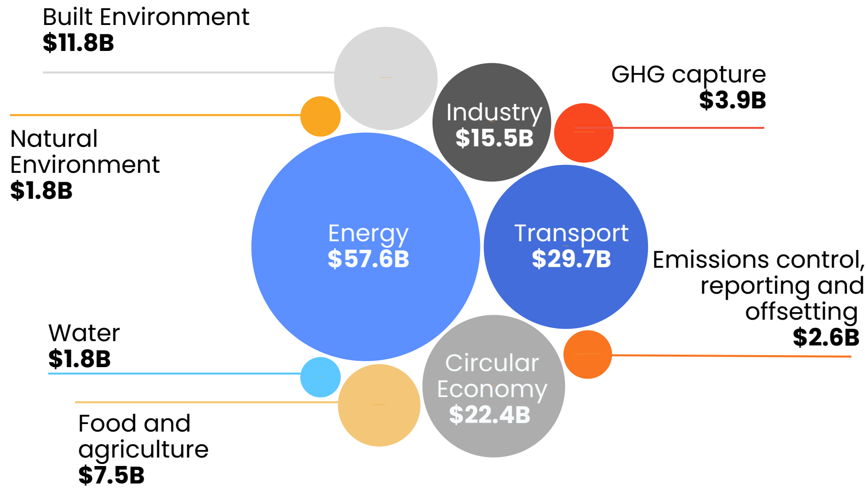
The figures show funding in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. For details on environmental objectives, see the report's research methodology section.

Priority to energy and transport, other critical areas underfunded

The 2024 Climate Tech funding landscape highlights a strong focus on energy and transport, which together account for the majority of investments.

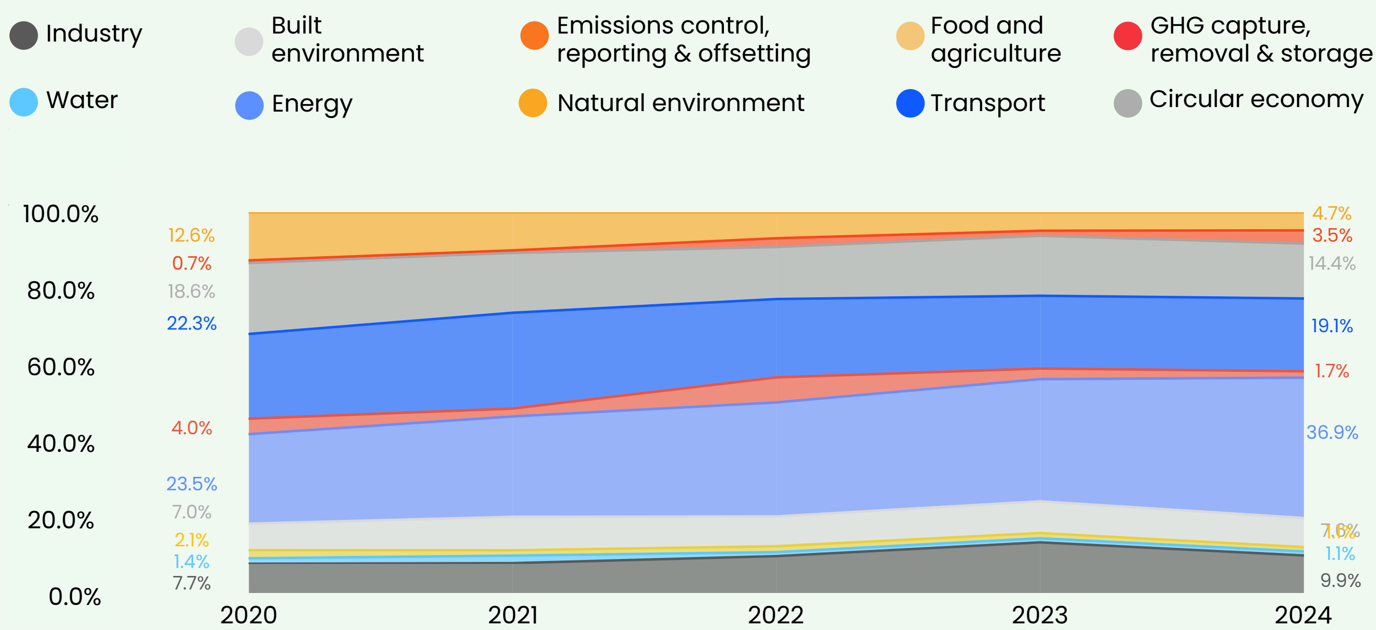
However, the persistent underfunding of critical areas like GHG capture, water solutions, and the natural environment raises concerns about addressing the broader spectrum of climate challenges effectively.

2024 global funding by climate change challenge area



Source: Net Zero Insights

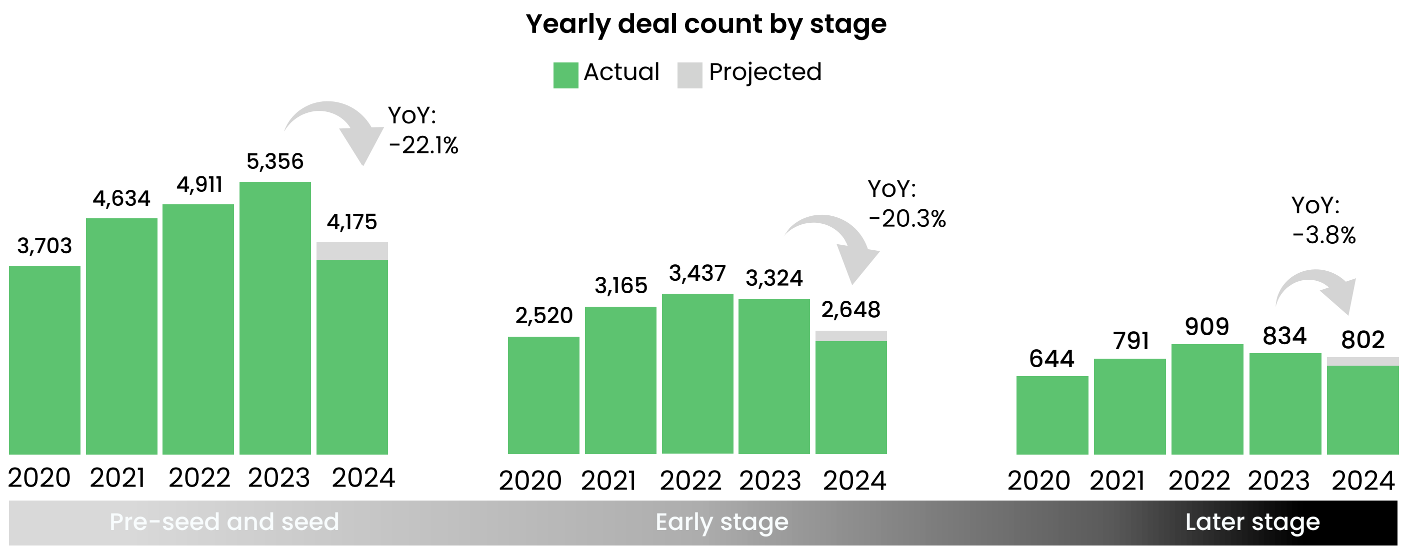
Yearly share of global funding by climate change challenge area



Source: Net Zero Insights

The figures show funding in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. For details on climate change challenge areas, see the report's research methodology section.

Deal activity drops across all stages with early stages hit the hardest



Source: Net Zero Insights

The figures show deal activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing.

Early-stage and later-stage Climate Tech deals experienced notable YoY declines in 2024 (-20.3% and -3.8%, respectively), reflecting a tightening funding environment. Pre-seed and seed-stage deals faced the steepest drop (-22.1%), signaling reduced investor appetite for early-stage risk.

Sectorally, energy funding remained resilient across all stages, benefiting from its proven scalability and investor confidence. GHG capture funding saw exceptional growth in later stages (+305.7%), indicating a growing focus on scaling carbon removal solutions as a critical climate mitigation strategy. Meanwhile, transport and food and agriculture experienced sharp declines across all stages, reflecting shifting priorities and potential concerns over these sectors' scalability or market readiness.

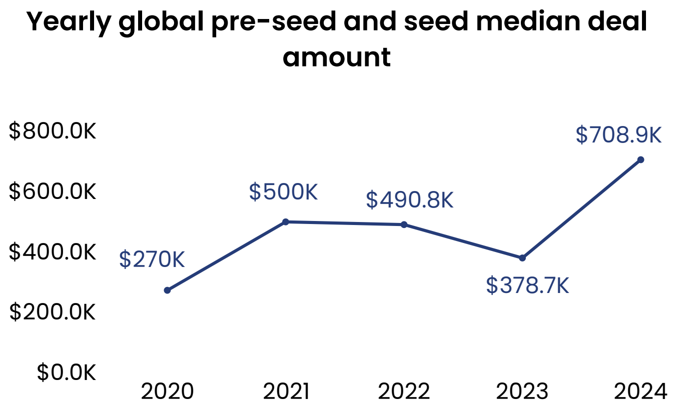
Another notable trend is the stark disparity in later-stage funding. While sectors like energy and GHG capture showed strength, industry funding plummeted by 60.8%, suggesting challenges in advancing industrial decarbonization solutions to maturity. This highlights a funding gap that could hinder progress in decarbonizing emissions-heavy industries, crucial for net-zero goals.

YoY funding variation by stage

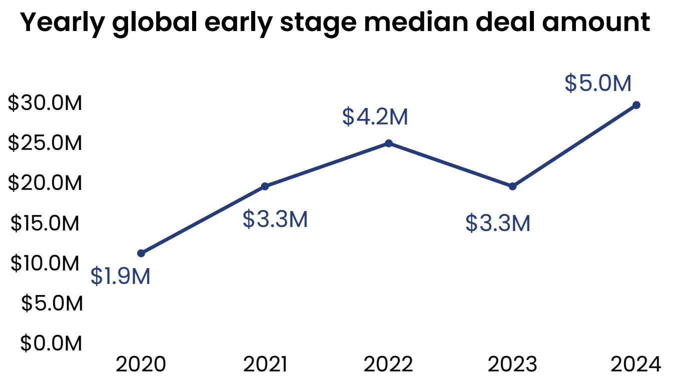
	Pre-seed and Seed	Early stage	Later stage
energy	26.8%	11.2%	12.5%
industry	14.8%	43.1%	-60.8%
transport	-32.8%	-32.4%	16.5%
food and agriculture	-38.3%	24.5%	-17.8%
circular economy	-20.0%	-39.3%	1.6%
built environment	-31.9%	25.0%	-35.7%
natural environment	-22.9%	-31.2%	3.0%
emissions control	-46.4%	-47.5%	-34.5%
GHG Capture	27.0%	11.0%	305.7%
water	-33.4%	35.6%	-6.6%

Larger early-stage deal amounts with stable late-stage ticket sizes

The YoY increase in the median ticket size for pre-seed/seed stage funding deals from 2023 to 2024 is approximately +87.2%. This reflects a significant uptick in the average size of funding rounds, indicating a recovery in capital availability and greater confidence in scaling early-stage ventures.



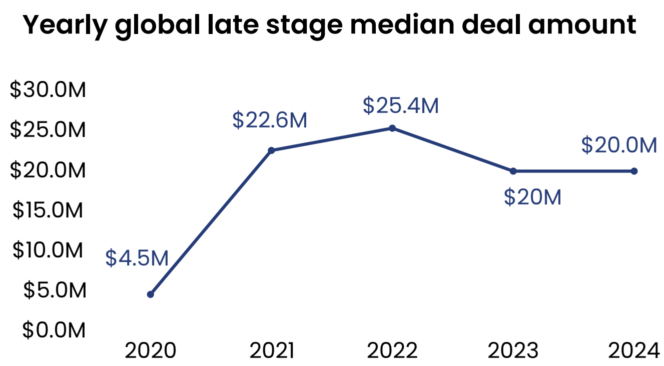
Source: Net Zero Insights



Source: Net Zero Insights

The 51.5% increase in the median deal amount for early-stage investments in 2024, despite a YoY decrease in deal count, reflects a shift toward larger funding rounds. This uptick can be attributed to the increase in average ticket size, as investors prioritize fewer, higher-value deals in a more cautious funding environment.

The slight increase in the median deal amount for late-stage investments from $20M in 2023 to $20M in 2024 helps explain nil YoY change in total annual funding, despite a decrease in deal count. This trend offsets the decrease in deal count, helping maintain stable total funding.



Source: Net Zero Insights

Top 2024 equity deals by stage

Top 2024 pre-seed and seed equity deals

Company	Amount	HQ Country	Tags
EvolutionaryScale	$142M	United States	#AI #Proteins
NATTERGAL	$52.2M	United Kingdom	#NatureRestoration
stargate hydrogen	$45M	Estonia	#GreenHydrogen
cusp.ai	$30M	United Kingdom	#MaterialDesign #AI

Top 2024 early stage equity deals

Company	Amount	HQ Country	Tags
IM MOTORS	$1.1B	China	#ElectricVehicle
PACIFIC FUSION	$900M	United States	#FusionEnergy
ENERVENUE	$308M	United States	#LongDurationEnergyStorage
black semiconductor	$273M	Germany	#Semiconductors

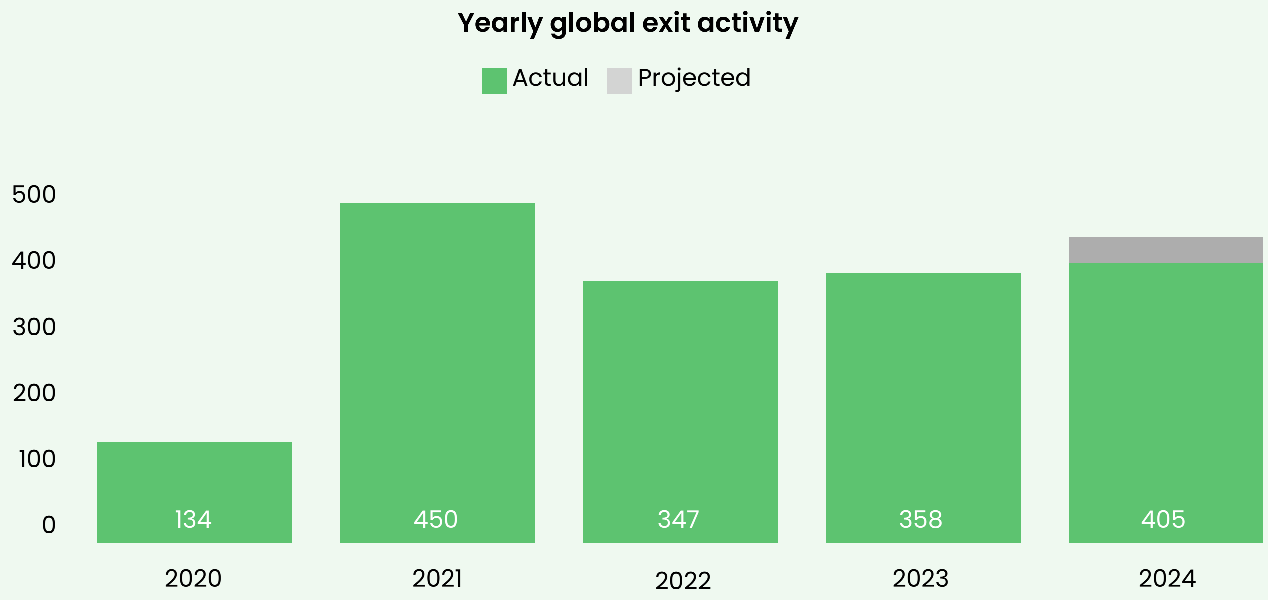
Top 2024 later stage equity deals

Company	Amount	HQ Country	Tags
INFINIUM	$1.1B	United States	#Electrofuels
twelve	$600M	United States	#CCUS
Crusoe	$500M	United States	#CleanComputing
X energy	$500M	United States	#NuclearEnergy

02 Exit activity

Examining exit activity reveals key insights for shaping early-stage company strategies and gauging the ecosystem's maturity. This section explores Climate Tech exit trends and success stories, offering a clear view of the evolving landscape.

Exit activity has slowly grown since 2022, but timelines prolonged



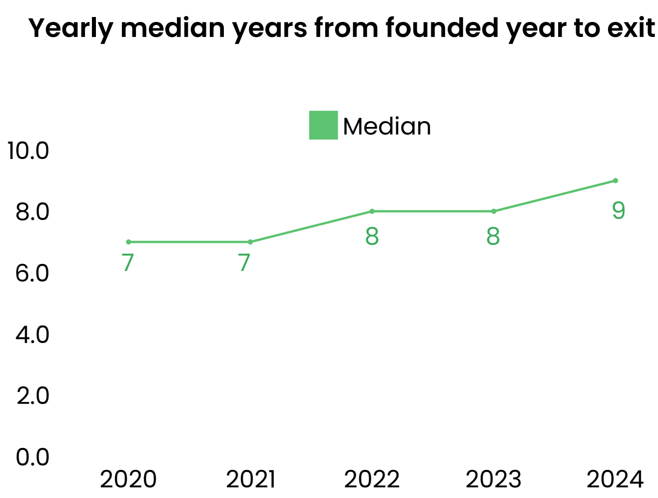
Source: Net Zero Insights

The figures illustrate exit activity in private Climate Tech ventures, while excluding venture and post-exit deals.

Global exit activity in Climate Tech has grown steadily since 2021, signaling a gradually maturing sector. This reflects startups' increasing readiness to scale through mergers, acquisitions, or IPOs, marking critical milestones in their lifecycle.

Investors are now supporting startups through longer growth cycles, requiring innovators to sustain momentum while scaling operations, refining technologies, and expanding market presence. For investors, these extended timelines delay returns, emphasizing the need for patience and strategic support.

This steady growth underscores confidence in the sector's long-term potential. As more startups transition from innovation to implementation, they are driving transformative solutions, solidifying Climate Tech's role in tackling global challenges.



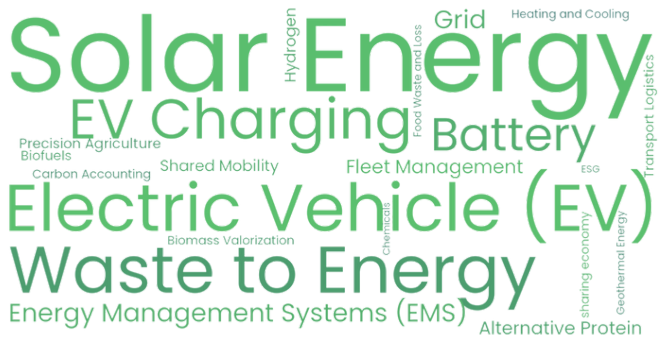
Source: Net Zero Insights

Solar, EV and waste-to-energy dominate exit activity

The prominence of solar energy, EVs, and waste-to-energy solutions in 2024 exit deals underscores the maturity and scalability of these technologies. Solar energy remains central to global decarbonization, while advancements in EV charging infrastructure and batteries are accelerating the shift to sustainable transportation.

These exits reflect strong investor confidence in solutions driving tangible climate impact and market readiness.

Top solutions by 2024 exit deals



2024 most notable exits

Company	Amount	Deal Type	HQ Country	Tags
Environmental Solutions Group	$2B	Acquisition	United States	#WasteManagement
OLA	$730M	IPO	India	#SharedMobility
ECO Special Waste Management Pte Ltd	$435M	Acquisition	Singapore	#IndustrialWaste
OWYN	$280M	Acquisition	United States	#PlantbasedProteins
Kent Renewable Energy Limited	$246M	Acquisition	United Kingdom	#BiomassPlant
Hyseteco	$217M	Acquisition	France	#HydrogenMobility
EVSE	$162M	Acquisition	Australia	#EVCharging
SOL	$100M	Acquisition	United States	#CarbonFootprint
convion	$80M	Acquisition	Finland	#FuelCells
ChargeNet	$64M	Acquisition	New Zealand	#EVCharging

Exit story

OLA ELECTRIC

Ola Electric is a two-wheeler EV manufacturer. The company also plans to open up its in-house manufactured 4680 lithium-ion EV batteries for commercial usage from 2025.

Funding rounds (in USD millions)

Timeline

May 2017

Headquartered in Bangalore, the tech hub of India, Ola Electric Mobility was founded in 2017. But only in 2021, the company created its manufacturing facility in the southern state of Tamil Nadu, where it produces its electric two-wheelers.

December 2018-January 2019

Founder Bhavish Aggarwal bought a 92.5% stake in the company at a valuation of about $1,400 to establish it as a separate entity from ANI Technologies, which operates its cab fleet company.

December 2020

Ola Electric signs an MoU with Tamil Nadu government at $323.9M for setting up the world's largest two wheeler factory.

March 2022

Ola Electric invests in Israel-based battery tech company *StoreDot* to incorporate and manufacture its XFC (extreme fast charging) battery technologies.

July 2022

The company announces an investment of $500M for building a battery cell R&D facility.

August 2024

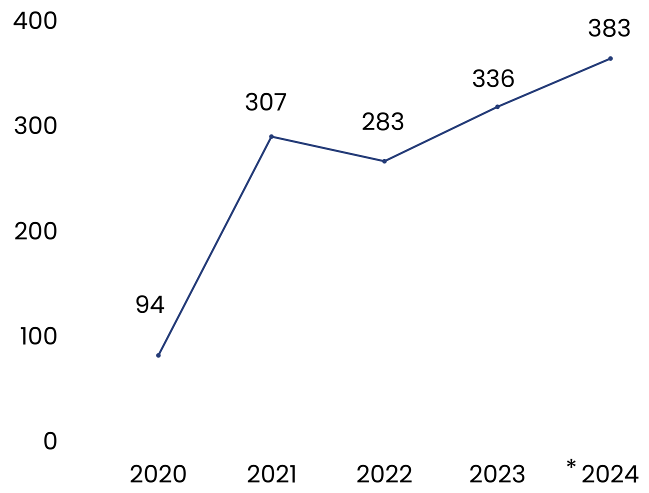
Ola Electric launches its IPO, raising $732.6M.

Merger & Acquisition activity hit record high in 2024

Mergers and acquisitions (M&A) in Climate Tech have grown steadily since 2020, reflecting rising market confidence and sector consolidation. M&A remains a reliable pathway for startups to scale by integrating with established players while delivering investor value.

Key drivers include ESG integration, regulatory incentives like tax credits and grants, and corporations' ambitions to lead in sustainability through innovative Climate Tech. Emission-intensive sectors such as energy and transport are expected to continue driving this trend.

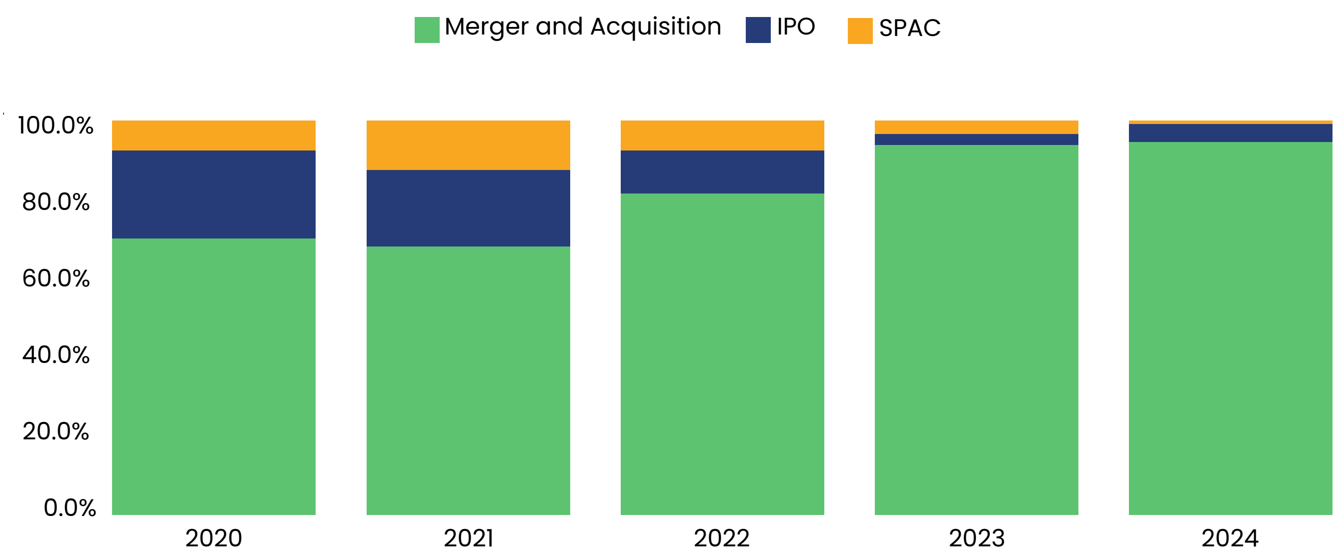
Yearly count of M&A deals



Source: Net Zero Insights

The figures depict M&A activity in private Climate Tech ventures, excluding IPOs, SPACs, other exit types, and post-exit deals. *2024 data is based on linear projections to year-end.

Yearly share of exits by exit type

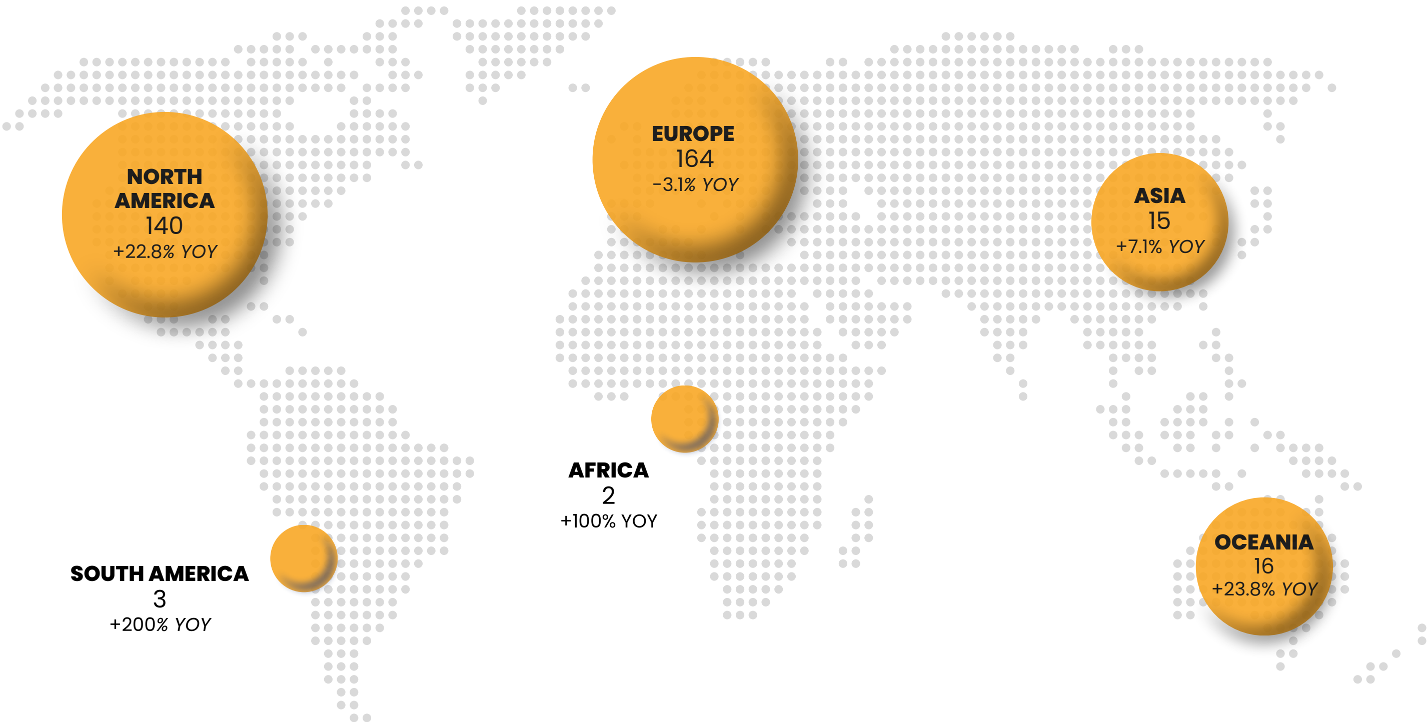


Source: Net Zero Insights

The figures illustrate exit activity in private Climate Tech ventures, including M&A, IPO and SPAC while excluding venture and post-exit deals.

Corporations drive Climate Tech M&A growth across regions

2024 M&A activity by continent



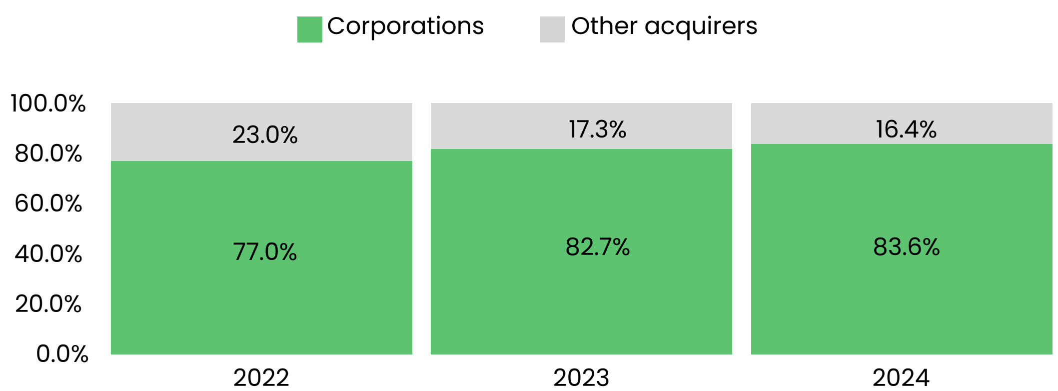
Source: Net Zero Insights

The figures illustrate M&A activity in private Climate Tech ventures, while excluding IPO, SPAC and other exit types, as well as venture and post-exit financing.

In 2024, Europe and North America led M&A activity, with Europe remaining dominant despite a slight decline and North America showing robust growth. Asia and Oceania are emerging as new hubs.

Corporations increasingly dominate M&A, leveraging acquisitions to integrate innovative technologies and bolster their Climate Tech portfolios.

Share of global M&As by acquirer type



Source: Net Zero Insights

The figures illustrate M&A activity in private Climate Tech ventures, while excluding IPO, SPAC and other exit types, as well as venture and post-exit financing.

03 Regional analysis

This section examines the distribution and impact of innovation hubs, focusing on the United States and Europe as the top-funded Climate Tech regions. It highlights key trends, strengths, and differences shaping their ecosystems.

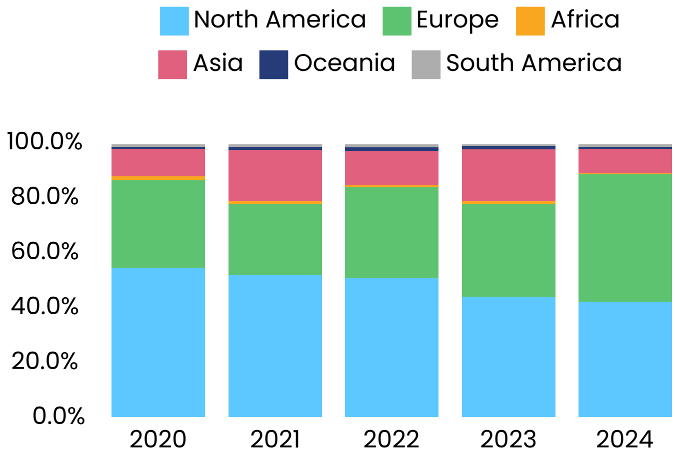
With non-dilutive, Europe leads in funding and deals in 2024

North America and Europe have long dominated Climate Tech funding, securing the majority of investments.

In 2024, Europe surpassed North America, driven by a surge in large debt rounds in the first half of the year. Meanwhile, North America retained its lead in early-stage equity funding, supported by strong regulatory incentives like tax credits and grants under the Inflation Reduction Act.

All other regions, apart from South America, saw a YoY decline in total funding.

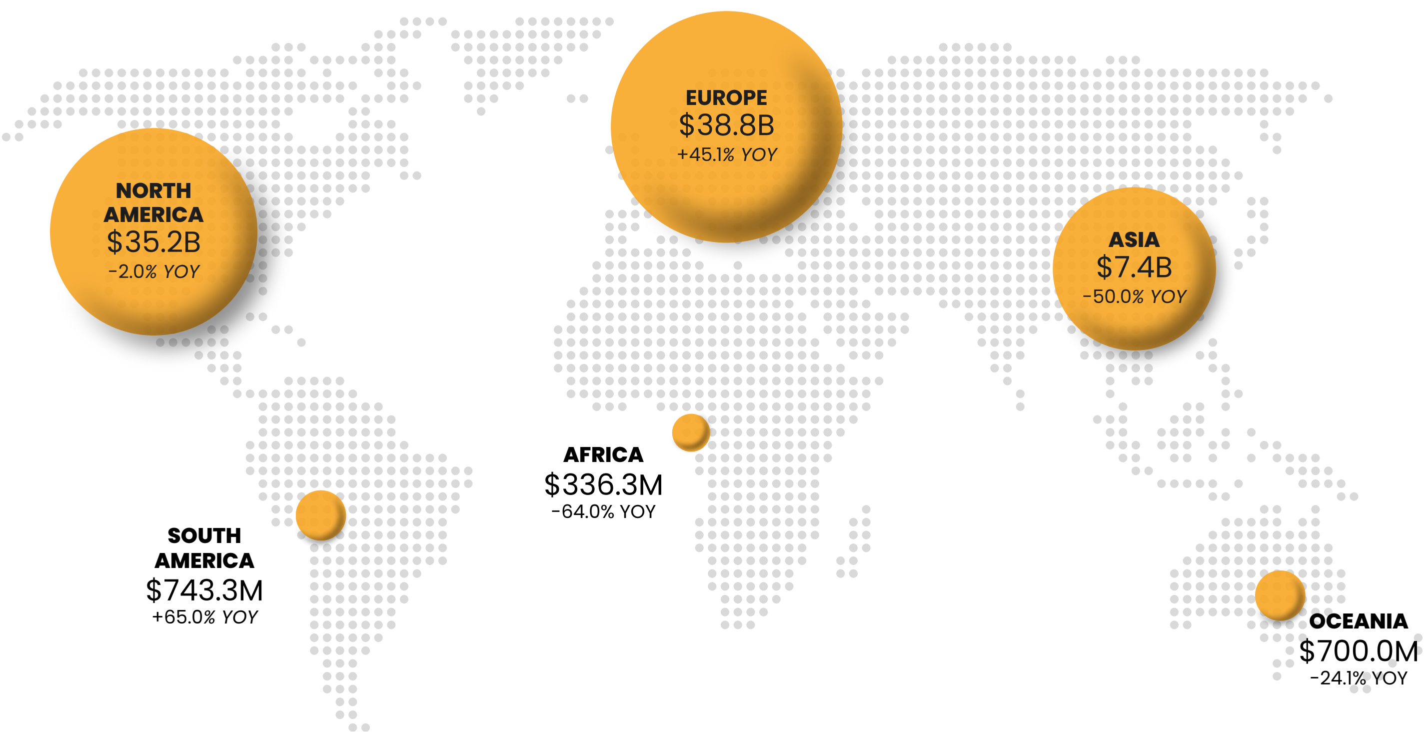
Yearly share of funding by continent



Source: Net Zero Insights

The figures show funding activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing.

2024 funding by continent

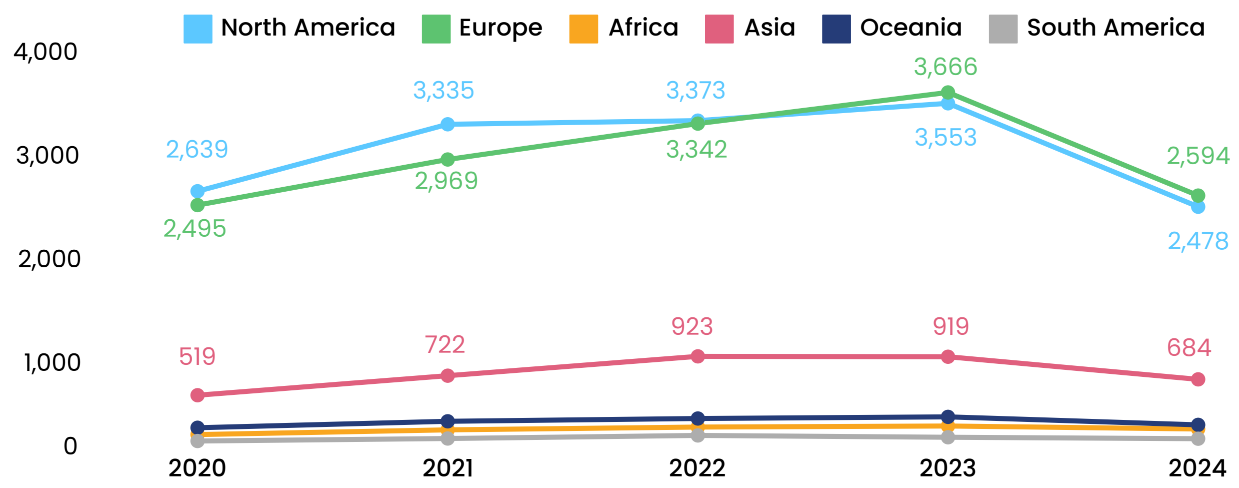


Source: Net Zero Insights

The figures illustrate funding activity in private Climate Tech ventures, covering equity, debt, grants, and other financial instruments, while excluding exits and post-exit financing.

2024 deal activity declines across all regions.

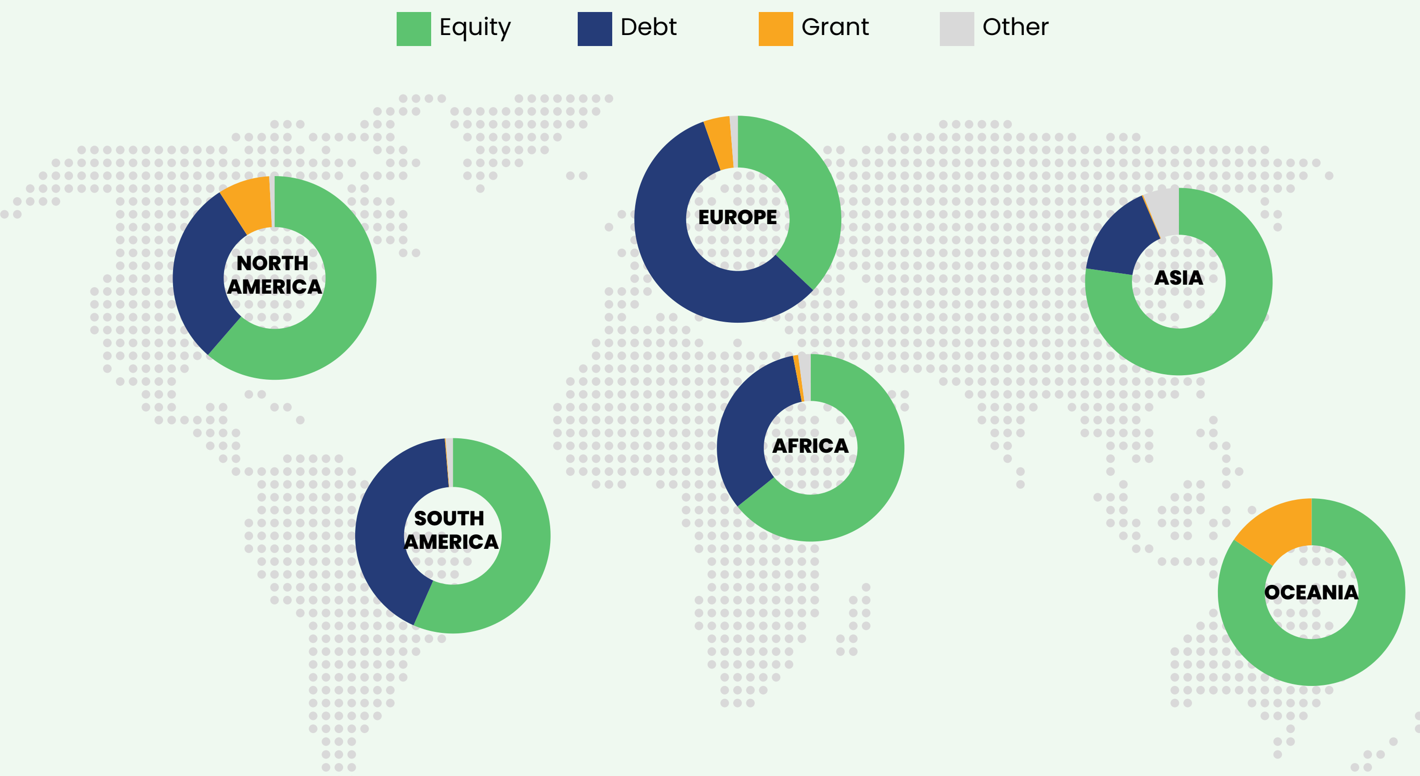
Global deals by continent



Source: Net Zero Insights

The figures show deal activity in private Climate Tech ventures, including equity, debt, and grants, but excluding exits and post-exit

2024 funding by financing type and continent

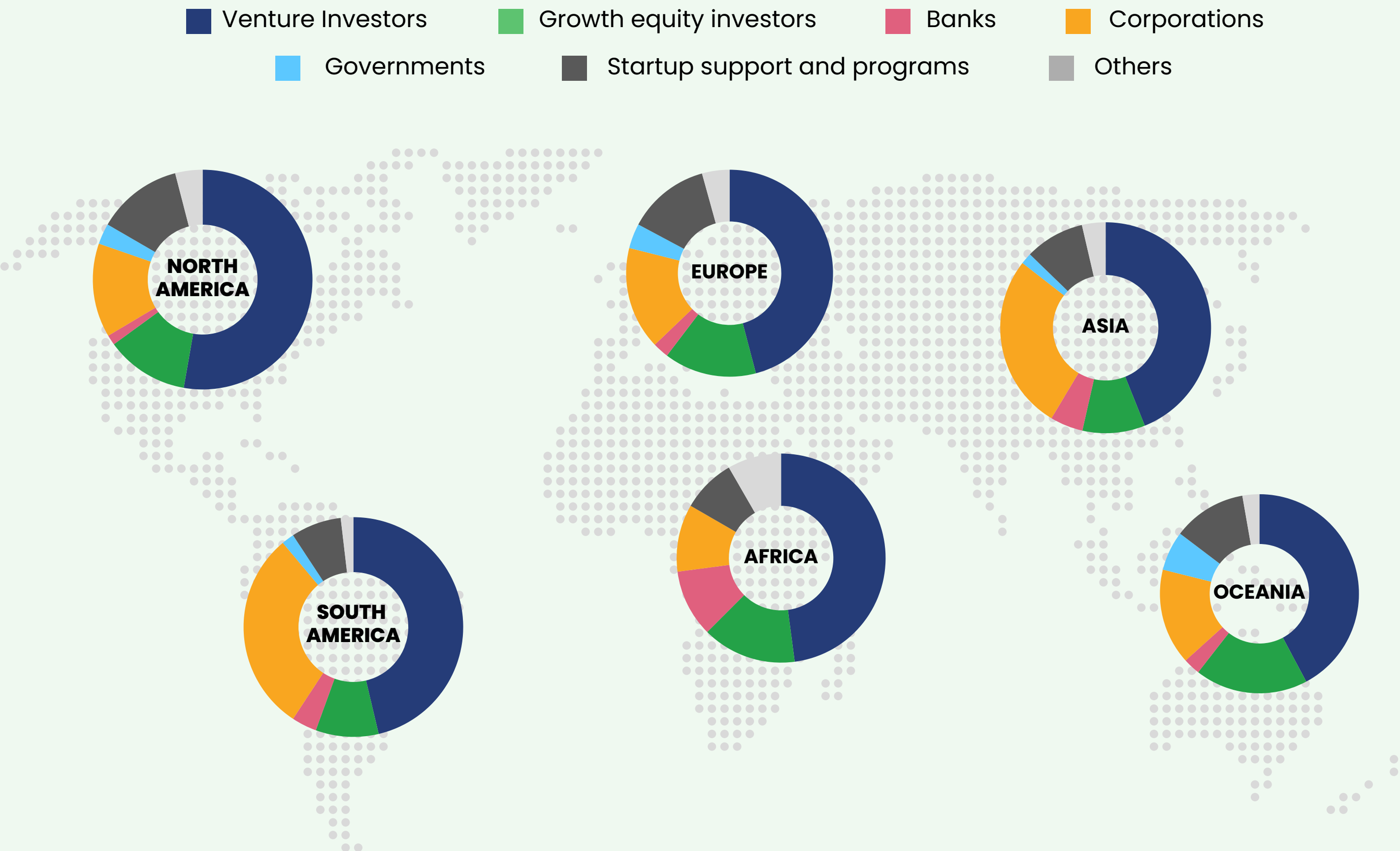


Source: Net Zero Insights

The figures illustrate funding activity in private Climate Tech ventures, covering equity, debt, grants, and other financial instruments, while excluding exits and post-exit financing.

Venture investors dominate with regional diversity in investor types

2024 active investor share by type and continent



Source: Net Zero Insights

The figures show funding activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. For details on investor categories, see the research methodology section.

Investor type proportions remain largely consistent between Europe and North America.

In 2024, venture investors dominate the global Climate Tech landscape, with significant diversification of investor types across continents. North America and Europe share similar investor type distributions, but North America sees a higher share of venture investors, driven by a stronger VC funding culture.

Corporations and government support are key drivers in emerging Climate Tech hubs

In Oceania, state-affiliated investors held the largest share, led by Australian and New Zealand government investments in energy and transport. In emerging markets like South America and Asia, corporations made up a larger share of investors in 2024 compared to established markets like Europe and North America.

US the top funded country, with Sweden and France following

Top 7 countries by funding (incl. non-dilutive) in 2024

ORGANIZATION HQ COUNTRY	FUNDING	YOY FUNDING VARIATION	DEALS
UNITED STATES	**$33.5B**	+3.1%	**2,314**
SWEDEN	$13.3B	+178.9%	149
FRANCE	$8.3B	+66.0%	268
UNITED KINGDOM	$6.5B	-5.2%	783
GERMANY	$3.4B	+8.1%	425
INDIA	$3.1B	-43.4%	304
CHINA	$1.9B	-63.8%	35

Source: Net Zero Insights

Mature hubs lead, emerging markets decline

The US dominates Climate Tech funding, with five of the top seven most-funded cities globally. Sweden and France show rapid growth, fueled by major giga-rounds in their capitals, while funding declines in China and India highlight shifting dynamics in emerging markets.

Top 7 cities in 2024 by funding (incl. non-dilutive) with 20+ deals

ORGANIZATION HQ CITY	FUNDING	DEALS
STOCKHOLM	**$8.1B**	55
LONDON	$4.8B	**326**
HOUSTON	$2.1B	68
DENVER	$1.6B	33
OAKLAND	$1.4B	33
NEW YORK CITY	$1.3B	120
SAN FRANCISCO	$1.2B	116

Source: Net Zero Insights

Europe leads 2024 funding with debt giga-rounds, US stays steady

Europe leads Climate Tech funding in 2024 with $38.88 billion (+45.1% YoY), significantly outpacing the US's $33.5 billion (+3.1% YoY), driven by large debt giga-rounds. Both regions saw sharp declines in deal counts, with Europe down 24.1% and the US down 20.8%, reflecting a contraction in deal activity. Active investor numbers also shrank, dropping 12.5% in Europe and 14.7% in the US, indicating a tightening investor landscape.

United States

	2024	YoY
Total funding	$36.5B	+12.3%
Deal count	1,541	-20.4%
Investors	1,975	-14.7%

Europe

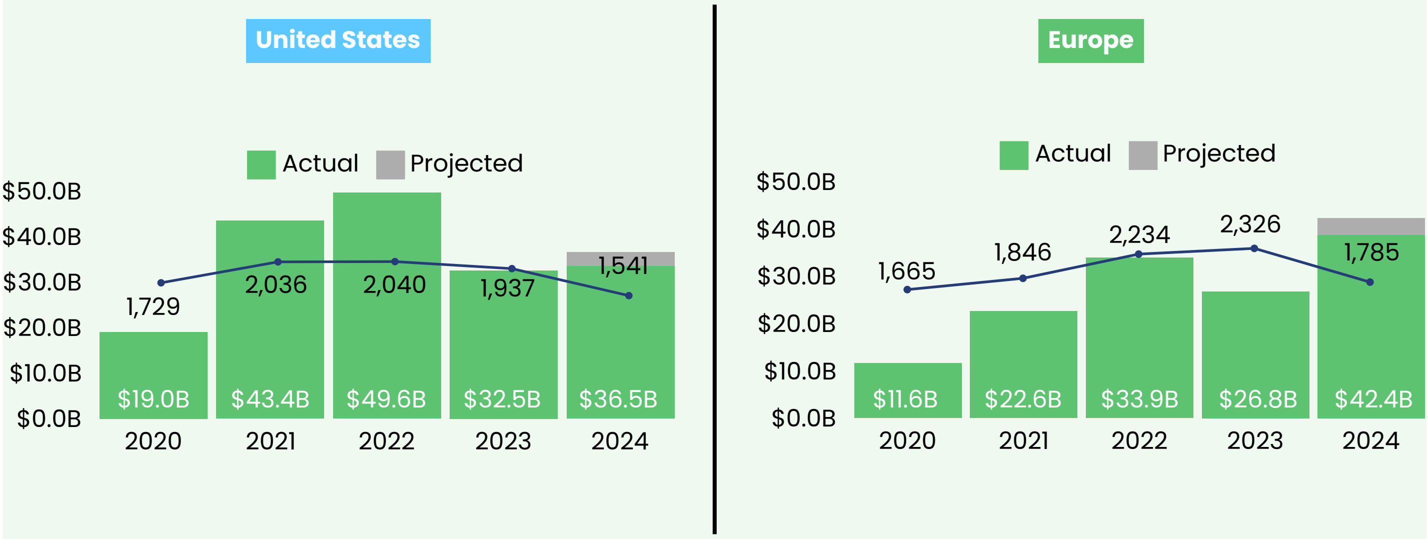
	2024	YoY
Total funding	$42.4B	+58.2%
Deal count	1,785	-23.3%
Investors	2,751	-12.5%

Source: Net Zero Insights

The figures show funding and deal activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. 2024 data incorporate end-of-year projections.

Both markets are shifting toward fewer but higher-value investments, driven by a more concentrated investor base compared to 2023.

Yearly funding and deal activity

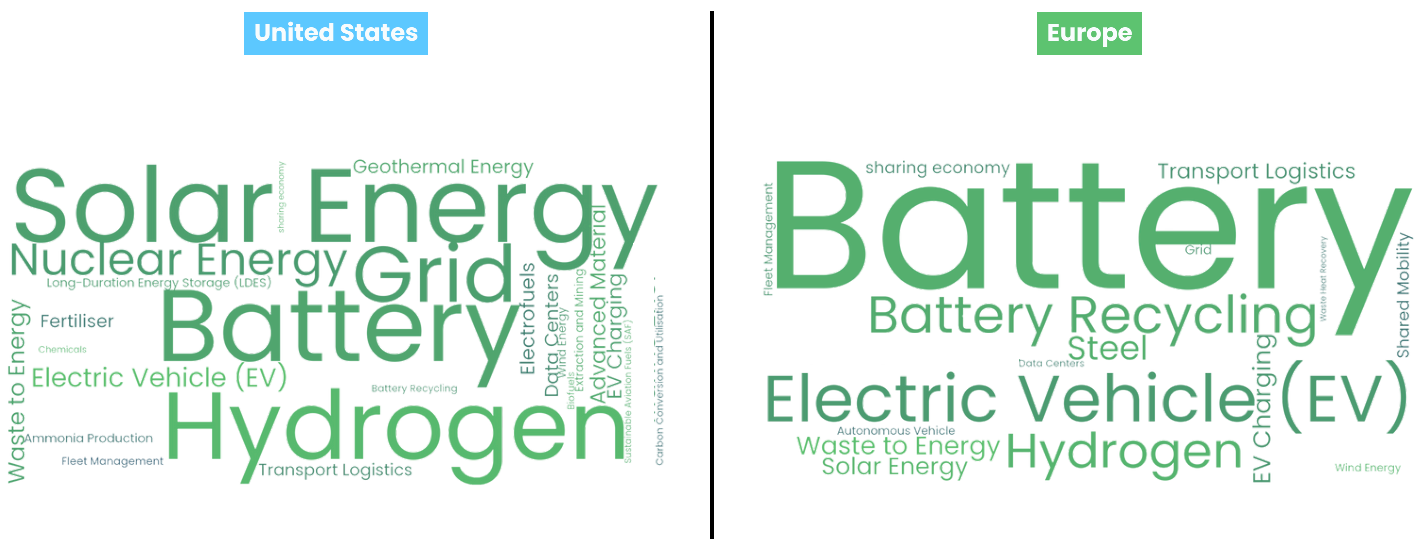


Source: Net Zero Insights

The figures illustrate funding and deal activity in private Climate Tech ventures, covering equity, debt, grants, and other financial instruments, while excluding exits and post-exit financing. 2024 deal count incorporates end-of-year projections.

Despite a decline in deal counts across both regions, this year, the US and Europe are set to reverse 2023's downward funding trend. Europe, in particular, is projected to experience a substantial funding increase, driven by large debt rounds supporting the scale-up and adoption of battery solutions. The 2024 funding landscape in Europe reflects a more risk-averse strategy, with a notable rise in late-stage funding for energy, transport, and circular economy solutions. In contrast, the US shows a more balanced growth approach, combining investments in mature technologies with significant funding for breakthrough innovations such as hydrogen and nuclear energy. This dual focus underscores its commitment to fostering both immediate and long-term Climate Tech advancements.

2024 most funded solutions

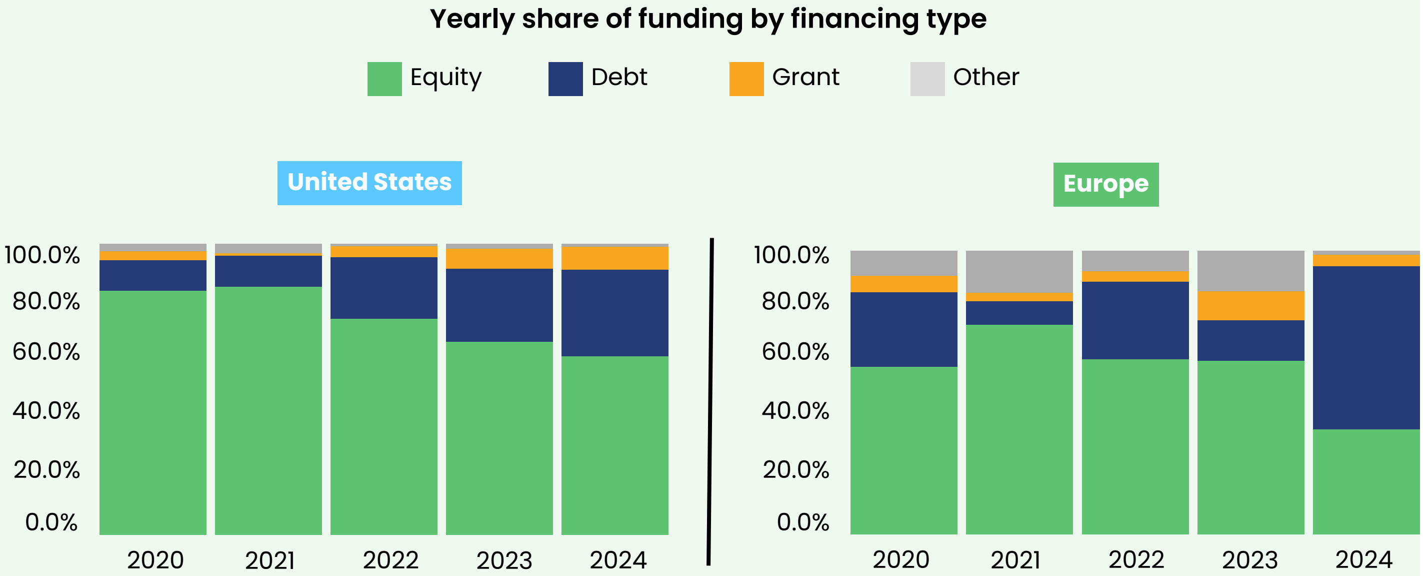


Source: Net Zero Insights

Despite fewer deals, both regions sustained 2023 equity funding

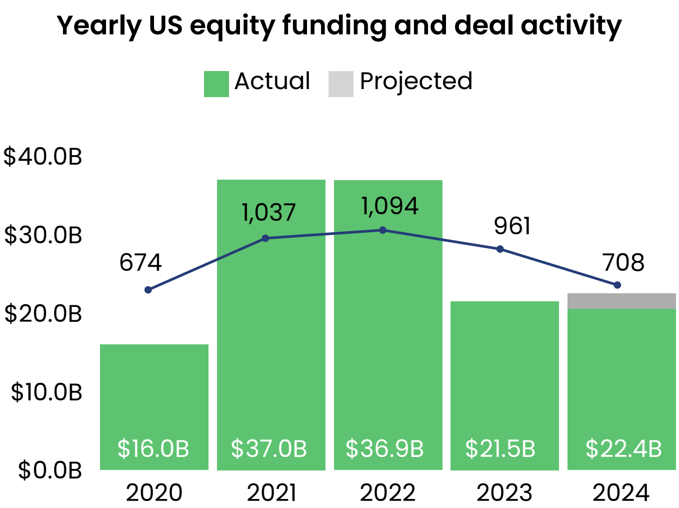
Both markets have seen an increased focus on capital diversification, with strong debt financing, particularly in Europe. Additionally, contrary to the global trend, equity investment showed signs of stabilization in 2024 following a sharper slowdown in 2023. However, deal activity declined in both markets, with a sharper drop in the US (-26% YoY) compared to Europe (-19% YoY).

The consequence for the US, where a strong culture of equity financing is driven by a heavy reliance on venture capital for high-growth, high-risk innovations, was having fewer startups receiving funding this year. While the US sustained equity funding levels in 2024, Europe saw a slight decline, largely offset by the strong presence of debt financing.

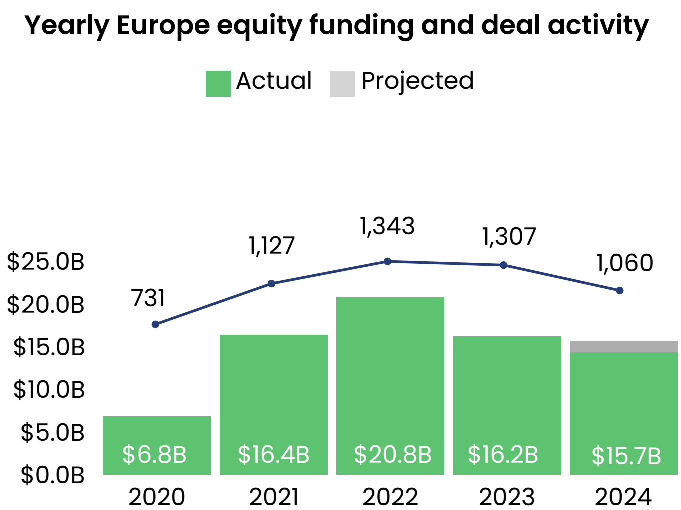


Source: Net Zero Insights

The figures show funding activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing.



Source: Net Zero Insights



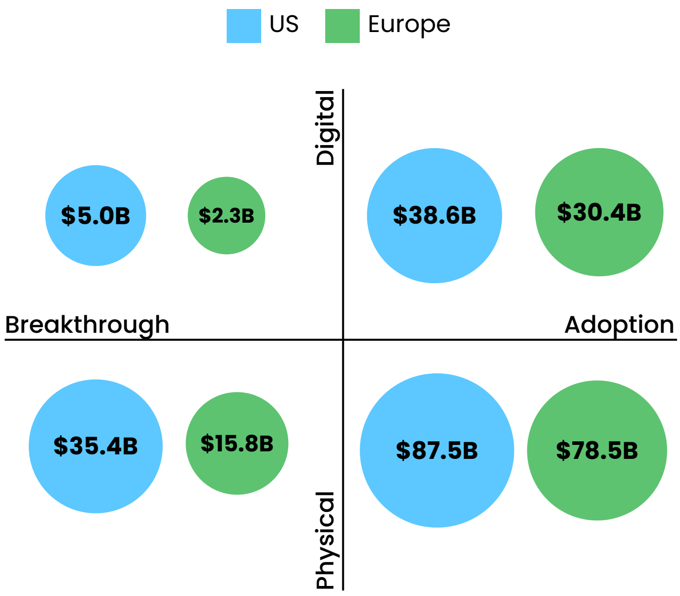
Source: Net Zero Insights

The US excels in breakthroughs, while Europe thrives in adoption

The United States leads in breakthrough innovations, investing heavily in pioneering technologies with long R&D timelines. Between 2020 and 2024, the US allocated $35.4B to physical breakthroughs and $5B to digital breakthroughs, totaling $40.5B—more than double Europe's $18.1B investment during the same period. This positions the US as a global leader in developing next-generation solutions.

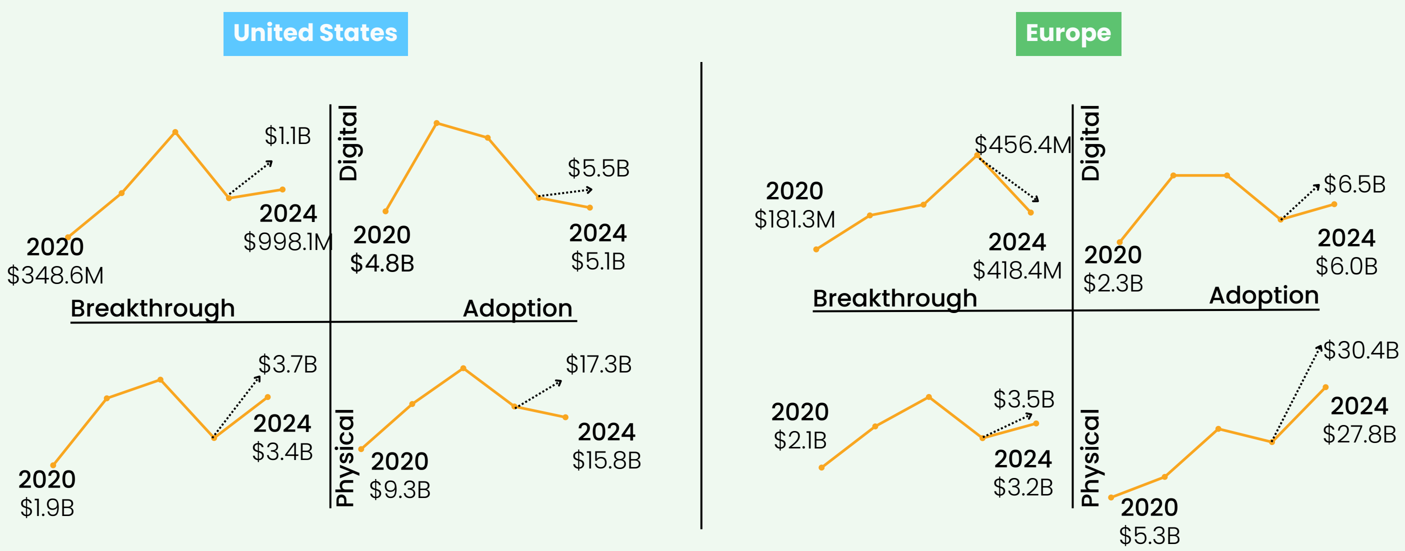
Meanwhile, both the US and Europe have made significant investments in adopting proven innovations. The US directed $87.5B toward physical adoption and $38.6B toward digital adoption, while Europe channeled $78.5B and $30.4B, respectively, showcasing strong commitments to scaling established technologies.

Last 5 years funding (2020–2024) by region across PI framework



Source: Net Zero Insights

Yearly funding by region across PI framework



Source: Net Zero Insights

The figures show funding in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. Analysts manually categorized all deals over $70 million from the past five years for consistency.

Diverging equity funding strategies in the US versus Europe

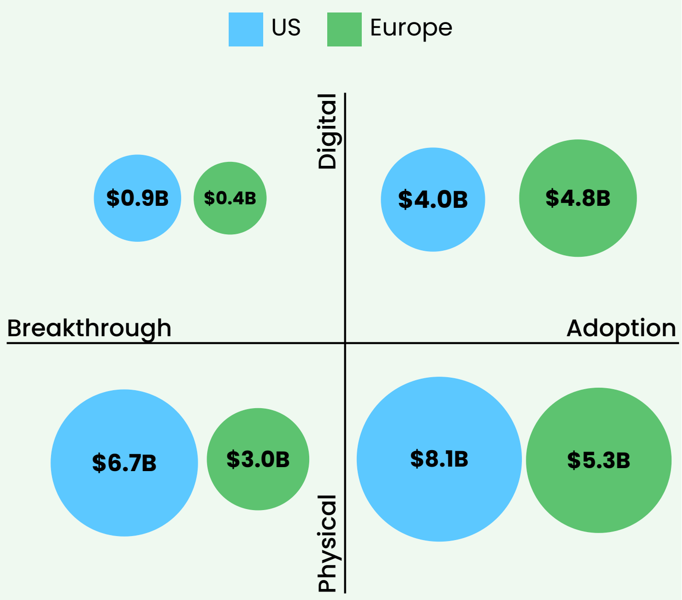
In 2024, equity funding in the US and Europe prioritized the adoption of physical solutions.

In 2024, about 40% of equity funding in the US and Europe targeted adoption-stage physical solutions.

This reflects two trends: investors' preference for physical over software solutions in Climate Tech and investors' cautious approach to breakthrough solutions due to uncertainties amid a significant equity funding downturn.

In the US, adoption-stage funding for physical solutions was only 21% higher than breakthrough-stage funding, compared to a 76% in Europe, highlighting US VCs' relatively stronger focus on breakthrough-stage solutions.

2024 equity funding across PI framework



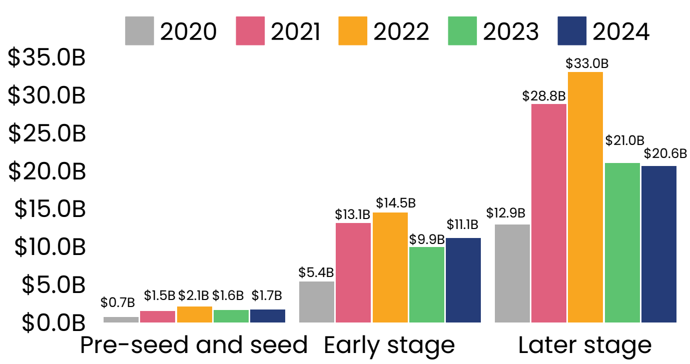
Source: Net Zero Insights

The figures show equity funding in private Climate Tech ventures, excluding debt, grants, other instruments, exits, and post-exit financing. For details on the product-innovation framework, see the research methodology section.

In 2024, Europe's later-stage funding peaked at $24.1B, driven by a few large debt giga-rounds. In comparison, US late-stage funding held steady at $20.6B, well below its 2022 peak of $30B.

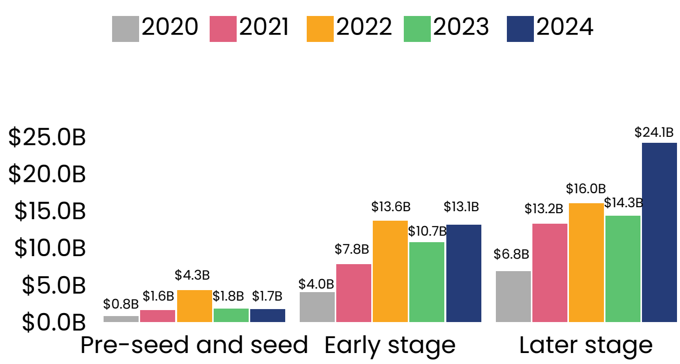
Contrary to global trends, early-stage funding experienced significant growth in both regions, with the US rising to $11.1B in 2024 (+12.1% YoY) and Europe to $13.1B (+22.4% YoY). Pre-seed and seed funding stayed steady at 2023 levels.

Yearly US funding by capital stage



Source: Net Zero Insights

Yearly Europe funding by capital stage

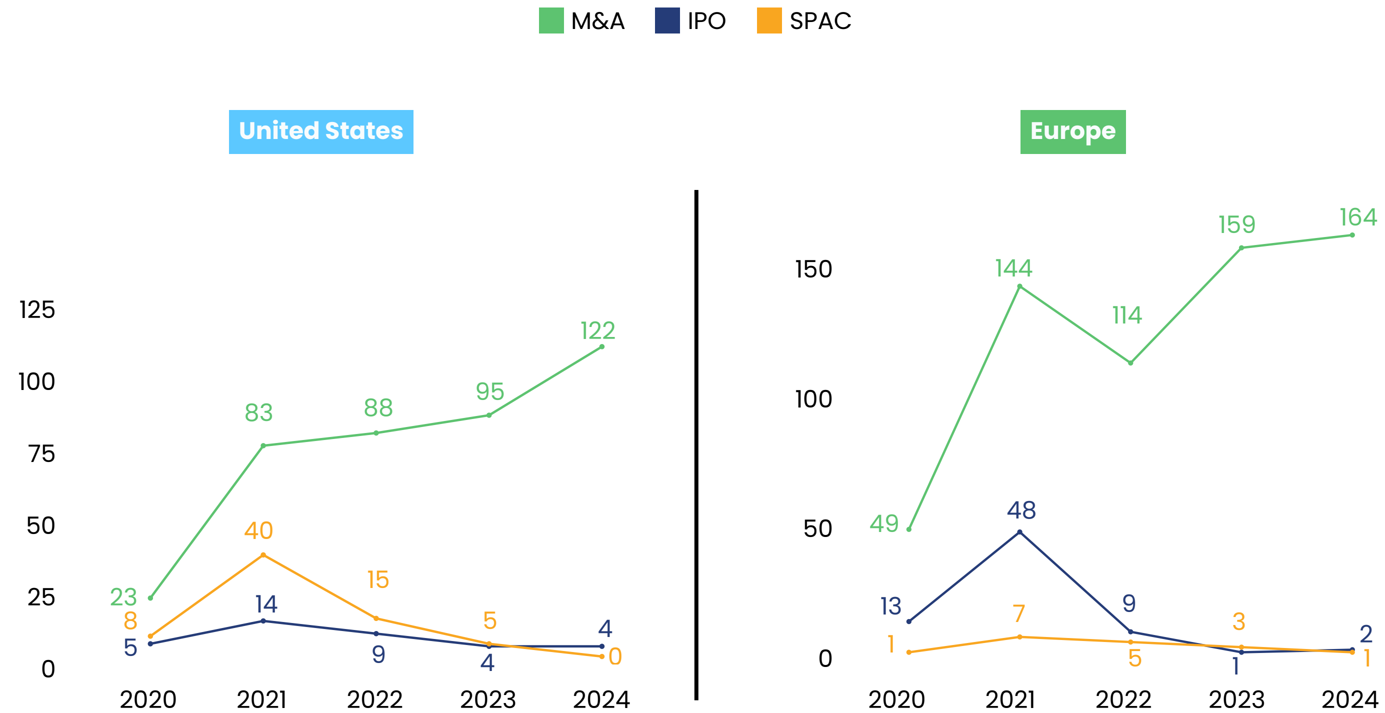


Source: Net Zero Insights

M&As lead in both regions, with Europe showing greater activity

M&A remains the dominant exit strategy in Climate Tech, with Europe leading at 160 deals in 2024, surpassing the US, which reached 121 after recovering from a dip in previous years. IPO activity has sharply declined in both regions, with the US showing almost no public market exits in 2024, reflecting ongoing challenges in leveraging public offerings. SPAC exits have become nearly nonexistent, highlighting a clear shift toward M&A as the preferred and most reliable exit route for Climate Tech ventures.

Yearly exit activity by region



Source: Net Zero Insights

The figures illustrate exit activity including IPOs, SPACs, mergers and acquisitions of private climate tech ventures, while excluding venture and post-exit financing. The 2024 data in this chart are actual as of November 30th 2024.

Regional exit trends reflect contrasting sentiments between European and North American investors

Our 2024 survey reveals a sharp contrast in investor sentiment on the exit environment. Pessimism is higher among North American investors, with 71% expecting further deterioration, compared to 44% of European investors.

04 Thematic and trends analysis

This section highlights key topics of the year, enriched by industry leader insights, and standout startups across climate change challenge areas, offering a clear view of progress and opportunities.

Case Study

Technological trends in Climate Tech

Overview

As the world confronts the urgent need to address climate change, a wide variety of technologies have emerged to help decarbonize sectors like energy, industry, and transportation. From hydrogen and energy storage to carbon capture, utilization, and storage (CCUS), the global focus concentrate now towards these Climate Tech solutions. However, a deeper look reveals that focusing on individual technologies may not be the most effective strategy. Instead, understanding market needs and aligning solutions to address specific challenges can ensure long-term, scalable, and cost-effective outcomes.

This analysis explore the importance of adopting a broader, more flexible approach to the deployment of Climate Technologies, which involves prioritizing market demand, creating stable investment conditions, and avoiding short-term hype cycles that may lead to unsustainable growth or disillusionment.

1,500+

number of Climate Tech solutions categorized within the Net Zero Insights taxonomy.

> 45%

of investors who participated in the SOCT 2024 survey highlighted CCUS, hydrogen, and energy storage as the emerging Climate Tech trends and technologies they are most excited about.

> 50%

of ecosystem supporters who responded to the SOCT 2024 survey identified energy storage as the sector most in need of support from ecosystem contributors. Notably, two-thirds of these responses came from representatives of governmental organizations.

This case study was developed by synthesizing key insights from the aggregated perspectives of stakeholders who participated in the State of Climate Tech 2024 survey, as well as findings from a focus group discussion that brought together diverse viewpoints on the topic.

Additionally, the analysis was enriched by data gathered from the Net0 Platform and supporting desk research.

Energy storage plays an important part in stabilising the grid by storing renewable energy for use during low-generation periods. CCUS enables emissions-intensive industries, like cement and steel, to manage CO2 that's otherwise hard to abate, while hydrogen serves as a versatile energy solution for challenging-to-electrify sectors, including heavy-duty transport and certain industrial processes.

However, scaling these technologies faces challenges, including economic hurdles, inadequate policy and regulatory frameworks in certain regions, and concerns about safety and reliability. A key barrier is accessing capital for the first commercial demonstrations, which is essential to establish feasibility and performance, particularly for resource-intensive technologies like hydrogen. While modular and stackable designs can help reduce upfront capital costs, obtaining initial funding remains a major hurdle.

For example, hydrogen is critical for decarbonizing industries like steel and fertilizer production, yet it faces infrastructure challenges, such as CO2 storage and transportation, despite supportive policies like the US 45Q tax credit.

A balanced approach that integrates investment in scaling and commercialization alongside tech innovation is needed to achieve cost-effectiveness and resilience. Addressing sector-specific adoption challenges—such as those in agriculture and land use that require distributed, tailored approaches—is also essential. By defining clear use cases and filling knowledge gaps across sectors, stakeholders can create a unified approach to climate action and foster collaboration that accelerates the integration of main technological trends, driving toward a low-carbon economy.

A collaborative effort

A big challenge in scaling new Climate Technologies lies in the gap between innovation teams and core business units in large corporations, which may be hesitant to disrupt established processes. Even after successful pilot tests, moving to large-scale adoption requires significant investment and commitment from industry leaders. Mismatched expectations between startups and industrial players regarding scale and volume exacerbate the challenge, as traditional industries tend to adopt new technologies more gradually, influenced by regulatory frameworks and cautious innovation approaches.

Sectors with high visibility or under regulatory scrutiny, such as automotive and luxury goods, are more proactive in adopting sustainable solutions to meet public expectations. In contrast, traditional sectors like construction tend to adopt technologies more slowly and in a compliance-driven manner. Public-private partnerships, along with the engagement of major industrial players, are essential for driving the integration of green technologies across value chains and addressing the challenges of scaling.

Government involvement plays a pivotal role in catalyzing investment by acting as both a partner and customer for early-stage technologies.

This approach provides a long-term commitment and reassures investors of market reliability. Additionally, balancing regulation to foster demand while avoiding excessive market expectations is crucial for both mainstream and niche market adoption. Targeting specific sectors like aviation, where early-stage technologies can gain traction, could offer a clear entry point and help capture investor attention across diverse industrial applications.

There has been a growing investor engagement in Climate Tech, however, converting this interest into tangible investments remains a challenge. This is primarily due to the difficulty in aligning the language around technological readiness, commercialization, and adoption. Climate Technologies, especially those in emerging sectors like CCUS and hydrogen, often don't fit traditional investment models due to their multidisciplinary nature and long-term development cycles. Tailoring investment models and metrics to better evaluate these technologies could help boost investor confidence.

Bridging the gap between technological progress and market readiness is key. As technologies move from being perceived as “exotic” to more mainstream, investors will begin to understand them in the same way they do established tech like batteries, solar, and wind.

Avoiding tunnel vision

The Climate Technology landscape is evolving rapidly, but a critical question remains: how can we ensure these technologies make a meaningful, long-lasting impact? While technologies like hydrogen, CCUS, and energy storage are often touted as solutions to decarbonization, they do not represent a one-size-fits-all fix for every climate-related challenge.

Each Climate Technology offers unique advantages and limitations depending on the context. For instance, hydrogen may be a promising solution for decarbonizing certain industrial processes but may not be the best choice for other applications due to cost and energy inefficiencies.

The growing popularity of hydrogen and carbon capture technologies, often driven by subsidies and hype, highlights **the risk of a "tunnel vision" approach,** where these technologies are overemphasized without assessing their true potential in different settings.

One of the most powerful takeaways from discussions among industry experts is the importance of **aligning technologies with the specific problems they address.** Rather than broadly advocating solutions like energy storage or hydrogen, we must focus on market demands across sectors. For instance, some industries may need base-load power from nuclear energy, while others benefit from energy storage or hydrogen for decarbonizing heavy industries.
Clear case studies showcasing these fits will enable more strategic investment and deployment.

A holistic approach to Climate Technology deployment is necessary — focusing on market needs rather than overemphasizing any single solution.

Long-term vision and signals

For the successful deployment of Climate Technologies at scale, long-term, patient capital is crucial. The industry needs more institutional investors, like pension funds, to engage with emerging technologies, providing the necessary funding to drive large-scale development.

A focus on long-term market stability—rather than speculative short-term returns—will ensure that technologies such as energy storage, carbon capture, and hydrogen have the time and support they need to mature and become viable solutions for decarbonization.

This is also enabled by **creating stable demand signals.** Ensuring that there is a stable, growing demand for these emerging technologies is paramount. This is where governments and policymakers have an essential role to play.

By creating long-term incentives through procurement policies, tax credits, and clear market signals, they can foster a conducive environment for technology developers and investors.

Stable market demand will help companies secure funding and confidently deploy their solutions, enabling technologies to mature and become cost-competitive over time.

Avoiding boom-bust cycles

History shows that overhyping specific technologies or sectors can lead to boom-and-bust cycles, where market enthusiasm leads to rapid investment, followed by disillusionment when expectations fail to meet reality. This has happened before with electric vehicles and lithium production, where short-term media hype led to volatile investment trends. A better approach would be to ensure that emerging technologies avoid these cycles, instead being subject to steady, data-driven investment, grounded in the long-term outlook for decarbonization.

Creating stable demand signals and long-term incentives is crucial to securing continuous investment in emerging technologies. Mobilizing patient, long-term capital will support the maturation and scaling of these technologies, avoiding the pitfalls of speculative, short-term trends

Summary

The future of Climate Technologies lies not in focusing narrowly on a single solution, but in leveraging a diverse toolkit of solutions that are tailored to the specific challenges and demands of different sectors.

The climate crisis requires coordinated, forward-thinking actions from governments, investors, and industry players. Only by adopting a flexible, holistic approach can we ensure that the technologies of today and tomorrow will truly meet the needs of the planet and its inhabitants.

Startups to Watch

Built environment

Sublime Systems	United States	#Materials
AIRA	Sweden	#HeatPumps
Quilt	United States	#HeatPumps
Materrup	France	#Concrete
Monumental	Netherlands	#Construction
Luxwall	United States	#EnergyEfficiency

Circular economy

Syre	Sweden	#TextileRecycling
samsara eco	Australia	#PlasticRecycling
ZwitterCo	United States	#WastewaterReuse
Cyclic Materials	United States	#REEsRecycling
PRISM Worldwide	United States	#TyreRecycling
Solarcycle	United States	#SolarRecycling

Emissions, control, reporting and offsetting

greenly	France	#CarbonFinance
Doconomy	Sweden	#Carbonfootprint
Carbonfact	France	#CarbonAccounting
Climate Impact X	Singapore	#CarbonCredits
retraced	Germany	#Traceability
Atlas Metrics	Germany	#ESGreporting

Energy

Renew Home	United States	#Energy management
Kaluza	United Kingdom	#EnergyPlatform
Sion Power	United States	#Batteries
Marvel Fusion	Germany	#FusionEnergy
Reverion	Germany	#Biogas
Eavor	Canada	#Geothermal

Food and agriculture

Formo	Germany	#Dairy
ElicitPlant	France	#ResilientCrops
Cradle	Netherlands	#ProteinEngineering
Heura	Spain	#Alternative Protein
InnerPlant	United States	#CropMonitoring
Ohalo	United States	#ResilientCrops

GHG capture, removal and storage

Graphyte	United States	#CarbonRemoval
44.01	United Kingdom	#CarbonStorage
MissionZero	United Kingdom	#DAC
carbyon	Netherlands	#DAC
Planetary	Canada	#OceanAlkalinity Enhancement
280earth	United States	#DAC

Industry

Halo Industries	United States	#SiCWafers
Magnus Metal	Israel	#MetalAlloys
Molten	United States	#HeavyIndustry
Calyxia	France	#MicroCapsules
unspun	United States	#Textile
iSABers Materials	China	#Semiconductors

Natural environment

Arbol	United States	#ClimateRisk
MuonSpace	United States	#Satellites
LiveEO	Germany	#EarthObservation
reforestAction	France	#Reforestation
XOcean	Ireland	#OceanData
OroraTechnologies	Germany	#WildfireDetection

Transport

启源芯动力	China	#BatterySwapping
Windrose	China	#ElectricTrucks
Electra	France	#FastCharging
Vela	France	#SailingCargo
HIF	Chile	#Electrofuels
moove	Netherlands	#EVFinancing

Water

Hydraloop	Netherlands	#Greywater Recycling
OceanWell	United States	#WaterDesalination
Puraffinity	United Kingdom	#WaterTreatment
SewerAI	United States	#SewerSystems
geodesic	Spain	#WaterTreatment
epiccleantec	United States	#WaterRecycling

Case Study

Strategic role of Corporations in Climate Tech

Overview

Corporations play a pivotal role in the Climate Tech ecosystem, contributing vital resources and expertise to drive innovation and commercialization. By providing funding, technical guidance, and access to markets, they serve as key enablers for startups working on transformative solutions.

Despite these strengths, corporate involvement is often limited by challenges such as slow decision-making processes, risk aversion, and economic constraints, which can hinder their ability to effectively support emerging technologies.

This case study examines the unique value corporations bring to Climate Tech, the obstacles they face in scaling their impact, and the strategies needed to overcome these barriers, fostering greater collaboration and innovation within the sector.

60%

of surveyed investors identified growing corporate commitments as a key driver of future Climate Tech growth.

$32.2B

represents the total value of deals involving corporations in 2024, making up approximately 35% of the forecasted $92B invested.

54%

Corporations responding to the SOCT 2024 survey emphasize that, beyond funding, providing technical support and advisory services is key to helping Climate Tech startups succeed.

This case study was developed by synthesizing key insights from the aggregated perspectives of stakeholders who participated in the State of Climate Tech 2024 survey, as well as findings from a focus group discussion that brought together diverse viewpoints on the topic.

Additionally, the analysis was enriched by data gathered from the Net0 Platform and supporting desk research.

Strengths of Climate Tech corporate-startup collaborations

Corporations bring several strengths to the table in Climate Tech:

- **Capital and long-term investment**: Corporations possess the financial capacity to invest in high-risk, long-term ventures that are essential for the development of Climate Tech solutions. Unlike venture capitalists, who often demand rapid returns, corporations can afford to adopt a patient approach, enabling investments in hardware and deep-tech innovations with extended timelines.

- **Sector expertise and know-how**: Corporations, particularly those in energy, manufacturing, and other climate-related sectors, have the expertise necessary to help startups navigate the complex regulatory and technical landscapes. Startups often need guidance on standards, compliance, and market dynamics, which corporations can provide. This knowledge accelerates the scaling process for startups, helping them avoid costly pitfalls.

- **Speed and market validation**: Corporations can act as critical partners for startups by providing early validation for new products. By offering test facilities, customer networks, and market feedback, corporates help startups refine their solutions, facilitating faster entry into the market. This partnership reduces the time it takes for startups to transition from development to deployment.

Challenges in corporate-startup collaboration

Corporations face several challenges in integrating Climate Tech into their existing operations, primarily stemming from the mismatch between the rapid pace of innovation in the Climate Tech space and the slower adoption cycles typical of large corporate environments. A key challenge is the **slow corporate decision-making processes,** driven by bureaucracy and a preference for proven technologies, often create a mismatch between the fast-paced nature of startups and the more cautious, slower decision cycles of large corporations. This can hinder timely collaboration and innovation, preventing startups from getting the support they need to scale.

Another challenge is the **financial viability** of certain Climate Tech solutions. Many of these innovations, particularly in hardware and deep-tech, require substantial upfront capital and extended timelines to become commercially viable. Without strong government support and incentives to offset the initial costs, corporations may find it difficult to justify investing in technologies that don't offer immediate returns.

Government incentives play a crucial role in de-risking these investments and encouraging corporations to pursue Climate Tech solutions that may otherwise lack strong economic value in the short term.

Finally, **risk aversion and a focus on short-term returns** hinder corporate investment in transformative technologies, especially those that challenge existing business models. Many corporations **struggle to anticipate when emerging technologies will mature and deliver tangible benefits**, further delaying their commitment to disruptive innovations. This hesitation limits the adoption of promising solutions and slows progress toward sustainability goals.

Overcoming these barriers requires **a shift toward long-term strategic thinking**, emphasizing sustainable innovation over immediate financial returns. Corporations must embrace calculated risks and commit to investments that, while not offering quick rewards, hold significant long-term potential.

Recommendations for overcoming challenges

To overcome these challenges, the following strategies can be adopted:

- Leveraging government support: government policies and financial incentives, such as the US. Inflation Reduction Act (IRA), can help offset the risk for corporations by supporting the initial costs of investing in Climate Tech. Such policies make it easier for corporations to take risks on technologies that may not generate immediate returns but have the potential for significant long-term impact.

- Adopting a cultural shift in corporations: corporations need to shift their mindset and embrace long-term, risk-tolerant investment strategies. By adopting a more flexible, forward-thinking approach, corporations can invest in disruptive technologies that align with sustainability goals.

- Encouraging cross-corporate collaboration: surprisingly, collaboration with competitors can also accelerate innovation. By pooling resources and sharing risks, multiple corporations can support early-stage startups more effectively. Co-investment models, where competitors invest in a startup together, reduce individual risk and increase the likelihood of success.

Leveraging government incentives, fostering a culture of long-term investment, and engaging in collaborative co-investment models are essential to driving progress in the Climate Tech sector.

Best practices for effective partnerships

Successful partnerships in the Climate Tech space between corporations, startups, and other stakeholders thrive on collaborative strategies and adaptable processes. Transforming competitors into allies fosters innovation, particularly when Corporate Venture Capital (CVC) teams engage in knowledge-sharing and co-investment. Collaboration across different CVCs is equally beneficial, enabling pooled capital resources and coordinated support for startups, which increases the likelihood of successful innovation.

To overcome the persistent barrier of slow corporate decision-making, involving operational team members early and ensuring that top management acknowledges the strategic value of climate innovation are crucial.

Corporations should be encouraged to take calculated risks on breakthrough technologies rather than limiting investments to market-ready solutions. Startups, in turn, should prioritize showcasing their offerings' performance, cost-effectiveness, and customer value, alongside their sustainability benefits, to align with corporate objectives.

For larger corporations, mergers, acquisitions, or dedicated venture funds can serve as impactful tools to drive Climate Tech advancement. Medium-sized firms, however, may benefit from focusing on strategic partnerships. Structures such as CVC arms, accelerators, and cross-CVC collaborations offer versatile and scalable models for companies of all sizes to support and grow Climate Tech innovations effectively.

Summary

Climate Tech is complex but holds tremendous potential, and its commercial deployment remains a significant challenge. Strategic partnerships are vital, with corporates testing and providing actionable feedback to startups, guiding them in the right direction. Achieving meaningful impact requires a collective mindset shift, extensive collaboration, and substantial capital from both corporations and governments. This journey toward scalable Climate Tech solutions cannot be accomplished in isolation.

Corporations play a vital role in advancing Climate Tech by providing patient capital, sector expertise, and market validation. However, to accelerate progress, they must overcome challenges such as slow decision-making, risk aversion, and economic viability concerns. By embracing a long-term, risk-tolerant investment strategy, collaborating with competitors, and leveraging government incentives, corporations can significantly enhance their impact on Climate Tech. Ultimately, fostering deeper collaboration across the innovation ecosystem—comprising startups, corporations, investors, and governments—will be key to scaling transformative technologies and accelerating the transition to a sustainable future.

Case Study

Public-private partnerships in advancing Climate Tech

Overview

Public-private partnerships (PPPs) are emerging as a cornerstone for accelerating innovation and adoption in the Climate Tech sector. By harnessing the unique strengths of both the public and private sectors, these collaborations address complex challenges while driving scalable and impactful solutions.

While the public sector provides regulations, incentives, and critical infrastructure, the private sector contributes financial agility and innovation. Together, they form a dynamic force capable of overcoming barriers such as regulatory complexities, funding gaps, and geopolitical pressures.

This case study, informed by insights from a focus group of industry experts, delves into how well-structured PPPs leverage complementary strengths to accelerate the commercialization, deployment, and scalability of Climate Technologies.

$34.2B

represents the total value of funding activities involving government and state-backed entities in the last 5 years (2020-2024).

50%

50% of the ecosystem contributors interviewed in the SOCT 2024 survey considered public-private partnerships as the most effective collaboration for advancing Climate Tech solutions.

80+

Governmental entities were involved in Climate Tech funding activities during the year 2024.

This case study was developed by synthesizing key insights from the aggregated perspectives of stakeholders who participated in the State of Climate Tech 2024 survey, as well as findings from a focus group discussion that brought together diverse viewpoints on the topic.

Additionally, the analysis was enriched by data gathered from the Net0 Platform and supporting desk research.

The Role of PPPs in advancing Climate Tech

Collaborative PPPs enable risk-sharing in climate investments and increase funding opportunities, particularly through subsidies and incentives that drive tech adoption.

The public sector sets the foundation for Climate Tech growth by establishing regulations, offering financial incentives, and investing in critical infrastructure. For example, initiatives like the European Commission's EIC accelerator illustrate how public sector support actively bolsters climate efforts. This program's focus on urban resilience helps cities better manage the effects of climate risks such as floods and droughts, while providing crucial support to vulnerable populations like the elderly.

Climate Tech also requires specialized infrastructure that supports public goods such as renewable energy, water management, and EV charging networks. These projects often need public sector involvement to help mitigate long-term risks and align with international frameworks, which can harmonize cross-regional objectives and promote collaboration.

The private sector excels in financial agility and technological innovation. Companies introduce novel solutions and adapt to market demands more quickly than public entities. When coupled with public sector support, private enterprises can mitigate long-term environmental risks, ensuring solutions are both sustainable and scalable.

However, experts noted that overly stringent regulations could inadvertently stifle innovation. A balanced regulatory approach ensures rapid deployment while safeguarding environmental integrity.
Through well-structured PPPs, both sectors can establish a supportive environment that enhances the scalability and impact of Climate Tech solutions, making sustainable progress toward global climate goals.

Overcoming barriers to Climate Tech deployment

Scaling Climate Tech across markets faces various interconnected challenges, with geopolitical dynamics, inconsistent policy support, and customer adoption obstacles among the most prominent. Geopolitical tensions, particularly in Europe, often shift public and political attention toward national security over climate priorities. This redirection of focus can reduce interest and long-term commitment to climate initiatives, with recent conflicts further emphasizing energy independence and resilience as national security priorities. Consequently, policies that initially promoted climate resilience now often blend environmental and defense goals, potentially complicating the widespread adoption of Climate Technologies.

The success of Climate Tech initiatives, particularly through public-private partnerships (PPPs), is heavily dependent on stable, long-term policy support. Changing political administrations can disrupt climate agendas, underscoring the need for policies resilient to political shifts. Consistency in policy is crucial to maintaining affordable and resilient energy infrastructure, which ensures that Climate Tech projects are protected from political volatility.

Moreover, market-specific challenges include limited customer adoption and funding constraints, particularly beyond early-stage investments. Specialized infrastructure, such as renewable energy grids and EV charging networks, demands substantial investment and risk mitigation. Public involvement in funding and aligning these projects with international standards promotes scalability and cross-border collaboration.

Main barriers to Climate Tech deployment include:

- Infrastructure challenges
- Policy instability
- Geopolitical and public awareness hurdles

Innovative financial mechanisms in PPPs

Effective financial mechanisms are crucial for supporting Climate Tech, particularly through innovative funding models and targeted public interventions. Government funding is most impactful when it strategically supports high-need areas and large utility players with significant public assets. For example, public procurement policies mandating low-carbon materials and recycled content help drive demand for sustainable solutions, positioning the government as a major market influencer.

Blended finance models, which combine public and private funds, offer a promising approach by distributing risks. Programs like the European Innovation Council (EIC) play a key role by matching funds for startups, covering up to 50% of funding needs, and attracting private follow-on investments. This approach not only mitigates risks for private investors but also facilitates the growth of high-potential Climate Tech ventures. Similarly, government-backed funds can provide critical support for pioneering but high-risk projects, absorbing part of the financial risks to incentivize private sector involvement.

Flexible funding models, such as grants, are critical for early-stage Climate Tech startups that require a longer development runway. Expanding grants to fully cover costs for early-stage projects can help these companies advance without the pressure of meeting typical venture capital timelines. In the US, for instance, some grants fund up to 100% of early-stage climate initiatives, offering a sustainable blueprint that can be adopted globally to support climate innovation.

Consistent and aligned policy support is also key, particularly in smaller countries and regions like the EU, where harmonized policies can prevent competitive disparities. For example, clear policies in hydrogen projects across EU member states can create a balanced market landscape, fostering healthy competition and supporting Climate Tech adoption at scale. As Climate Tech financing evolves, these mechanisms highlight a progressive learning curve toward addressing global climate challenges more effectively.

Strengthening PPPs in Climate Tech

Expanding financing options beyond traditional venture capital and grants is essential to scaling Climate Tech infrastructure. Alternative financing methods, such as project financing and partnerships with alternative lenders, provide a diversified financial ecosystem capable of supporting climate projects across local, regional, and national scales. These additional pathways broaden the funding landscape, enabling projects that require long-term investment and significant capital. Integrating multiple funding sources, including both public and private capital, is key to developing effective blended finance models that accommodate the wide range of risk levels and project scopes in Climate Tech. Blended finance enables flexibility and adaptability, making it possible to address complex and diverse needs across the Climate Tech ecosystem.

A comprehensive approach combining policy and financial mechanisms is crucial for successful Climate Tech adoption. This includes a mix of mandates, such as carbon pricing, alongside incentives and public guarantees to mitigate investment risks.

Public guarantees, as proposed by entities like the European Commission, create a stable environment that encourages private investment by offering a layer of risk protection. A balanced policy approach that combines incentives with regulatory support is more effective in advancing decarbonization and promoting the adoption of various Climate Tech solutions, supporting a more sustainable transition across sectors and regions.

Fostering Collaboration through Knowledge Sharing

Breaking down information silos and enabling cross-border knowledge exchange is key to accelerating global Climate Tech adoption and amplifying the impact of PPPs.

Summary

To harness the full potential of Climate Tech, actionable steps must strengthen PPPs:

1. Policy Alignment: Establish resilient, harmonized policies that support long-term growth.
2. Financial Innovation: Expand blended finance models and alternative funding mechanisms to diversify capital sources.
3. Awareness and Education: Promote public understanding of Climate Tech’s economic and environmental benefits.
4. Collaboration Networks: Facilitate platforms that encourage cross-sector partnerships and knowledge sharing.

With cohesive efforts, PPPs can unlock the potential of Climate Tech, driving solutions to address urgent environmental challenges while fostering sustainable economic growth. This multifaceted approach provides a roadmap for leveraging collective strengths to achieve large-scale decarbonization and resilience.

Case Study

Singapore’s green playbook

Overview

As climate change accelerates, nations play a critical role in promoting sustainability. Singapore, a proactive global citizen, is committed to decarbonizing its economy and advancing global climate goals. With innovative policies, smart regulations, and strategic partnerships, Singapore is driving new sustainability opportunities for businesses in Southeast Asia and beyond.

Singapore as a gateway to Asia’s Green Economy

Southeast Asia is projected to become the world’s fourth-largest economy by 2030. The region holds significant renewable potential, with 16 terawatts of solar energy and 1 terawatt of wind energy, along with nearly a quarter of the world’s natural carbon sinks, including the largest stock of blue carbon. However, substantial efforts are needed to fully realize this potential and achieve net-zero emissions by 2050.

Singapore is central to this vision, providing a trusted base for companies committed to decarbonization and regional climate goals. The Sustainable Jurong Island - an initiative involving 100 leading companies from the energy, petrochemical, and specialty chemicals sectors - demonstrates Singapore’s leadership in forging alliances for green growth.

With its strategic location, skilled talent, and supportive business environment, Singapore is well-positioned to drive green growth and impact. Its infrastructure and forward-thinking policies support businesses on their sustainability journeys while contributing to the global green transition. By collaborating locally and internationally, Singapore aims to build a more sustainable and resilient future.

By 2050

SIngapore’s commitment to achieve net-zero emissions

$2.7B

invested in equity funding in Southeast Asia over the past 5 years (2020-2024), with a 38% CAGR.

National Sustainability Initiatives

Singapore integrates environmental stewardship with economic growth through a comprehensive approach. Central to this is the Singapore Green Plan 2030, a roadmap that aims to position the nation as a global leader in sustainability. The plan is built on five main pillars: City in Nature, Sustainable Living, Energy Reset, Green Economy, and Resilient Future. For businesses, the Green Plan 2030 opens up opportunities to adopt practices that lead to sustainable development and growth, from green building designs to energy-efficient manufacturing processes.

The Finance for Net Zero Action Plan further reinforces Singapore's position as a global hub for green finance. Empowering companies' decarbonization journey by funding their sustainability initiatives and scaling efforts to generate long-term value, it offers incentives such as green bonds, loans, and Environmental, Social, and Governance (ESG) investing.

A component of Singapore's multi-pronged decarbonization strategy is the National Hydrogen Strategy, which positions low-carbon hydrogen as a critical element in the country's energy transition. Hydrogen, as a versatile energy carrier, is set to complement other low-carbon energy sources in Singapore's power mix, including solar energy and imported electricity. This approach is part of a broader effort to address emissions from various sectors while Singapore continues investing in a diverse range of measures such as enhancing energy efficiency, deploying renewable energy technologies, and advancing sustainable practices in energy-intensive sectors. By 2050, hydrogen could supply up to half of Singapore's power needs, playing a pivotal part in decarbonizing hard-to-abate industries like petrochemicals, shipping, and aviation.

The national strategy focuses on five crucial areas: experimenting with advanced hydrogen technologies, investing in research and development, pursuing international collaborations, undertaking long-term infrastructure planning, and supporting workforce development. These efforts aim to accelerate hydrogen adoption across various sectors, offering businesses new avenues to decarbonize their operations.

For example, the upcoming Keppel Sakra Cogen Plant, Singapore's first hydrogen-ready power plant, is set to significantly lower carbon emissions in the power sector. Such initiative exemplifies Singapore's proactive approach to driving innovation and sustainable energy solutions.

Facilitating public-private partnerships

Singapore's proactive approach to sustainability is characterized by strong industry partnerships, ensuring companies have the resources, expertise, and support needed to transition to greener practices.

Central to this is the Singapore Carbon Market Alliance (SCMA), a collaboration between EDB and IETA (formerly the International Emissions Trading Association). The SCMA empowers Singapore-based companies to access high-quality Article 6 carbon credits by enhancing their knowledge of carbon markets and connecting them with leading international carbon credit suppliers to meet corporate climate goals. Other noteworthy case studies illustrate Singapore's collaborative spirit. Climate Impact X (CIX), a joint venture between major financial players DBS, Standard Chartered, Singapore Exchange and state investor Temasek, leverages Singapore's robust regulatory and financial infrastructure to build credibility and trust in the global carbon market.

The city's transparent governance and strong market mechanisms amplify CIX's efforts, aligning with Asia's broader emissions reduction goals.

Similarly, Neste's sustainable aviation fuel (SAF) initiative with Changi Airport Group marks a significant step in reducing aviation emissions. EDP Renewables' involvement in the SolarNova program, which deploys solar photovoltaic systems across public housing and government buildings in Singapore, advances the city's solar energy ambitions.

These initiatives balance long-term environmental goals with immediate energy needs, reinforcing Singapore's leadership in sustainable urban development and setting a model for other cities.

Renewables

Singapore is committed to a major renewable energy transition. With 95 per cent of its power coming from imported natural gas, the country aims to import up to 6 GW of low-carbon electricity by 2035, addressing land limitations for solar power. To this end, the Energy Market Authority (EMA) is actively trialling and approving renewable energy imports from Cambodia, Indonesia, and Vietnam, while also expanding initiatives in solar, energy storage, and hydrogen.

Singapore's commitment to advancing renewable energy is supported by a dynamic network of project financiers. Major banks, such as DBS and UOB, play a major part in this transition.

DBS has pledged to reduce its absolute financed emissions in the oil and gas sector by 92 per cent by 2050 from 2020 levels, with renewables financing now constituting about half of its power portfolio. Meanwhile, UOB's Sustainability-Linked Advisory, Grants and Enablers (SAGE) programme, in collaboration with EnterpriseSG, offers SMEs preferential loan rates for meeting sustainability targets, helping them tackle challenges such as high costs and limited expertise.

Additionally, investment institutions BlackRock and Temasek have raised S$1.4 billion through Decarbonization Partners to invest in critical decarbonization technologies such as clean hydrogen and low-emissions battery recycling, further accelerating the shift to renewable energy.

Singapore is also leading the charge in advancing renewable energy technologies through strategic partnerships with top technology providers. Schneider Electric's collaboration with EDB has led to the creation of NaviX Solutions, offering full lifecycle management for enterprise power and cooling assets via an Infrastructure-as-a-Service model.

These efforts are complemented by substantial investments from both local and international developers. French multinational utility company Engie formed a joint venture with ComfortDelGro (CDG) to enhance electric vehicle charging infrastructure in Singapore. Meanwhile, Singapore-based Vena Energy is advancing a 125MW solar project in Queensland with Amazon, reinforcing Singapore's vital role in global renewable energy initiatives.

02

The stakeholders voice

01 Innovators

This section profiles this year's innovators, highlighting notable founders and their startups. It shares their insights on challenges, future prospects, and emerging opportunities, showcasing the individuals shaping Climate Tech innovation.

Profile of 2024 innovators

Overview

Innovators that secured funding in 2024 were almost evenly distributed between those developing predominantly physical solutions and those focusing on digital innovations.

While physical solutions remain slightly more prominent among funded organizations, the leading enabling technology in 2024 was digital—particularly artificial intelligence (AI).

The electricity sector secured the highest overall funding, fueled by substantial investments in batteries, hydrogen, and solar energy, which were among the top-funded solutions of the year.

53% hardware

53% of organizations funded in 2024 are predominantly focused on developing solutions centered around physical innovations.

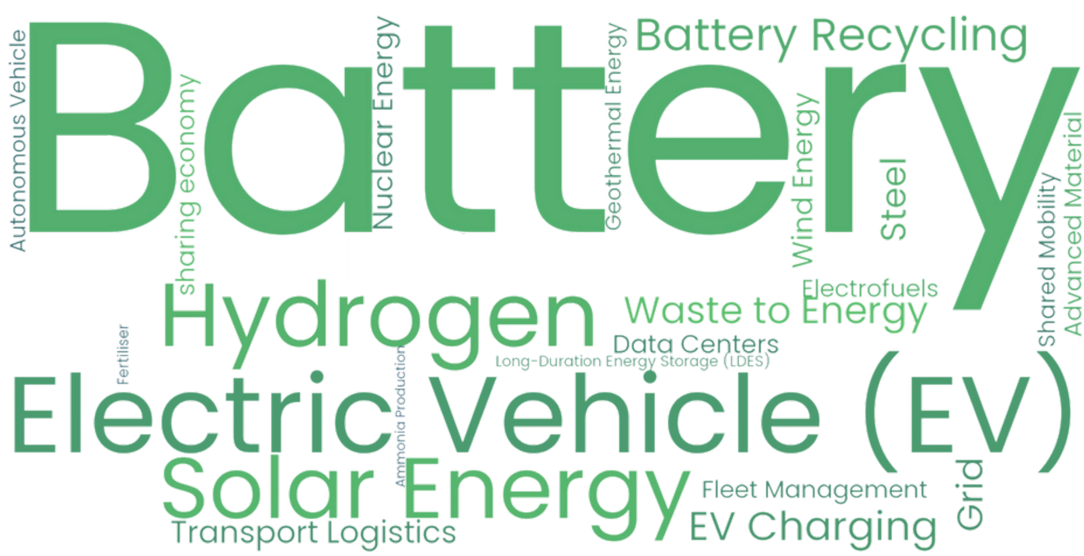
Artificial Intelligence (AI)

emerged as top enabling technology, receiving the highest funding among in 2024.

Electricity

was the top most funded sector in 2024.

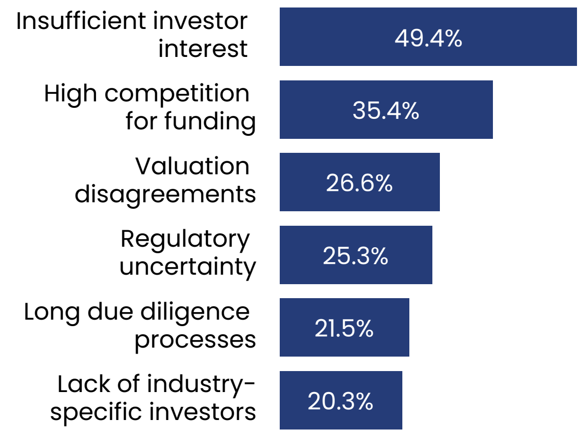
2024 top funded solutions



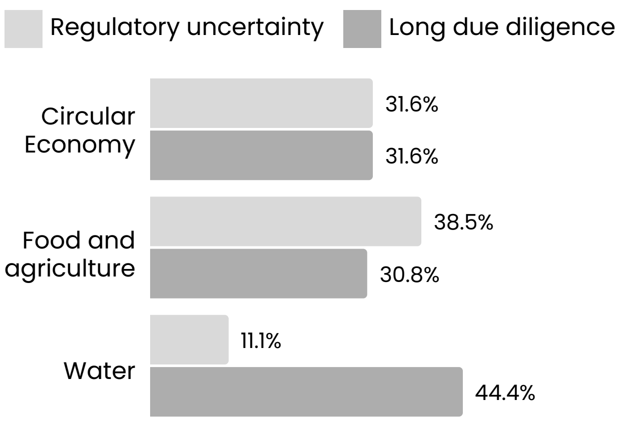
Source: Net Zero Insights

Fundraising challenges and opportunities
from NZI's SOCT 2024 survey

What are the main challenges faced by innovators in raising funding ?



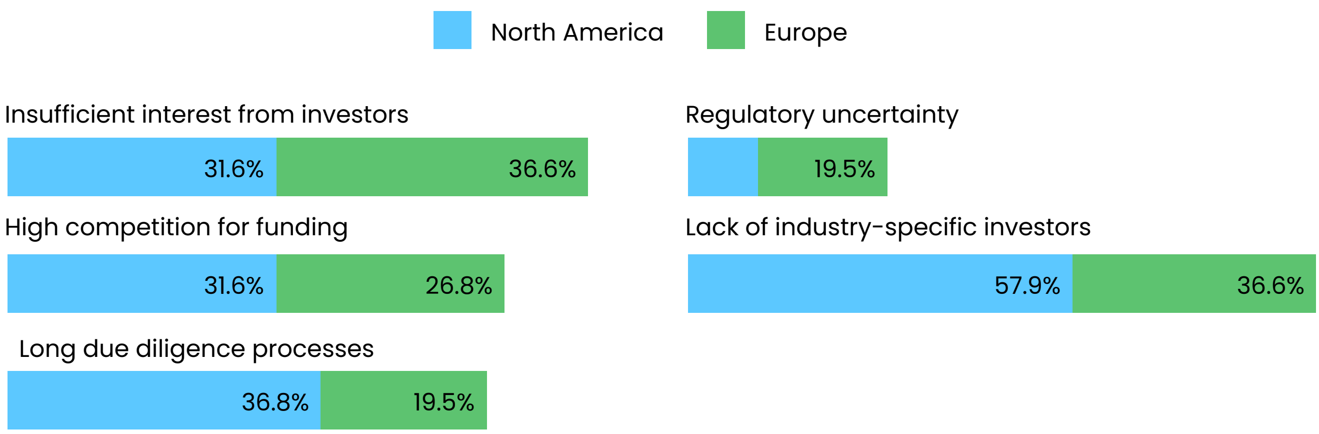
Which sectors face due diligence and regulatory challenges as main obstacles?



Among surveyed innovators, the most common funding challenge was the lack of industry-specific investors, followed by limited interest. Circular Economy innovators reported this challenge most frequently (75%), compared to lower rates in Natural Environment (33%) and Food and Agriculture (37%). Regionally, 58% of North American respondents cited this issue, compared to just 37% of European innovators.

Long due diligence processes was mentioned as a challenging factor by nearly double the percent of North American innovators than by Europeans. Regulatory uncertainties and lengthy due-diligence processes were major challenges in the Food, Agriculture, and Water sectors due to extensive government oversight.

How do these challenges differ across the most funded regions?



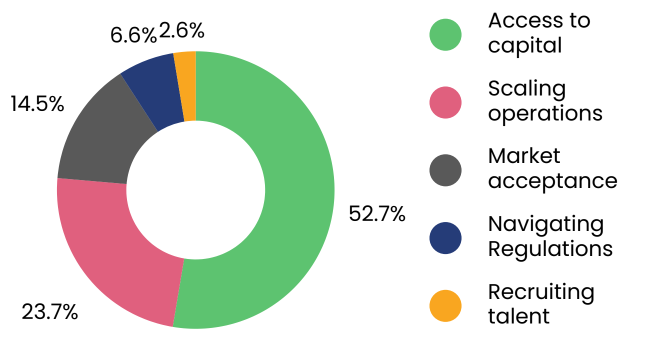
Fundraising challenges and opportunities
from NZI's SOCT 2024 survey

Innovators with hardware solutions frequently cited a lack of industry-specific investors as a challenge, while insufficient investor interest was the most common issue among software-focused innovators. This reflects the funding disparity between hardware and software solutions in recent years and highlights the perceived scarcity of hardware-specialized investors.

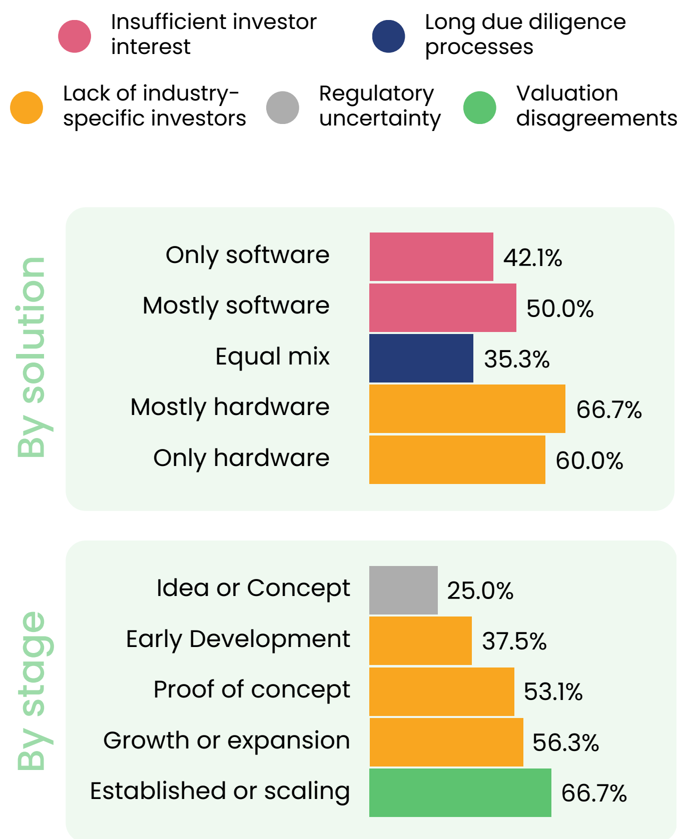
The lack of industry-specific investors was the most common challenge across developmental stages, except for the idea/concept and established/scaling stages. In the established/scaling stage, valuation disagreements were reported by two-thirds of respondents, underscoring heightened risk perceptions in this phase.

Investor and funding-related challenges, such as accessing capital and scaling operations, were consistently identified as the biggest hurdles in the current market environment. In contrast, talent recruitment and regulatory concerns were the least cited challenges among innovators.

What are the biggest challenges currently faced by your company?



What is the biggest challenge innovators face in the current market environment?



Thought Leader Interview

Desolenator

Desolenator is revolutionizing industrial water security through breakthrough solar-powered desalination technology. Our flagship SP40 system delivers what traditional desalination cannot: a modular, affordable, and zero emission solution producing 1,000 m³ of customized water daily while eliminating toxic brine discharge.

Our patented technology delivers unmatched advantages by capturing 4x more solar energy than conventional systems, integrating with waste heat streams, enabling 24/7 off-grid operations, flexibly treating all sorts of saline water (from seawater to contaminated wastewater), delivering ultrapure or custom-mineralized water, and achieving zero liquid discharge through our integrated ZLD module.

Alp Katalan
Business Development Manager

Alp is a Business Development Manager at Desolenator, a pioneering desalination startup delivering affordable, sustainable, and pure water to water stressed enterprises and communities. He is working on accelerating the commercialisation of Desolenator's solution, and building strategic partnerships.

Alp has spent his entire career working on climate and innovation, from running government-backed clean energy accelerators and advising businesses on how to reduce their carbon footprint at the Carbon Trust, to investing in early stage startups at the Climate Tech VC World Fund.

He holds an MSc in Sustainability, Enterprise and the Environment from the University of Oxford's Smith School.

How does Desolenator's solar thermal desalination technology address global water scarcity, particularly in regions facing severe water stress?

As we are targeting industrial water users operating in water-stressed regions, our solution can perfectly serve key water-intensive sectors (such as advanced manufacturing and controlled environment agriculture). Our modular, decentralized solution is versatile and customizable, offering a resilient, high-quality freshwater source to reduce water risk and prevent plant shutdowns, capacity loss, and supply chain disruptions. By solving industrial water stress, we also avoid scenarios of tense water competition between municipal and industrial/agricultural water users in scenarios of increasing climate-induced droughts.

We help industries build resilience against water scarcity affordably and sustainably. Our low-maintenance solution, free of membranes, consumables, or toxic chemicals, offers a low cost of water. With waste heat integration or our patented solar thermal panels, we also power plants with renewable energy while avoiding toxic brine production.

With your flagship pilot plant delivered to Silal, what have been the key takeaways from developing your solution to meet the local water demands?

Our flagship pilot plant was commissioned in August of 2024, and is an enhanced system providing custom remineralised water for greenhouse crop irrigation and clean cooling. Our pioneering project partners are Silal (the UAE's national Food & AgTech company) and its parent company ADQ (Abu Dhabi's sovereign wealth fund). This project demonstrates how using brackish (i.e. saline) groundwater to grow crops in the desert can boost the UAE's food and water security. It builds on key lessons from our first pilot plant, optimizing technology, refining protocols, and improving supply chain management.

The learnings allowed us to build the plant in less than half the time of our first pilot.. We are now looking forward to building on this massive success to scale our FOAK commercial plant.

What value do your investors bring beyond funding, and how do they support Desolenator's strategic growth?

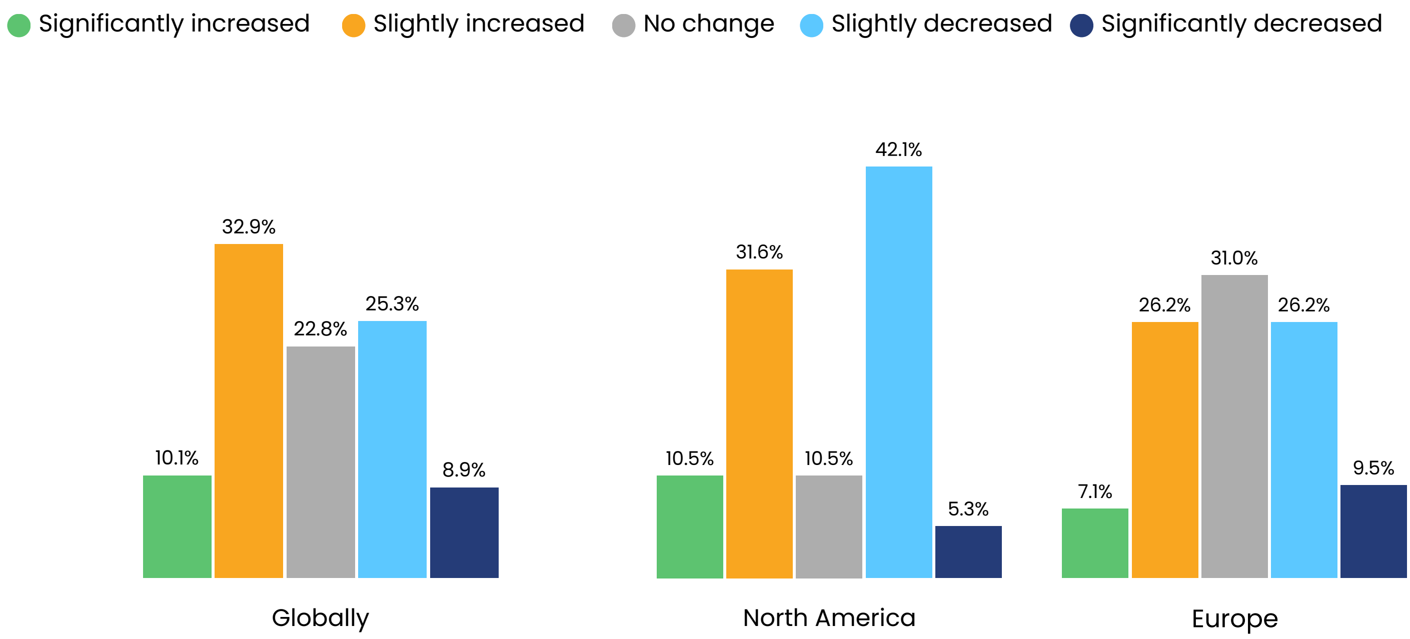
What we have always looked for in investors are (a) shared values in terms of securing the water-energy nexus, and (b) the value-add they can bring beyond capital. Although water tech funding has historically been more limited than Climate Tech (although this is starting to shift), we've successfully gathered an incredible group of backers. For example, our latest investor, Hexagon AB (via R-Evolution), has helped us develop digital tools to deploy plants faster, cheaper, and smarter. They are also enabling AI-driven Digital Twin simulations to optimize plant design and shorten sales cycles. Similarly, Katapult Ocean (a leading early stage ocean VC) connects us with key stakeholders and events, while our experienced Angel backers open doors to potential customers and partners, sharing invaluable industry knowledge.

What major milestones are you aiming for with your recent funding, in terms of both technology and market expansion?

A key milestone is securing an offtake agreement for our first-of-a-kind (FOAK) largescale plant. We're working with world-leading blue chip companies operating in water stressed areas to help them secure their industrial water assets. It's crucial to partner with industry pioneers focused on sustainable innovation. We're also consolidating and streamlining our supply chain for scaling, using pre-commissioned, containerised solutions to improve efficiency and reduce costs. Finally, driving sales through regional champions will further extend our reach and help us develop cutting edge projects globally.

Market trends and sentiment
from NZI's SOCT 2024 survey

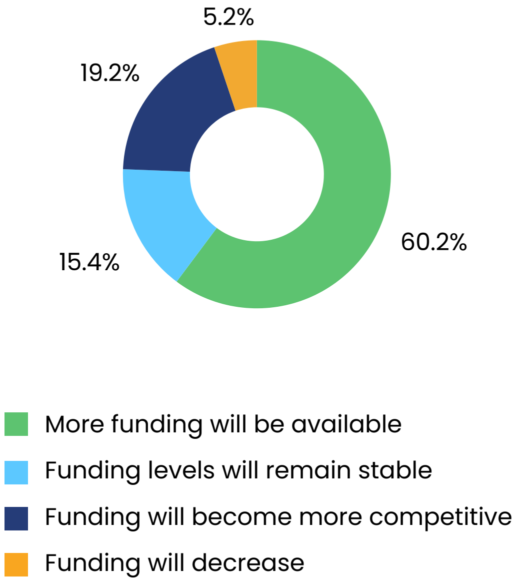
How has investor interest in Climate Tech changed in 2024 as compared to the previous years?



Globally, more innovators see rising investor interest in Climate Tech in 2024 compared to previous years. Nearly 60% remain optimistic about increased funding availability over the next 1–3 years.

European innovators largely perceive investor interest as steady in 2024, while North American innovators show more polarized views, with higher proportions seeing interest either increasing or decreasing. Notably, 11.7% more North American innovators report a decline in funding compared to their European counterparts.

How do innovators expect funding to change in the next 1–3 years?



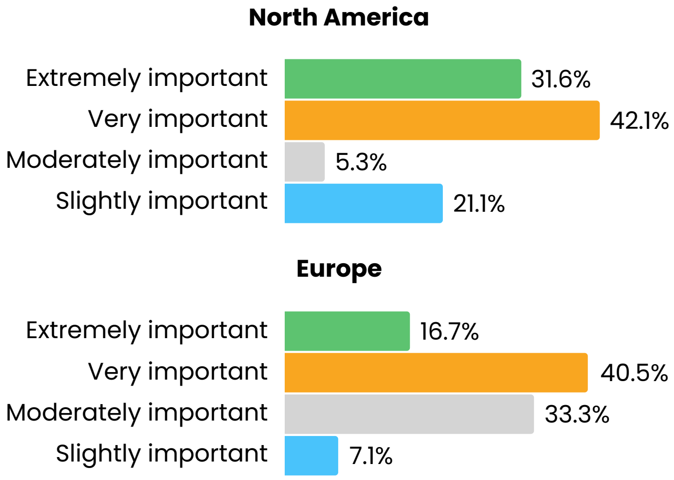
Market trends and sentiment
from NZI's SOCT 2024 survey

Regionally, innovators' views on the role of public policy and government support align closely with their perceptions of the market environment and investor interest.

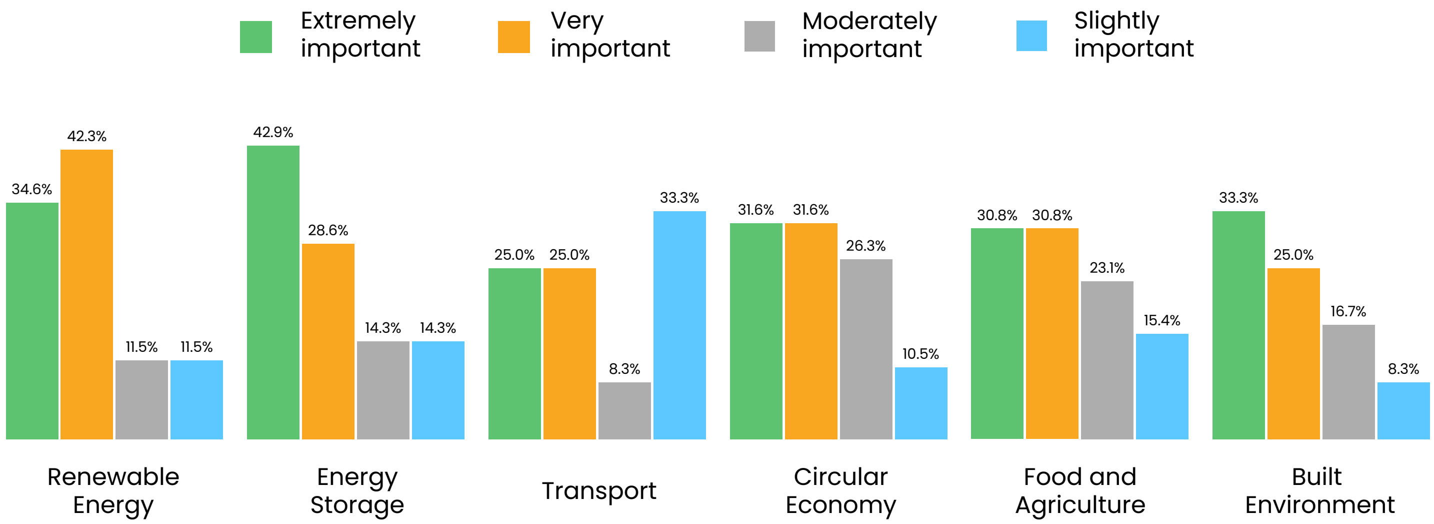
According to the survey, nearly 75% of North American innovators rated public policy or government support as extremely or very important, compared to 56% of European innovators. This disparity reflects regional differences in funding sources, challenges, and investor sentiment. For example, 48% of North American and 35% of European innovators reported a slight or significant decline in investor interest in Climate Tech in 2024.

Additionally, only 7% of European innovators received government grants, compared to 16% in North America. Public policy and government support are particularly critical in sectors with strong regulatory oversight such as energy and agri-tech. Emerging sectors like circular economy and built environment also heavily rely on government support due to long development timelines and delayed ROI, which deter private investors.

How important is government support or public policy to your company's success?



How does the importance of public policy or government support differ across sectors in determining a company's success?



Thought Leader Interview

Mykor develops and scales the next generation of building insulation that considers both human and planet health.

Mykor's products respond to the new regulations addressing the issues of the current insulation on the market; requiring better fire safety and lower embodied carbon.

Our products offer an alternative to conventional insulation with low embodied carbon and higher performance. With one single product, we tackle 18% of global carbon emissions.

Olivia Page
Co-Founder and CEO

Olivia Page, CEO and Co-Founder of Mykor. Originally from London, with a background in architecture, Olivia has been Portugal based since 2017 and fluent in Portuguese.
Some of her previous work include developing the one of first large scale 'hempcrete' residential projects in Portugal and developing innovative interior tiles from olive oil industry residues for French conglomerate LVMH.

Her award winning research prior to Mykor has been internationally recognised for her bio-based approach to achieving sustainability within construction. Since founding Mykor in 2022, she leads her team of 16 across the UK research lab and Portuguese pilot factory.

With Mykor, she aims to decarbonise the construction industry by developing and scaling the next generation of insulation that considers both planet and human health.

Mykor has seen a steady growth curve since 2022. How have you managed to maintain this growth trajectory in a highly competitive ecosystem?

The key to maintaining our growth trajectory lies in the unique value proposition (USP) of our product and technology. We stand out not only for the innovation and intellectual property behind our process but also for the exceptional performance of our product in construction applications. This combination of cutting-edge technology and proven results has positioned us as a leader in addressing a significant gap in the market. While the construction industry is highly competitive, there is a growing demand for sustainable and high-performance solutions like ours, and our ability to meet this demand with a superior offering has been instrumental in driving our continued growth.

What role do strategic partnerships with corporations play in a startup's growth? And how should startups approach maintaining these relationships for long-term success?

Strategic partnerships with corporations play an absolutely vital role in a startup's growth. In our case, our entire business model and scale-up plan are intentionally designed around such partnerships to leverage existing expertise, infrastructure, and commercial opportunities. These collaborations allow startups to accelerate their development, access resources that would otherwise be out of reach, and validate their solutions in real-world markets.

For long-term success, startups should approach these relationships by clearly demonstrating the value they bring to the table, while being realistic and avoiding overpromising. It's equally important to have a strong understanding of your intellectual property (IP) and a well-thought-out strategy for protecting it. This clarity not only builds trust but also ensures that negotiations with corporate partners are more productive and mutually beneficial.

Timing also plays a crucial role—engaging corporations at the right stage is key. We've found the sweet spot to be near the MVP stage, where the product has sufficient validation to inspire confidence but isn't so early that it lacks direction or feasibility for scaling.

What are the challenges connected to the financing of breakthrough solutions? What capital sources are needed for those companies to enhance their appeal to VCs?

Financing breakthrough solutions is challenging, particularly from TRL 9, which has also been coined, the Commercial Inflection Point (CIP), by Elemental Impact. To get funding at this CIP stage, it is important to have a good pilot stage from early on to prove the scalability of the process combined with commercial traction such as offtake agreements. The financing ideally comes at this stage from a combination of public, VC and debt finances. For this, the offtake agreements need to be well drafted, which is hard to get with larger players. Therefore, our route to market strategy is to go to more of the midsize players who are often more able to sign offtakes but this takes off the reliance on just one or two large clients.

What advice would you give to Climate Tech startups looking to innovate in a similar sector? Any tips for how to stand out to attract funding?

Prioritize scalability alongside MVP development, supported by advisors or a seasoned team to validate both based on practical realities and visionary goals. Seek non-dilutive funding early to maintain flexibility, as venture capital often demands stricter timelines and focuses on scaling rather than development. Focus on innovations that fill major market gaps with few competitors, targeting high-impact, complex challenges over easy wins. Lastly, recognize the transformative potential of hardware—it is poised to drive the substantial environmental changes the world urgently needs.

Exit activity
from NZI's SOCT 2024 survey

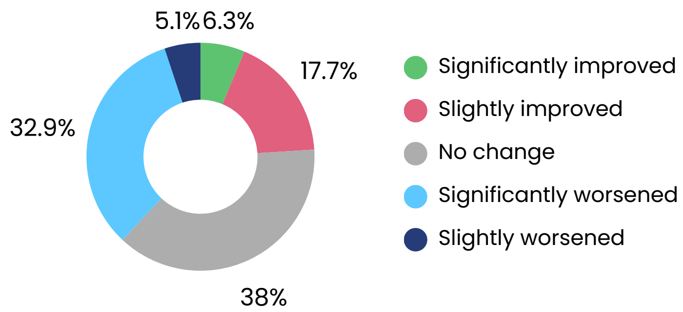
Innovators view exit environment as neutral but worsened compared to previous years

Most innovators in the State of Climate Tech 2024 survey viewed the current exit environment as neutral, though more respondents found it unfavorable than favorable. Over one-third of respondents believed the exit environment in 2024 had worsened compared to previous years.

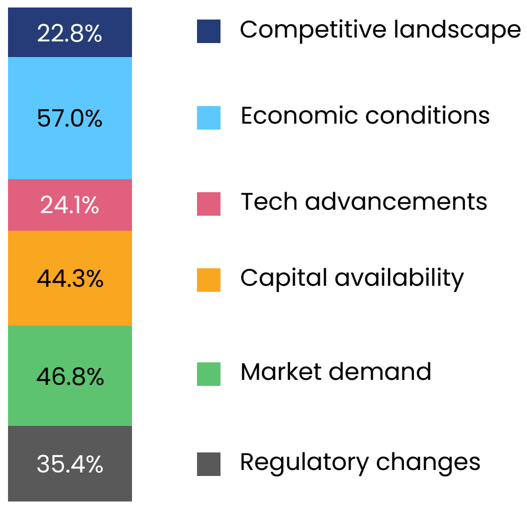
Technological advancements were cited as a key factor influencing the exit environment by 56% of Circular Economy and 55% of Industry sector innovators, compared to 15–25% across other sectors.

For most innovators, funding and market-related factors such as economic conditions, capital availability, and demand were the primary drivers of the exit environment. Only 23% cited "competitive landscape" as a factor, indicating limited concern about market competition shaping exit decisions.

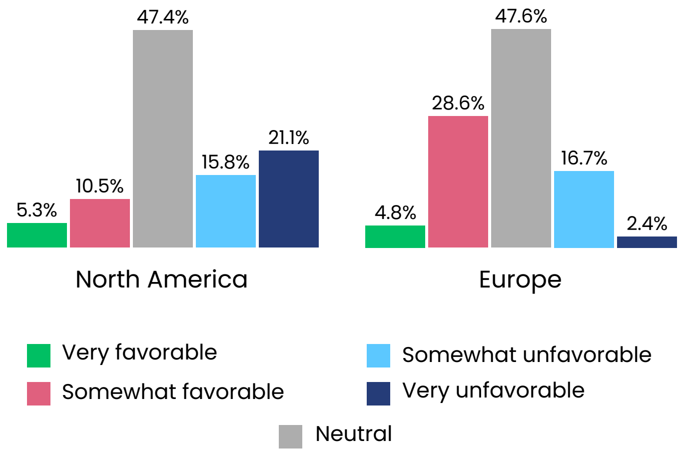
How has the 2024 exit environment changed compared to previous years?



What are the main key factors influencing the current exit environment?



How do you assess the current exit environment for Climate Tech startups?



Thought Leader Interview

SunGreenH2™ has developed and is commercialising an innovative hydrogen electrolyser technology specifically designed to overcome the high costs and energy demands associated with green hydrogen production (hydrogen produced by splitting water molecules with renewable electricity). Green hydrogen has enormous potential as a clean energy solution, yet traditional electrolysers rely heavily on expensive precious metals and consume significant energy, making green hydrogen up to five times more costly than hydrogen produced from fossil fuels. This price barrier prevents wide adoption of green hydrogen for decarbonising transportation, manufacturing, and energy storage. SunGreenH2™ has radically reduced capex by 50%, opex by 20% and eliminated precious metals in its systems.

Tulika Raj
CEO and Co-Founder

Tulika is an experienced commercial leader who has led $2B worth of transactions including fundraising and large project investments. She has hands-on experience of developing and growing renewable energy companies, coupled with more than 10 years of experience in clean energy investments (from solar to wind developers at growth to exit stages), at major energy companies and renewable funds.

Tulika holds a bachelor's degree from Indian Institute of Technology, Delhi and an MBA from the London Business School. After a long career working for companies like Schlumberger, BP, Denham Capital and Octopus Investments, in 2020 she co-founded SunGreenH2.

How does your proprietary technology accelerate advancements in green hydrogen production, and what challenges do you face in scaling it up?

The company addresses a critical challenge in the transition to a net-zero economy: the high cost and inefficiency of green hydrogen production. Traditional electrolysis relies on expensive precious metals like platinum and iridium, making green hydrogen economically unviable for many industries. Additionally, these systems consume significant amounts of energy, further driving up operational costs and limiting green hydrogen's accessibility as a clean energy solution.

SunGreenH2's strategy aims to make green hydrogen accessible and economically viable at scale. By leveraging proprietary, PGM-free nanostructured electrodes, that are durable and flexible in performance, SunGreenH2™ doubles hydrogen output while reducing energy consumption by 20%, significantly lowering production costs. This technology enables industries, utility companies, and municipalities to integrate green hydrogen into their operations as a cost-effective alternative to fossil fuels, reducing their carbon footprint and meeting regulatory requirements.

What role do strategic partnerships with corporations play in your startup growth strategy?

SunGreenH2™ systems are modular, scalable, and easily integrated with renewable energy sources like solar and wind, making them adaptable for various scales, from large industrial applications to decentralised hydrogen generation for mobility and power-to-X use cases.

Through partnerships with Naturgy and further testing with Enerjisa and PLN, SunGreenH2™ has demonstrated its PGM-free nanostructured electrodes, which double hydrogen output while reducing energy consumption by 20%, significantly cutting costs compared to traditional electrolysis methods. By eliminating reliance on precious metals and optimizing energy use, SunGreenH2™ makes green hydrogen production more affordable and scalable, enabling industries to reduce fossil fuel dependence and achieve decarbonization goals. This proven innovation ensures reliable, cost-effective solutions for advancing a sustainable, low-carbon future.

How critical is non-dilutive funding in your funding strategy, and what role do participation in accelerator and incubator programs play in supporting your growth and development?

SunGreenH2™, launched in 2020 with industry and government grants from Singapore and Australia, is well-funded and supported by strong technology, engineering, and manufacturing teams at its commercially focused headquarters.

The company has earned accolades like the BloombergNEF Pioneers Award 2023 and participated in top accelerators, including Shell Startup Engine (Singapore), AWS Clean Energy Accelerator 3.0 (USA), and Petronas FutureTech 3.0 (Malaysia), gaining grant funding, industry collaborations, and investor connections.

Having completed seed and bridging rounds from 2022–2024, SunGreenH2™ plans a Series A in 2025 to drive market expansion and scale manufacturing capabilities.

What does SungreenH2's roadmap to commercialization look like, and what are the key milestones along the way?

SunGreenH2™ creates scalable, modular systems for seamless integration with renewable energy sources, supporting applications from industrial energy to decentralized hydrogen generation. Its design ensures rapid deployment, easy integration, and scalability, reducing customer risks.

With strategic partnerships and global pilot projects in Europe, Asia, and Australia, SunGreenH2™ validates and refines its technology. From its Singapore HQ, the company plans 2025 expansion into India and the Middle East, lowering adoption costs and helping decarbonize key sectors to drive a sustainable energy transition.

02 Investors

This section profiles this year's leading Climate Tech investors, highlighting the most active across investor types and detailing their sector focus and interests. It also shares their perspectives on challenges, future trends, and opportunities, offering insights into the key drivers of Climate Tech investment.

Profile of 2024 investors

Overview

Despite experiencing a decline in their share over the past three years, venture investors remain the driving force in Climate Tech. In 2024, nearly half of these investors participated in three or more deals. While a significant number of venture investors are bringing new players to the field, growth equity investors lead with the largest share of first-time investors.

Meanwhile, equity investors are shifting their strategies. Their activity in early-stage deals is tapering off, pointing to a preference for fewer, larger funding rounds—a move that reflects a more focused and concentrated investment approach.

Europe has emerged as a powerhouse, not only leading in total but also standing out as the most active region for investor activity, narrowly edging out North America.

Venture investors

was the most top investor category in 2024 for most **frequent investors**, with 47% participating in three or more deals.

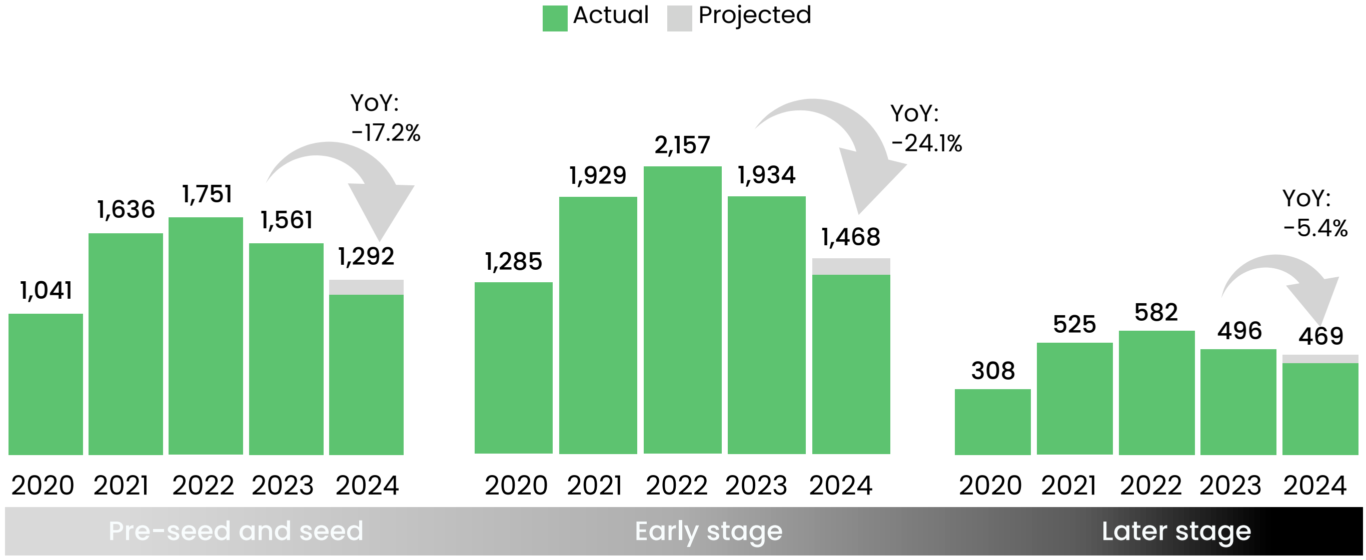
Growth equity investors

comprised the largest share of **first-time investors** in 2024, accounting for 35%. This was followed by corporations (27.6%) and venture investors (21.7%).

Europe

is home to the most active investors in 2024, surpassing North America by a tight 7% margin.

Yearly equity deal count by stage

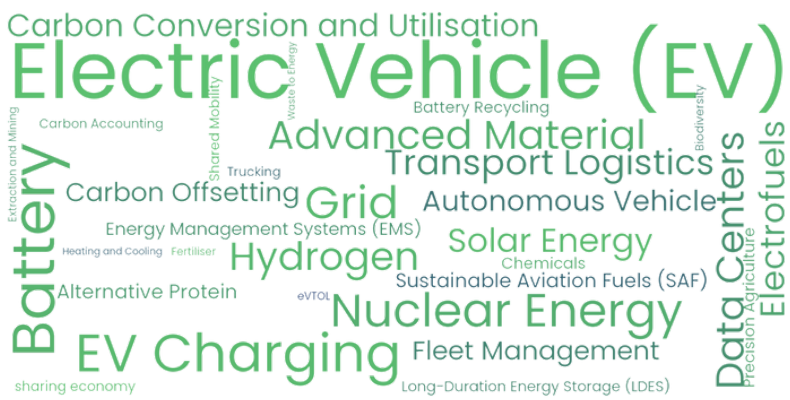


Source: Net Zero Insights

The figures illustrate equity deal activity in private Climate Tech ventures, while excluding debt, grants, and other financial instruments, exits and post-exit financing.

Most notable VCs

2024 most funded solutions by VCs



2024 most notable VCs

Lowercarbon Capital	Breakthrough Energy Ventures	Antler
36 Deals	33 Deals	19 Deals
United States	United States	Singapore

HTGF	Speedinvest	S2G Ventures
19 Deals	17 Deals	16 Deals
Germany	Austria	United States

Thought Leader Interview

Future Energy Ventures (FEV) invests in digital and digitally enabled technologies and business models that have the potential to redefine the future energy landscape.

From its hubs in Berlin, Palo Alto and Tel Aviv, Future Energy Ventures looks for both investments and scaling opportunities for its portfolio investments globally.

FEV invests initially 1-10 million euros in Series A and B start-ups in three decarbonisation themes: Future Energy, Future Cities and Future Technology. 40% of the funds are invested initially and 60% are available for follow-up investments.

Jan Lozek
Founder, Managing Director & Managing Partner

Jan is the founder and managing partner of Future Energy Ventures (FEV). He leads a team and experienced investors dedicated to backing scalable, world-class startups that accelerate the energy transition and contribute to decarbonizing society.

With offices in Berlin, Palo Alto, and Tel Aviv, Future Energy Ventures primarily invests between 1-10 million euros in Series A and B stage startups.

Jan is proud to be part of a global community of visionaries and thought leaders committed to building a more sustainable and resilient future.

Future Energy Ventures has a strong focus on digital and digitally-enabled solutions for the energy transition. What specific technologies or trends are you most interested in right now for future investment, and why?

We aim to invest in three key areas of energy transition within the energy and energy-adjacent fields: Future Energy, Future Cities and Future Technologies.
• Future Energy includes investments in start-ups across the energy supply chain, including smart generation, storage, distribution, and energy management. This covers technologies like energy storage, virtual power plants, smart grids, micrograms, demand response, energy monitoring, and efficiency solutions.
• Future Cities includes investments in start-ups in the broader energy ecosystem and adjacent industries like smart buildings, mobility, and urban infrastructure. This includes technologies such as building management systems, smart meters, e-mobility, smart charging, P2P trading, and smart city services.
• Future Technology investments focus on startups in areas outside the core energy sector that could create disruptions in the energy value chain. This includes energy-enabling technologies, cybersecurity and cyber resilience, IoT, AI, big data, robotics, drones, NLP, and AR/VR.

What challenges do you see in decarbonization and grid optimization, and how do your portfolio companies address them with their innovations?

We don't have time to save our planet, but perceptions and priorities are changing. There is no global consensus and no common focus to solve the climate challenge as quickly as we can together. Geopolitical issues, such as tensions between Russia, China, and the West, as well as conflicts in Lebanon, Gaza, and Ukraine, often overshadow Climate Technology. Despite this, we've developed promising technologies for grid management, renewable integration, and economically viable business models that reduce emissions. Progress is being made, but the challenge is meeting the net-zero timeline.

How does FEV support startups in the energy transition, and what key resources and expertise does it provide to help them scale and succeed?

FEV invests in companies driving the energy transition and carbon reduction, focusing on businesses with strong management, clear missions, and growth potential. We take lead roles, reserve 60% of capital for follow-on investments, and often secure board seats. Our approach includes:
• A specialized strategy refined since 2016.
• A strong track record in business development and technology support.
• A diverse team and global network for scaling and business evaluations.

We help startups with proof of concept, pilot projects, and commercial contracts, addressing key needs like business expertise, technical support, market access, facilities, and talent. FEV supports portfolio companies by:
• Matching startups with pazrtners and providing scaling expertise.
• Offering access to functional experts and resources.
• Facilitating industry access through strategic collaborations.

Looking ahead, what are FEV's goals for the next few years, and how do you plan to expand your impact on the energy ecosystem?

• Delivering Fund I with 3x and significant GHG Emission reduction
• Delivering Fund II with 3x and significant GHG Emission reduction
• Setting up Fund III
• Impact potential > 10 Gigaton of emission reduction
• Network Ecosystem > 120.000 people and 60.000 organisations

Thought Leader Interview

Orion Industrial Ventures (OIV) is the venture arm of Orion Resource Group, an $8 billion global investment firm headquartered in New York. Orion Resource Partners group was founded in 2013 and manages strategies that span the liquidity spectrum for critical minerals and materials.

OIV invests in early- and growth-stage technology companies that enable sustainable and economic supply of critical raw materials (CRMs) and in new energy solutions to support industrial decarbonization.

By accessing the know-how, markets, and assets across the Orion Group, OIV aims to add value to its portfolio companies and help them de-risk and scale to potentially become leading industrial companies of the future.

Mark Frayman
Managing Partner, Orion Industrial Ventures

Mark joined Orion in late-2022 to establish OIV. Prior to joining Orion, Mark was Head of BHP Ventures, where he led the initial design, establishment and scale-out of BHP Ventures' investment platform and portfolio.

Mark has previous experience in mergers and acquisitions investment banking, venture capital, and asset management, and degrees in Law (First Class Honours) and Commerce from the University of Melbourne.

Tell us about Orion Industrial Ventures (OIV). What's the purpose of this fund and what is your investment thesis?

CRMs are essential for global decarbonization, urbanization, and future growth and we need a clean, green, and "domestic" supply of these materials.

For example, wind and solar are 7x and 37x more copper-intensive than fossil electricity and EVs are expected to double the demand for lithium and copper.

Artificial Intelligence (AI) is further accelerating demand. The IEA predicts that AI could double the demand for energy for data centres, requiring more copper and silicon for AI chips.

Today, China controls the supply chains of many of these raw materials. This means that we face an unprecedented supply-side deficit at the very same time as we've got scarcity.

To solve this, we invest in solutions across the supply chain: from mineral exploration to extraction and recycling. We also invest in new technologies that support industrial decarbonization, such as new forms of green baseload power (geothermal, nuclear).

What are some unique challenges you face in investing in solutions for critical minerals?

The sector is relatively new and often misunderstood by generalist climate-tech investors. Despite contributing 30% of global emissions, heavy industry has received a little more than 10% of climate tech dollars.

This creates an attractive opportunity for OIV but it also means many companies in our sector have been deprived of capital.

Integrating new technology into mine sites is complex, as continuous "24/7 always-on" systems do not handle disruption well. This means that we need to screen for start-ups with solutions that can integrate into existing systems, operate as a standalone system, and have a clear, near-term and measurable pay-off to minimize friction for adoption.

Many business models we see are capital intensive and don't fit the traditional VC framework. These businesses struggle to obtain funding for "first-of-a-kind" (FOAK) plants, and need to think creatively about their business models to overcome this and meet our return criteria.

Finally, we see many technically-strong founders in our landscape, but at times they aren't sufficiently commercial to penetrate and "sell" to a large mining company, which is an area where we spend time with our founders and collaborate with our colleagues across the Orion platform.

What type of Climate Tech companies are you particularly interested in at the moment, and why?

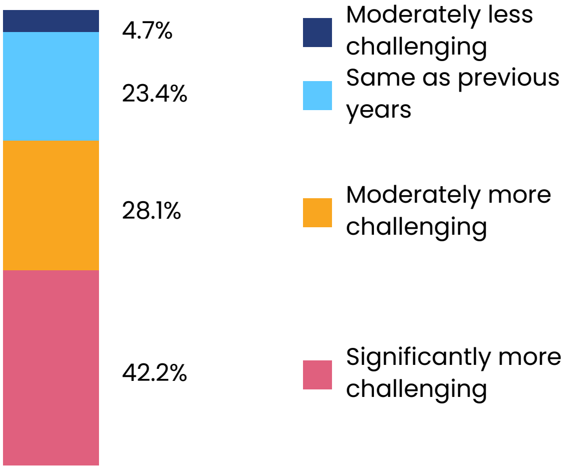
We see deal flow at both ends of the value chain – i.e. upstream exploration and end-of-life recycling, but it's in the middle we see the most untapped potential.

We are excited by the opportunity to leverage "platform technologies" such as AI and machine learning, advanced electrochemistry and/or synthetic biology to this part of the mining value chain. For example, we are looking at companies developing neural learning models to understand the subsurface, which has the potential to unlock efficiencies and extend mine lives.

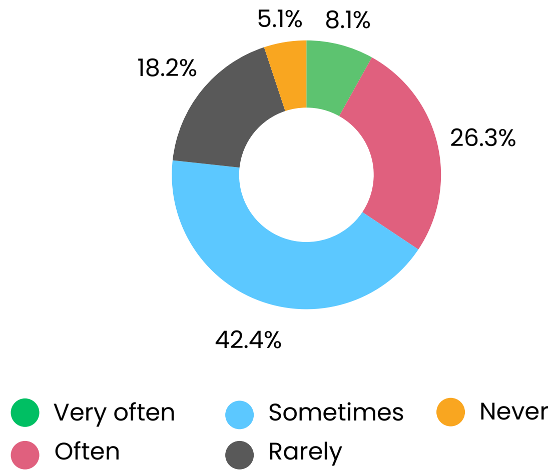
In the industrial energy solutions space, we are doing work on both geothermal and nuclear, both of which likely have tailwinds in the current environment and could provide necessary cheap, green baseload power.

Investment focus and strategy
from NZI's SOCT 2024 survey

How does the difficulty of fundraising in 2024 compare to previous years ?



How often do you struggle to find suitable Climate Tech investment opportunities?



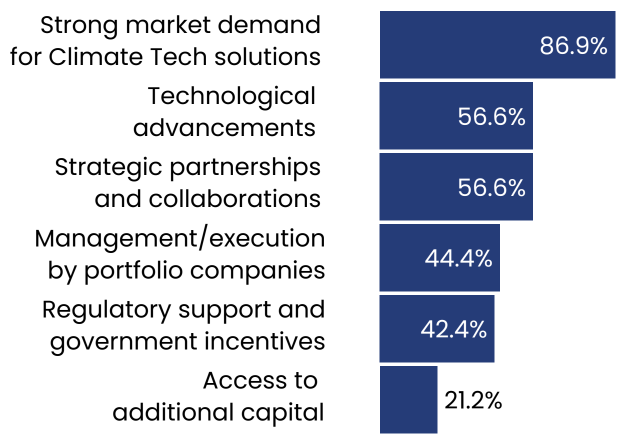
Investors Share Pessimism About 2024 Fundraising Environment

Across sectors, nearly 50% of investors in both conventional areas like renewable energy and emerging sectors such as GHG Capture and Circular Economy reported fundraising as moderately or significantly more challenging in 2024.

VCs and Early-Stage Investors Most Pessimistic

60% of Pre-Seed/Seed stage investors and VCs viewed the fundraising landscape as significantly or moderately tougher compared to previous years. Similarly, 60% of investors focused on both breakthrough and adoption innovations reported increased fundraising challenges in 2024.

What factors positively influenced your climate investment decisions in 2024?

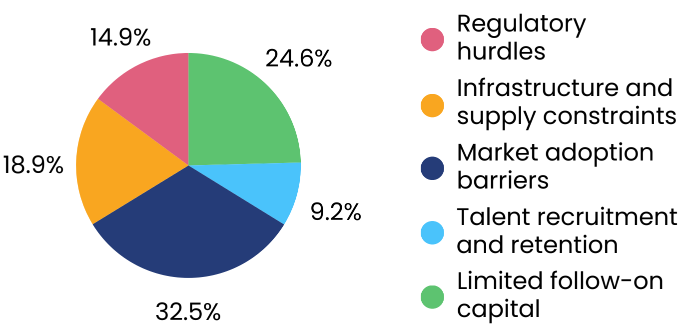


Growth and Later-Stage Investors Report Difficulties

50% of growth and later-stage investors struggled to identify suitable opportunities, compared to 33% of Pre-Seed/Seed and early-stage investors, highlighting differing challenges across investment stages.

Investment focus and strategy
from NZI's SOCT 2024 survey

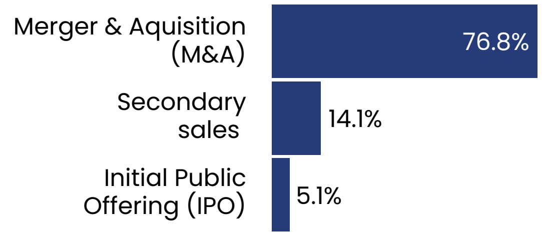
What were the main barriers to scaling your Climate Tech portfolio companies ?



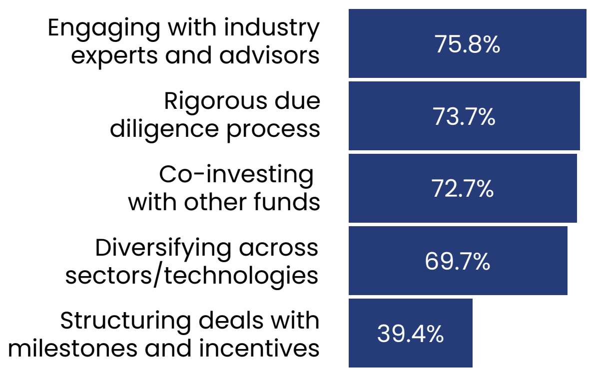
Market adoption barriers and limited access to follow-on capital were the most cited challenges among investors across all sectors, stages, and solution types.

Regulatory and compliance hurdles were particularly noted by investors in the Natural Environment (42%) and Water sectors (39%).

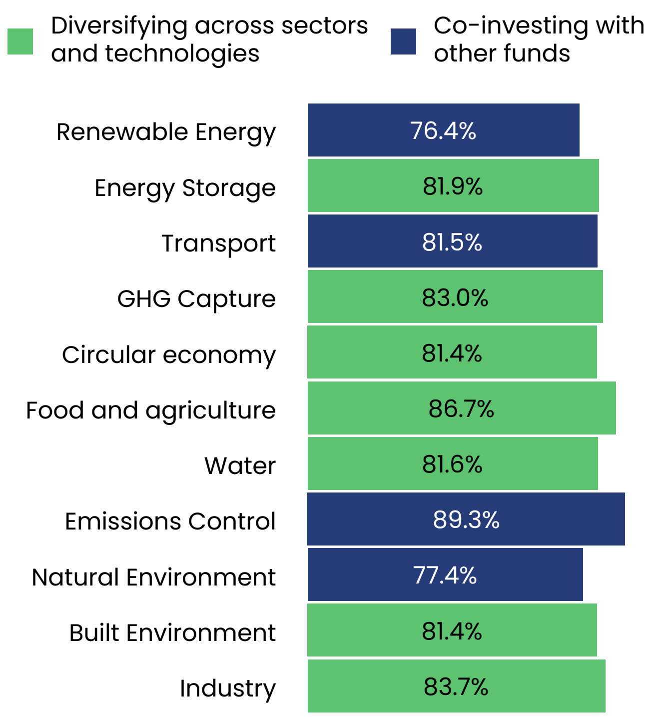
What is the most viable short-term exit strategy for Climate Tech?



How do you manage risks associated with Climate Tech investments?



What are the most common risk mitigation strategies by sector?



Rigorous due diligence processes was mentioned as a risk mitigation strategy by 100% respondents in North America and 67% respondents in Europe.

Thought Leader Interview

Climate First is a European investment firm dedicated to addressing the climate crisis by supporting exceptional startups for significant returns and substantial decarbonization potential. Its partners are experienced fund managers, venture investors, and scientists with over 50 years of combined experience, driving investment in Climate Tech. The firm's unique roadshow program has supported over 30 European Climate Tech founders, introducing them to key corporate partners and institutional investors within the firm's global network.

Headquartered in London, Climate First backs early-stage Climate Tech companies and actively supports their growth through its extensive connections with strategic corporations, leveraging its platform and partnerships to achieve Net Zero.

Nadav Steinmetz
Co-Founder & Managing Partner

Nadav Steinmetz is an early stage Climate Tech investor based in London. He is the Co-Founder and Managing Partner of Climate First, a European Climate Tech investment group.

Nadav backs exceptional companies driving decarbonization on a global scale. He is a former private equity investor with over a decade of experience in financial services, venture capital and real estate.

He also serves as an advisor, mentor and board member to numerous companies and non-profit organizations. Nadav holds a B.A. in Economics and Philosophy from Columbia University.

Climate First focuses on early-stage Climate Tech startups across industries like energy, mobility, and the built environment. How do you evaluate the decarbonization potential of these diverse sectors when selecting startups?

We back outstanding startups with significant decarbonization potential that will become the backbone of the 21st century economy.

We focus on scale, meaning that we assess the direct and indirect potential for CO2 mitigation (versus existing processes and technologies) and multiply this by the expected adoption and market opportunity. Direct mitigation would be weighed more heavily than indirect. This is a core part of our investment thesis and we seek to quantify these measures wherever possible.

Your portfolio covers various sectors, including sustainable fuels, recycling, and advanced materials. What common challenges do you observe in scaling these Climate Technologies, and how does Climate First help startups overcome them?

Common challenges in scaling climate and deep tech startups include initial high capital requirements, at times long R&D cycles, regulatory hurdles, and the need for strong industry partnerships. Climate solutions require cross-border collaboration with customers, supply chains and buyers often spanning multiple regions. Non-dilutive financing in the forms of grants is essential to validate the technology early on, while corporate partnerships are crucial for scaling.
At Climate First, we support startups by providing funding and strategic partnerships via our platform and global roadshows where we collaborate with leading corporations and multinationals. We connect them with industry experts and potential customers to accelerate adoption and scaling.

How does Climate First foster partnerships with corporates and public bodies to help startups accelerate market entry and achieve broader climate goals?

Corporate partnerships are critical in scaling Climate Tech companies. We foster strategic partnerships by connecting our companies with leading corporations and public bodies that align with their climate solutions. We facilitate pilot projects, joint ventures, investments and procurement agreements, helping startups validate and scale their technologies. Strategic corporations share valuable insights into facility operations, supply chain management, product testing, and more.

We have backed over 35 startups via our roadshow program where we select the leading European Climate Tech companies to meet, engage and explore potential partnerships with some of the world's largest players in energy, automotive, financial services and more.

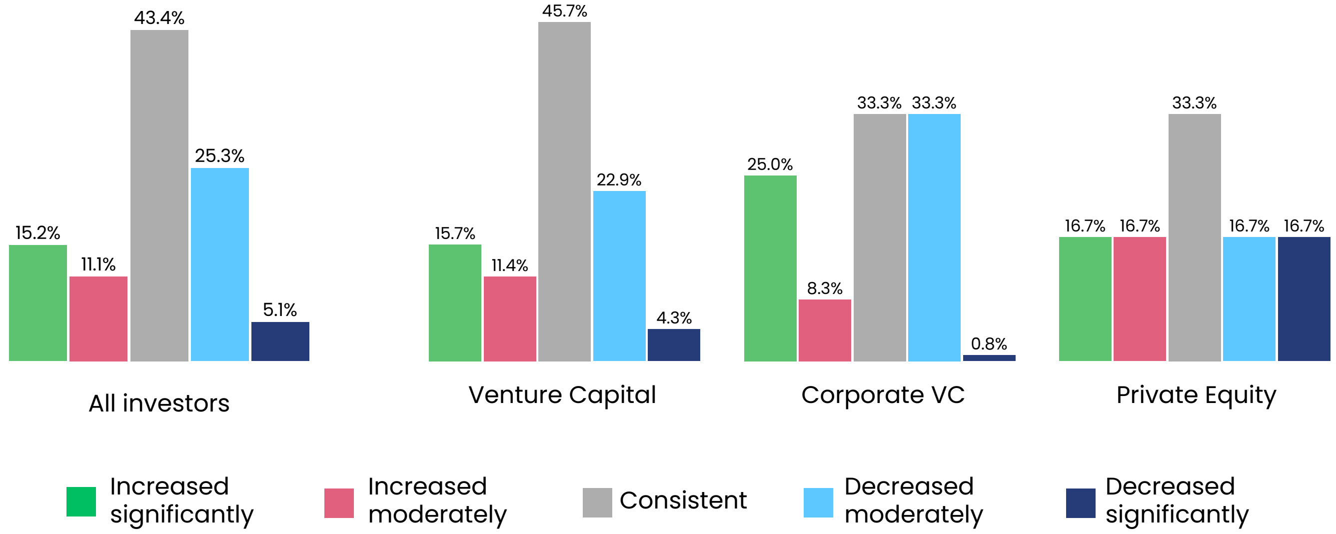
With an expanding portfolio of climate-focused investments, what emerging technologies are you most excited about, and how do you see them shaping the Climate Tech landscape in the coming years?

We anticipate major growth in the clean energy minerals and materials sector essential to global electrification and battery adoption. Technologies leveraging AI and machine learning to advance mineral exploration and extraction hold strong potential. Likewise, innovations in refining raw materials to battery-grade quality and in recycling to eliminate waste are increasingly vital.

We believe Climate Tech is at the heart of this industrial revolution, and we're committed to supporting visionary founders building transformative companies.

Capital deployment
from NZI's SOCT 2024 survey

How has the capital deployment in Climate Tech changed in 2024 as compared to previous years?

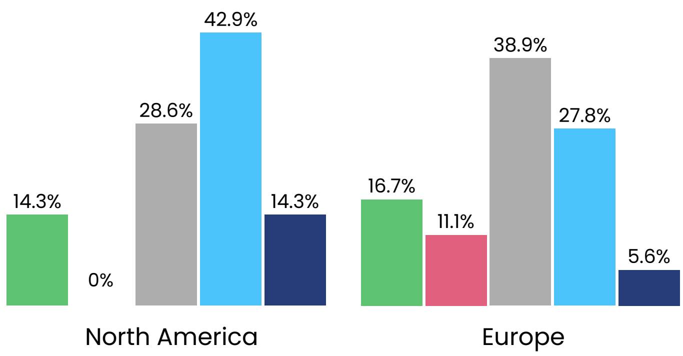


Overall, most investors viewed Climate Tech capital deployment in 2024 as consistent with previous years.

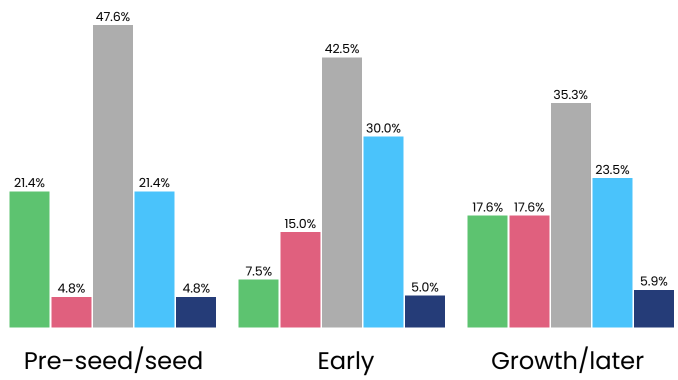
However, 30.4% of respondents reported a decrease, exceeding the 26.3% who noted an increase. Among Venture Capitalists, the views were nearly split, with 27.2% seeing a decrease and 27.1% an increase.

Survey responses indicate that pessimism about a downturn in Climate Tech capital deployment in 2024 was higher among North American investors (57.2% reported a decrease) than European investors. Early-stage investors were the most pessimistic by stage, with 35% observing a decline.

How do investors' views on capital deployment vary regionally?

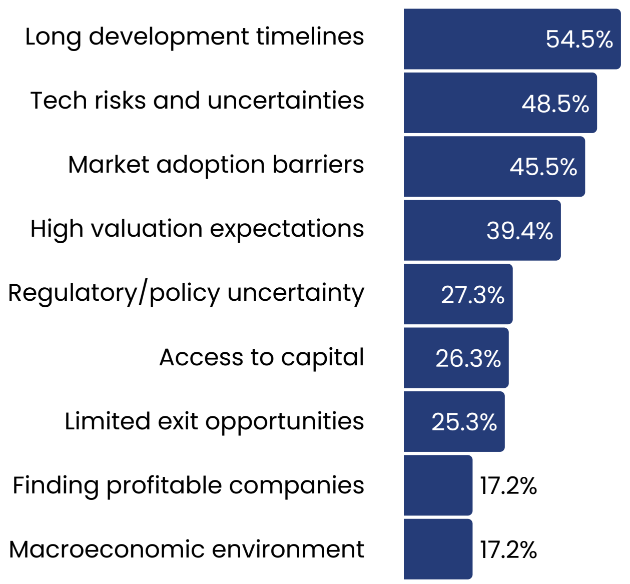


How do investors' views on capital deployment vary by stage?

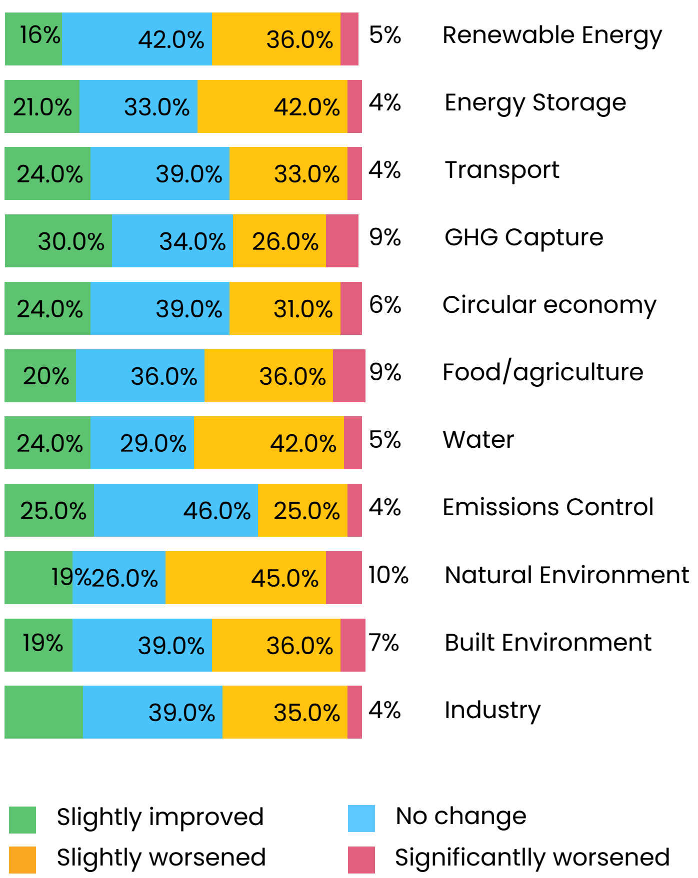


Capital deployment
from NZI's SOCT 2024 survey

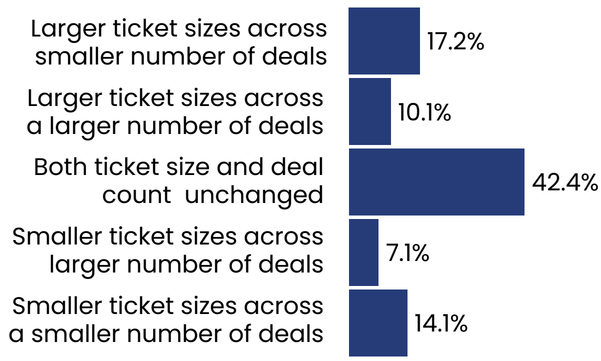
What are the biggest challenges faced in deploying capital?



How do investors view the current exit environment for Climate Tech startups?



How have capital deployment strategies shifted in 2024 from previous years?



Most investors haven't changed capital deployment strategy in 2024

Most respondents believed ticket sizes and deal counts remained unchanged in 2024 compared to previous years.

The second-most common response across all investor types, stages, and regions was "larger ticket sizes with fewer deals," reflecting the trend of capital concentrating on fewer transactions. This pattern was most pronounced among pre-seed and seed investors, with 26% reporting fewer deals paired with increased ticket sizes.

The biggest challenge in capital deployment was the long development timelines and extended return-on-investment horizons.

While regulatory and policy uncertainty ranked fifth overall, it was a major concern for Water and Natural Environment investors, with 32% in both sectors identifying it as their top challenge.
Technological risks and uncertainties were most frequently cited by pre-seed/seed stage investors and those focused on digital products.

Thought Leader Interview

E44 VENTURES | BUILDING CLIMATE BREAKTHROUGHS

E44 Ventures is a Climate-Tech fund investing in deep-tech, early stage ventures in EU and Israel that take climate action hands-on.

Based on our unique experience and expertise of serial entrepreneurs, scientists and business professionals working synergistically for one cause – creating a sustainable prosperous future for our planet.

Adi Vagman
Co-Founder & Managing Partner

Adi Vagman is a seasoned venture capitalist and serial entrepreneur with over a decade building startups and leading in large corporations.

Adi is an expert investment strategist who has founded and built successful Agri-Tech and food tech startups.

As an active board member, she leverages her mastery of fundraising and teambuilding to drive her investments from inception to commercialization. She continues to nurture the next generation as a senior mentor in the prestigious Sam Zell Entrepreneurship Program.

What are the biggest challenges in scaling climate solutions across sectors like energy and water, and how does E44 Ventures support startups in overcoming these hurdles?

Scaling climate solutions in sectors such as energy and water comes with complex challenges, including high initial capital requirements, regulatory hurdles, fragmented markets, and long adoption cycles due to entrenched traditional systems. These obstacles can be particularly daunting for early-stage startups aiming to disrupt established industries. E44 Ventures supports startups by providing not only funding but also strategic guidance and deep sector expertise. Our team of seasoned entrepreneurs and scientists collaborates closely with portfolio companies to navigate regulatory landscapes, streamline go-to-market strategies, and forge essential partnerships with industry leaders. This hands-on approach ensures that startups can effectively scale their innovative solutions and accelerate the path to impactful deployment.

E44 Ventures has backed high-risk, high-reward startups like Carbonade and Gigablue, which focus on carbon capture and ocean-based carbon removal. How do you balance risk and impact when investing in deep tech startups?

At E44 Ventures, we embrace a dual-pronged approach that balances risk with potential impact. We start by conducting rigorous due diligence to assess both the technical viability and the long-term scalability of a startup's solution. Our investment philosophy prioritizes startups whose innovations have the potential to create transformative change, even if they come with significant risk. To mitigate this risk, we engage closely with our portfolio companies, providing tailored support and leveraging our network to de-risk pathways to commercialization. Additionally, we diversify our portfolio across various sub-sectors in Climate Tech, allowing us to spread exposure while ensuring that each investment aligns with our mission to drive impactful solutions at scale.

With a portfolio spanning renewable energy, carbon capture, and water generation, how does E44 Ventures leverage strategic partnerships to accelerate commercialization across these sectors?

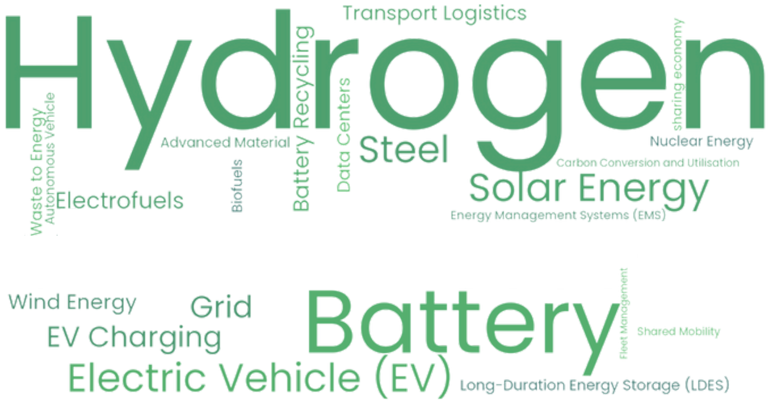
Strategic partnerships are at the core of E44 Ventures' approach to scaling innovations. We actively collaborate with research institutions, industry players, and government bodies to create a supportive ecosystem for our startups. These partnerships provide early-stage ventures with access to cutting-edge research, pilot opportunities, and co-development projects that accelerate product iteration and market entry. For instance, our alliances with energy conglomerates and infrastructure firms help startups expedite technology validation and unlock potential large-scale deployments. By acting as a bridge between innovators and established industry leaders, E44 Ventures helps startups overcome commercialization barriers and achieve rapid growth.

Looking ahead, which sectors or technologies is E44 Ventures most interested in for future investments and why?

E44 Ventures is focused on high-growth sectors with significant climate impact:
1. Data Centers Efficiency – Investing in energy-saving technologies to optimize data center operations and meet growing global demand sustainably.
2. Green Materials – Supporting innovations in sustainable materials to reduce carbon footprint.
3. Waste Sourcing & Circular Economy – Backing technologies that recycle, upcycle, and convert waste into valuable resources, driving the shift to a circular economy.

Most notable growth equity investors

2024 most funded solutions by Growth Equity Investors



2024 most notable Growth Equity Investors

Energy Impact Partners
17 Deals
United States

Business Growth Fund
14 Deals
United Kingdom

Invest-NL
10 Deals
Netherlands

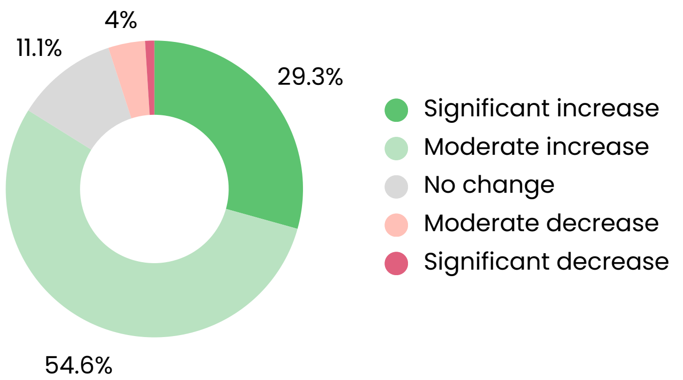
Blackrock
8 Deals
United States

Eurazeo
8 Deals
France

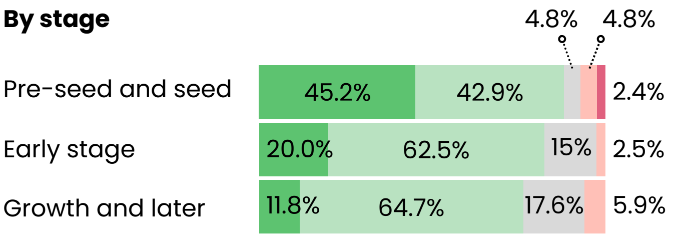
Closed Loop Partners
8 Deals
United States

Future outlook and opportunities
from NZI's SOCT 2024 survey

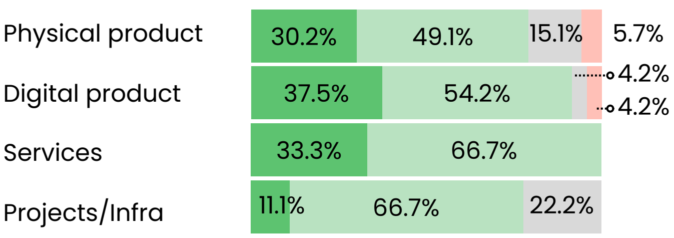
How do you expect the funding landscape to change over the next 1 to 3 years?



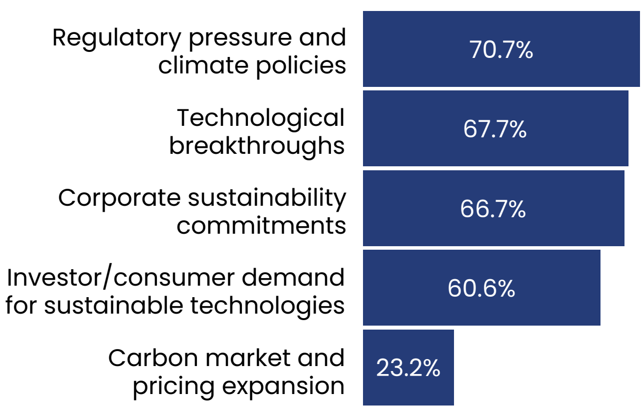
By stage



By solution type



What are the key drivers of future growth in Climate Tech investments?



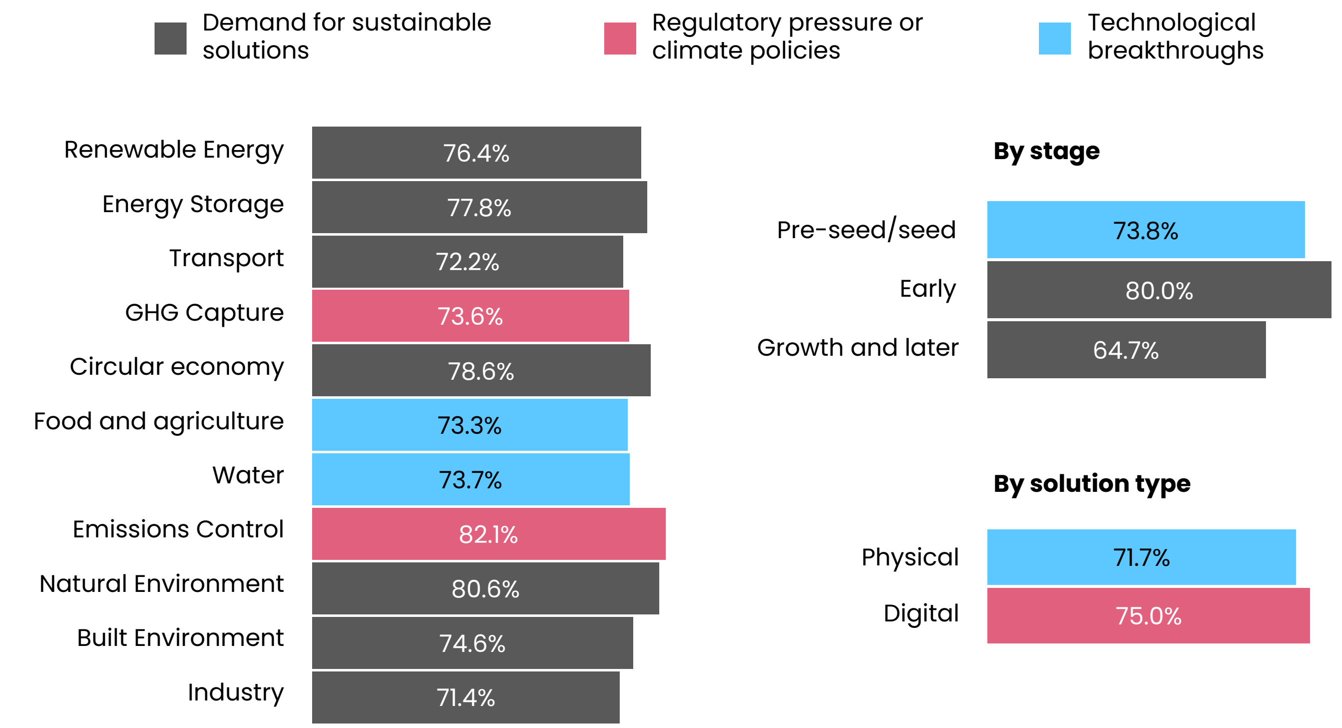
83% of surveyed investors anticipate a moderate or significant increase in funding over the next 1–3 years.

Pre-seed and Seed and early-stage investors showed greater optimism about the future funding landscape compared to growth and later-stage investors.

Regulatory policies drive Climate Tech investment

Over one-third of investors identified regulatory pressure and climate policies as key drivers of Climate Tech growth. This view was strongest among Emissions Control (82%) and Natural Environment (81%) investors, where government grants play a significant role.

What factors do most investors cite as primary growth drivers for Climate Tech, and what are their percentages?



Investors in emerging sectors emphasize the need for sector-specific expertise and stronger engagement with regulators and policymakers. A significant majority of investors in GHG Capture (70%), Circular Economy (76%), Water (76%), and Natural Environment (81%) identified a lack of sectoral expertise as a critical gap, highlighting the need to build specialized knowledge and skills.

Engagement with policymakers was also a top priority for sectors like Natural Environment and Built Environment, where regulatory collaboration is essential.

Market demand emerged as the most common growth driver across sectors. However, for capital-intensive and emerging sectors like GHG Capture and Emissions Control, regulatory pressure and policies were cited most frequently due to their long development timelines and limited investor appeal, underscoring the urgency for government support and sector-specific expertise in these fields.

What strategies will investors use to capitalize on future opportunities?

Thought Leader Interview

LOWERCARBON CAPITAL

Lowercarbon is a venture capital firm led by General Partners Chris Sacca and Clay Dumas. Building on the success of Lowercase Capital — the best-performing fund of all time in terms of return multiples — Lowercarbon invests in the most promising founders working in three key areas: (1) eliminating new emissions of CO_2 and other greenhouse gases, (2) removing at least a trillion tons of carbon already in the atmosphere, and (3) buying humanity, especially the most vulnerable, more time by actively cooling the planet.

Dr. Clea Kolster
Partner and Head of Science

Dr. Clea Kolster leads technical research, development, and scientific strategy at Lowercarbon Capital. Previously, she worked at E3, advising states and utilities on decarbonization, focusing on technologies like hydrogen, energy storage, carbon capture, and advanced nuclear. She co-founded a clean cooking fuel company in West Africa and has experience in clean energy investment across the continent. Dr. Kolster earned her Ph.D. in chemical engineering and energy policy from Imperial College London, with research on carbon capture systems, including time at Stanford. She's a technical advisor for Third Derivative and a founding fellow of On Deck's Climate Tech program. Fluent in five languages, Dr. Kolster enjoys skiing, sailing, and endurance cycling, having raised over $10,000 for refugee support by biking 300 miles from London to Paris.

How does Lowercarbon Capital assess the viability of novel business models, like battery-as-a-service and tokenized renewable energy, in emerging industries, and how does sustainability impact, including carbon reduction and resource efficiency, factor into your funding decisions?

The opportunity to generate positive climate impact is broad and large. At Lowercarbon, we've run the numbers and understand that in order to bring down the CO2 in the atmosphere, it will take three major priority items: (1) slashing new emissions of CO2 and other greenhouse gases to zero, (2) sucking up at least a trillion tons of carbon already swirling around the sky, and (3) buying all of humanity, including those most vulnerable, more time by actively cooling the planet. The markets have shown that it's too expensive to keep powering our economy with fossil fuels. Whether it's in construction with Sublime's low-carbon cement or in energy generation with Panthalassa's wave-energy generated data centers, innovation is actively displacing incumbent fossil fuels. Throughout practically all aspects of the global economy, we have seen alternatives to petrochemicals undercut 100-year-old industrial processes that are dirty and dangerous, all while providing national security, volatility hedging, and economic development advantages. Ultimately, we believe that the largest climate impact action we can back are companies that can effectively scale and decarbonize the economy by bringing better, faster, and cheaper to market.

Given Lowercarbon's investment in emerging technologies like fusion, what are some of the biggest technical or market-based challenges these companies face as they aim to scale their solutions?

Fusion energy is a burgeoning technology that, at its core, harnesses the universe's most powerful energy source: the same process that powers stars like our Sun.

By tapping into the strongest fundamental force of nature, fusion energy can generate clean, firm power with a practically unlimited supply of fuel. For over 60 years, publicly funded research has developed fusion technology with the goal of commercializing what would be the most abundant source of energy on Earth. The challenges this industry faces are largely technical, in the areas of achieving and sustaining ignition, plasma confinement, and plasma-facing materials. Over this time frame, key physics and engineering challenges in plasma, materials science, and other areas have benefited from key technology trends such as advances in computing and new materials like superconducting magnets, as well as cheaper lasers and capacitors, putting fusion technology on the same technical trajectory as semiconductors. In our portfolio today, we see Commonwealth Fusion Systems leveraging cutting-edge materials to confine plasma with the world's strongest high-temperature superconducting magnet, Zap Energy developing a novel stabilization system that addresses the historic challenge of plasma instability, Xcimer building upon semiconductor lithography excimer lasers to deliver a laser-driven fusion facility, and many other such cases. This incredibly fast rate of progress in fusion is why today it's no longer a matter of if this will happen; it's about when. At Lowercarbon, we back the founders and technologies that are demonstrating a clear path to building the first commercial fusion power plant in the world in the next 10 years. For many approaches, the physics is mostly understood, and the focus is on building the reactors, scaling up the power banks, and engineering solutions to harness the energy generated.

03 Corporations

This section spotlights corporations as key Climate Tech stakeholders, providing detailed profiles and insights into their views on current challenges and future collaboration with startups. It also highlights the most active corporate players driving innovation and investment in the sector.

Profile of 2024 corporate players

Overview

Corporations are playing an increasingly significant role in Climate Tech, with more new entrants joining and existing players ramping up their activity.

Europe and the US lead globally in the number of active corporations, while Asia is emerging as a rising player.

In 2024, the electricity sector dominated corporate acquisitions, reflecting the urgent need to adopt sustainable practices and transform traditional business models for a cleaner energy future.

850+

deals in 2024 involved at least one corporation, totaling over $20 billion in funding.

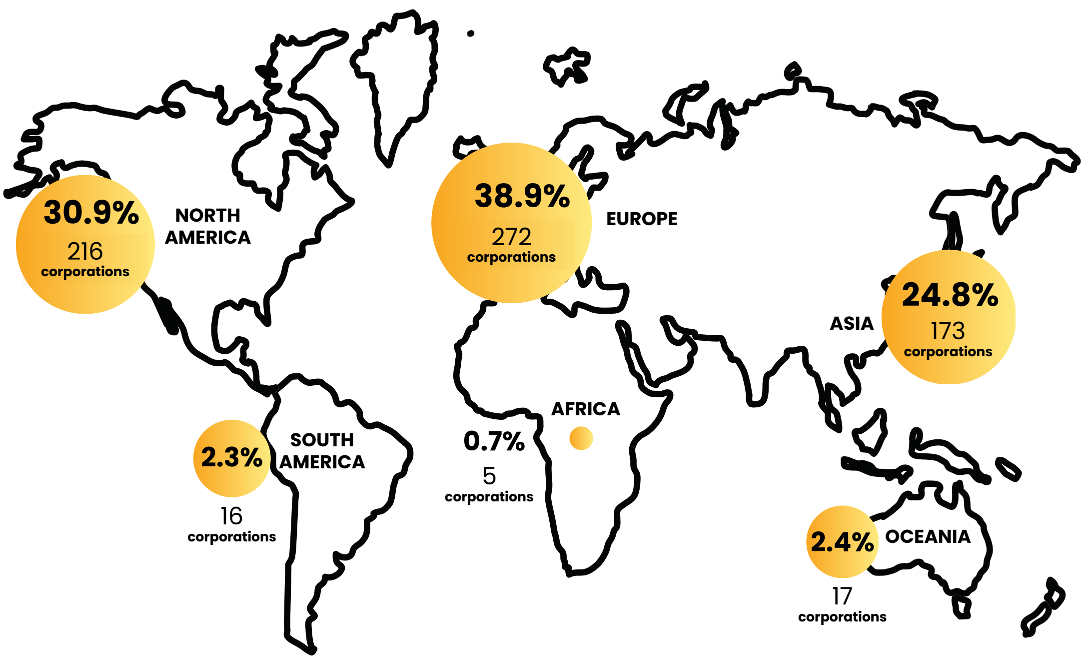
electricity

emerged as the sector with the highest number of corporate acquisitions.

Europe

is the home continent of the majority of active corporations in 2024.

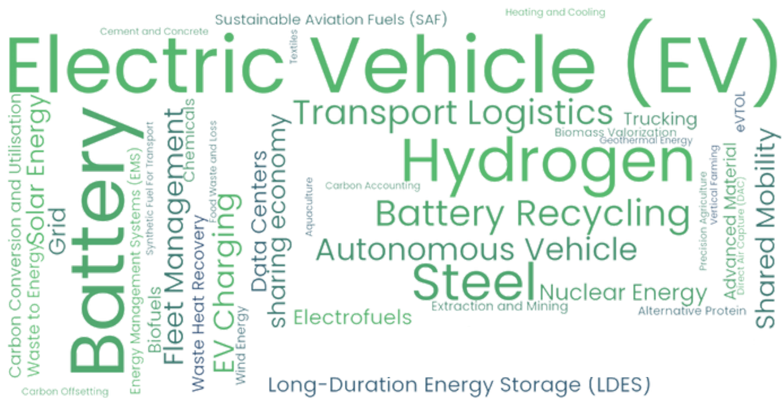
Corporations active in 2024 by region



Source: Net Zero Insights

Most notable corporations

2024 most funded solutions by Corporations



2024 most notable Corporations and Corporate Venture Capital (CVC)

Toyota Ventures 14 Deals United States	**Siemens** 13 Deals Germany	**Aramco Ventures** 11 Deals Saudi Arabia
Shell 10 Deals United Kingdom	**Gs Futures** 7 Deals United States	**Kddi** 7 Deals Japan

Thought Leader Interview

EDP is a global energy leader with a presence in four regions including Europe, North America, South America and Asia Pacific, and is committed with sustainability and innovation. EDP is on track to be carbon-free by 2025, all-green by 2030, and net-zero by 2040, investing more than €17 billion into the energy transition by 2026, and 97% of its energy generated is from renewable sources.

EDP Ventures is the Corporate Venture Capital of EDP Group, investing worldwide in early-stage tech startups focused on the energy sector. With a portfolio of 39 companies and more than €75 million invested (with a total 150M€ investment commitment), EDP Ventures supports innovations in renewable energy, grids, energy management and customer solutions that can support EDP's sustainability and decarbonization goals .

Frederico Gonçalves
Partner and Managing Director

Frederico has been a Partner at EDP Ventures, venture capital arm of EDP Group, since January 2022, having previously served as an Investment Director and Principal at the dame firm between 2010-2022. In addition to his role as Partner he also holds Board Member and Observer positions in various portfolio companies.

Previous to his tenure at EDP Ventures, Frederico spent 5 years in M&A/Corporate Development at the EDP Group.

His earlier experience includes roles in management consulting at Roland Berger and Diamond Cluster in Portugal, Belgium, and Brazil.

Frederico has a degree in Economics from Universidade Nova de Lisboa, MBA from IE Business School in Madrid and an Executive Course in PE/VC from Harvard Business School in Boston.

What technologies or trends is EDP Ventures prioritizing to support EDP's goal of becoming 100% green by 2030 and net zero in 2040?

Our investment strategy focuses on key technology trends to meet EDP's sustainability and decarbonization goals through specific "hunting zones" addressing major challenges. Aligning innovation with strategic priorities avoids costly missteps for the BU and Group, and some of our portfolio companies— such as Terabase, Splight, and Captura—reinforce this focus:

- Automation & Robotization: Terabase Energy's automated construction platform reduces costs and scales utility-scale solar.
- Industrial Heat Decarbonization: Electrified Thermal Solutions leverage renewable electricity for continuous industrial heat, benefiting sectors like food processing and enhancing grid flexibility.
- Grid Optimization: Splight's AI manages grid congestion, boosting grid capacity, reducing inefficiencies, and integrating more renewable energy.
- Carbon Removal: Captura's ocean-based CO2 removal creates growth opportunities for EDP, beyond carbon avoidance.
- These investments drive EDP's 100% green and net zero goals and advance the energy transition by tackling essential technical and economic challenges.

What challenges and opportunities do you see for scaling EDP Ventures' diverse climate solutions, from automation to ocean carbon removal?

Scaling climate solutions involves both challenges and opportunities. Key challenges include the cost and complexity of integrating new tech into existing infrastructure, regulatory delays, and slower market adoption due to operational shifts and perceived risks, such as transitioning to electrified industrial heat. Securing funding for large-scale projects is also difficult, though new financing models and support for First-Of-A-Kind projects are helping.

However, as technologies mature, costs decrease, automation boosts efficiency, and supportive policies emerge. Collaboration across the industry is also accelerating adoption by fostering knowledge-sharing and reducing costs.
By addressing these challenges and leveraging these opportunities, EDP Ventures supports innovative companies leading the energy transition.

How does EDP Ventures support its portfolio companies in the relationship with EDP´s Business Unit? Are Business Units involved in the investment process?

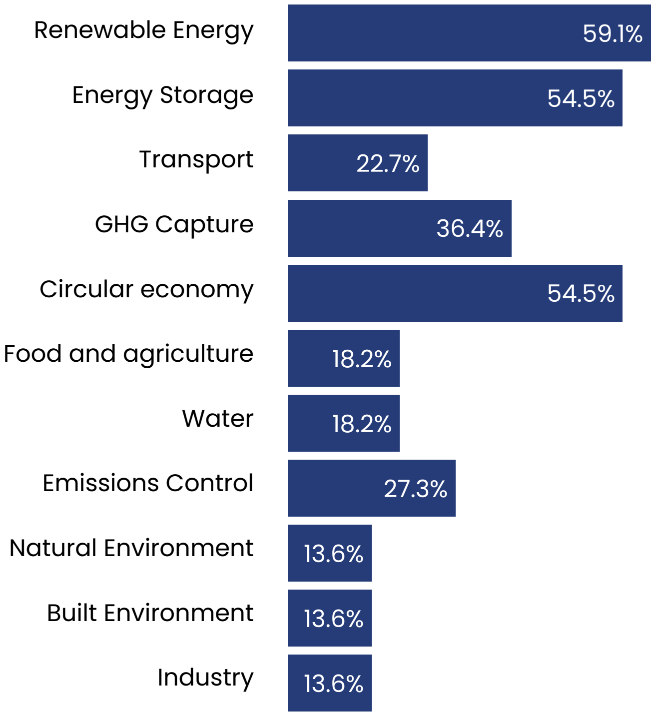
At EDP Ventures, we build strong connections between our portfolio companies and EDP's Business Units (BUs). From the start, BUs support the Ventures´ team to assess each company's strategic relevance, often through technical due diligence or pilot projects. As strategic investor, we target growth areas aligned with EDP's goals, through a team dedicated to capturing strategic value.
After the investment, we connect startups with key EDP contacts, assist with partnership negotiations, and streamline processes. Advisory boards, including BU directors, offer strategic guidance, helping startups scale within EDP.

How does EDP Ventures collaborate with other industry players and Corporate Venture Capitals to discover and co-invest in climate tech startups?

At EDP Ventures, we prioritize partnerships with VC and CVC firms, including utilities, to drive climate innovation. These collaborations help us identify startups, share insights, and co-invest in climate tech. Regular discussions allow us to explore opportunities and manage risks with CVCs. Also, through Free Electrons (EDP Innovation acceleration program in coordination with 6 other global power utilities) we partner with industry leaders to support startups, build pilot projects, and source dealflow. These partnerships enhance our support for startups and advance climate solutions.

Collaboration goals and strategies
from NZI's SOCT 2024 survey

Which sectors are most relevant to your corporation's Climate Tech investments?

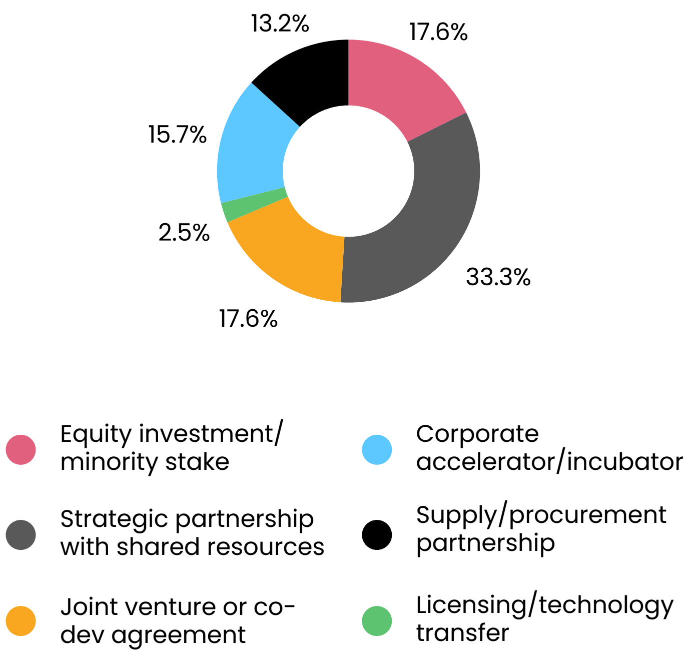


Among surveyed corporations, renewable energy, energy storage, and circular economy were identified as the most relevant sectors for Climate Tech investments.

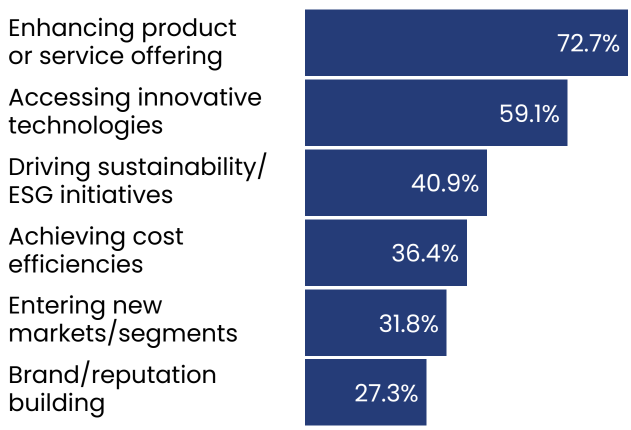
Sectors like Natural Environment and Built Environment, which attracted the least corporate investment interest, were also seen as those requiring the greatest government and policy support.

Collaboration models that pool resources and dilute financial risks were regarded as the most effective strategies, regardless of corporation size or sector.

Which collaboration models have been most successful for your organization?



What are the primary goals of corporations in partnering with Climate Tech startups?



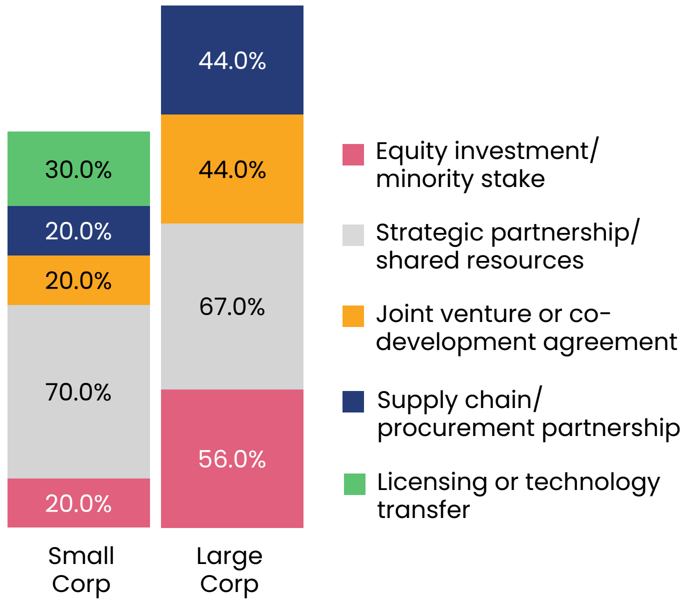
Collaboration goals and strategies
from NZI's SOCT 2024 survey

Large corporations favor joint ventures and co-development agreements, while small corporations prefer unilateral investments.

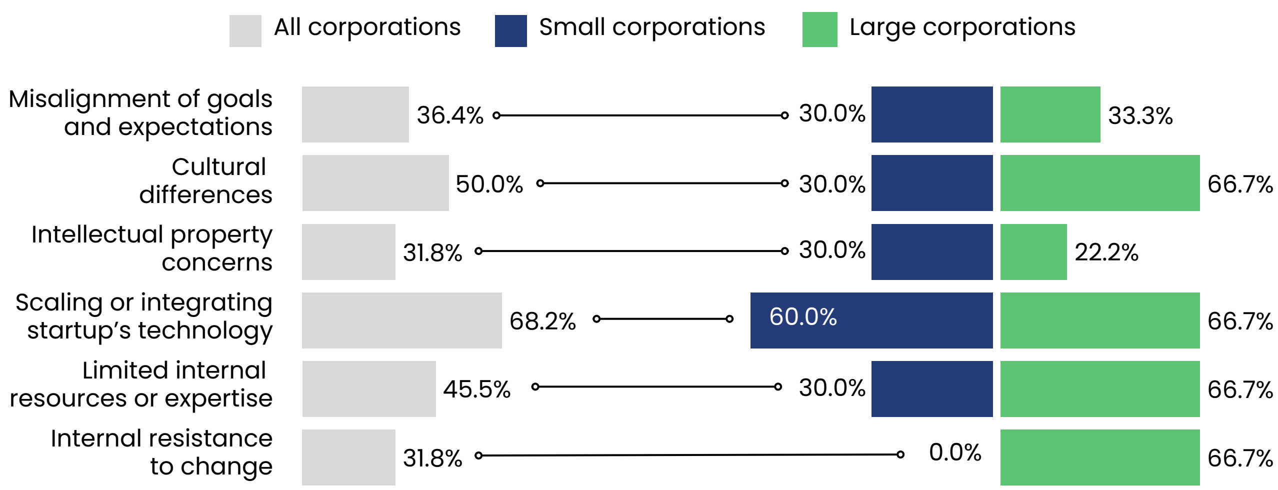
Responses revealed sharp disparities: more than twice as many large corporations cited joint ventures and equity investments/minority stakes compared to small corporations.

This underscores small corporations' preference for independent collaborations, contrasting with large corporations' inclination toward partnerships and joint ventures.

What are the most successful collaboration models for large vs small corporations?



What challenges has your organization faced when partnering with Climate Tech startups?



Challenges differ for small and large corporations in partnerships

Small and large corporations diverge most on challenges like cultural differences and internal resistance to change, reflecting the complex stakeholder dynamics and varying interests in larger organizations. Common challenges across sizes and sectors include capital constraints, misaligned goals, and scaling or integrating startup technology, cited by a majority of respondents regardless of company size or sector.

Thought Leader Interview

Cemex Ventures is the open innovation platform and the CVC arm of Cemex, based in Madrid, Spain. The firm seeks to partner with and invest in seed-stage, early-stage, and later-stage companies operating in ConTech and ClimateTech spaces applied to the built environment across North America, EMEA, LatAm, Middle East and APAC.

Ibon Iribar
Investment advisor

Ibon is an Open Innovation and Investment Advisor at Cemex Ventures, based in Madrid. He currently scouts for breakthrough technologies and solutions in the sustainability space applied to the built environment, plus oversees investments as well as other strategic partnerships. Ibon has a special interest in startups that effectively tackle the carbon footprint of the construction industry through circular business models, considering multiple levers in both the physical and digital worlds

Cemex Ventures focuses on revolutionizing construction logistics. What key trends are driving your investment strategy in this space?

Our investment strategy is driven by several key trends:

- Rapid urbanization: As cities grow quickly, there is increasing demand for efficient, sustainable construction solutions to meet urban needs.
- Scarcity of natural resources: With limited resources, there is a push for more sustainable practices, alternative materials, and improved resource efficiency in construction.
- Shift to renovation and retrofitting: rather than new builds, there's a growing focus on renovation and retrofitting, which is often more sustainable and cost-effective. We invest in solutions that enhance these processes.

These trends guide our investments in technologies that improve efficiency, sustainability, and scalability in the construction sector.

When assessing the scalability and sustainability of startups like Carbon Clean and Energy Vault, what specific factors do you consider to ensure their long-term success in the construction industry?

When assessing the scalability and sustainability of startups, we focus on several key factors:

- Commercial viability: Even if technologies are still under development, we look for signs they can offer real-world value in the construction sector, with potential for scalability and market adoption.
- Technology Readiness Level (TRL): We assess how close the technology is to commercialization. Technologies with proven prototypes or pilot projects are prioritized for their higher success potential.
- Partners and investors: A mix of financial and strategic partners is essential. Startups with strong backing show greater confidence and market access, increasing their chances of long-term success.
- Strategic fit with Cemex: We look for technologies that align with Cemex's goals and offer opportunities for developing use cases within our operations, ensuring real-world integration and value.

With a focus on last-mile delivery solutions for construction materials, what are the biggest challenges you see in transforming traditional logistics in the construction industry?

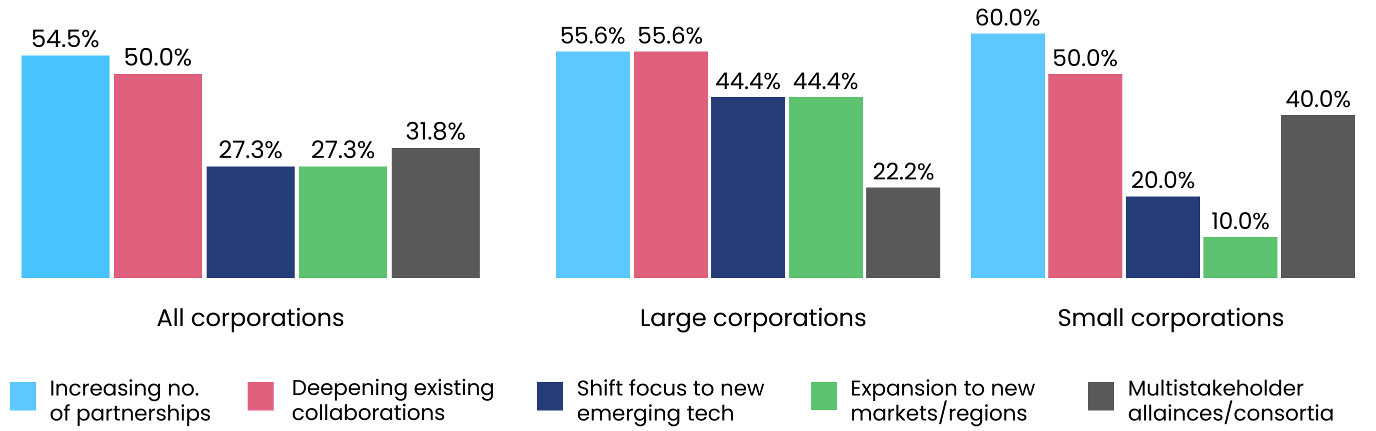
There are more and more people at urban areas, thus large construction developments are already being a megatrend in the industry. That means that delivering enough construction materials efficiently and on time is becoming a challenge. Besides, we are also going to start seeing scarcity of few natural resources such as natural aggregates or water, thus making sure the access to these components through a strong value chain is key for manufacturers and construction managers.

What are some examples of how you've integrated the solutions of your portfolio companies into CEMEX?

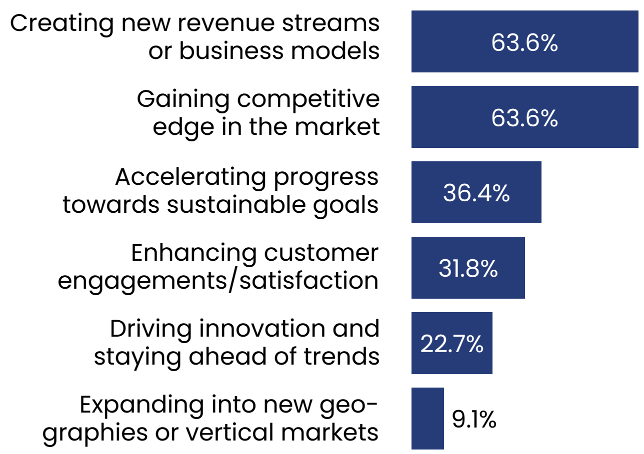
In addition to integrating several of our portfolio companies directly with Cemex, such as Vizcab, Cobod, and Synhelion, we are also working with other startups through our Leap Lab Acceleration Program. We select technologies and solutions that have the potential to address our internal needs, and then collaborate with the companies to develop use cases or pilot projects over the coming months.
Some examples of these companies include Introid, Movener, Mixteresting, Verusen, Waterplan, Cloud Cycle, AEInnova, Ception, Qsee, and Ruedata.

Future outlook and opportunities
from NZI's SOCT 2024 survey

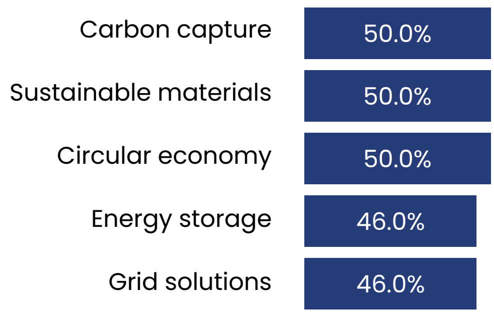
How do you see your organization's engagement with Climate Tech startups evolving over the next 1–3 years?



What key opportunities does your organization see in partnering with Climate Tech startups?



Which emerging solutions do corporations see as offering the greatest partnership opportunities?



Large corporations prioritize emerging technologies and market expansion

Expansion into new markets and regions is a priority mainly for large corporations, with only 10% of small corporations expecting such engagements in the next 1–3 years. Similarly, 20% of small corporations anticipate focusing on emerging technologies, compared to a greater emphasis among large corporations.

Emerging mtigation solutions offer key partnership opportunities

Most corporations see Climate Tech investments as a chance to drive innovation, aligning with their preference for mitigation-oriented solutions like CCUS (50%), more sustainable materials (50%) and Circular Economy (50%) as top choices for partnerships.

Thought Leader Interview

Schmidt + Clemens

The name Schmidt + Clemens has stood for high-quality special steel solutions for more than 145 years. Be it spun casting, static casting, investment casting or forgings, we will face your challenges head on.

The Schmidt + Clemens Group is active on the international stage. Besides its German headquarters in Lindlar-Kaiserau, S+C also has production facilities in Spain, the Czech Republic and Malaysia. Our distribution company in the US combined with more than 30 commercial agencies consolidate our global presence.

Dr. Ansgar Niehoff
Strategic Technology and Investment Manager

Ansgar Niehoff holds a PhD in Chemistry from the University of Technology in Braunschweig, Germany. He worked as Materials Researcher and Technology Manager for Smart Products and Head of Innovation & Tech Venturing at the REHAU Group, one of the biggest polymer manufacturers worldwide. There he also developed ten Circular Design Principles which should support Developers and Engineers to design products in a circular way. He is strongly connected with the tech startup Ecosystems in Europe and Israel and also serves as a pro bono mentor at the international Climate Tech Accelerator program Climate First. Since July 2024 he has taken on a new role as Strategic Technology & Investment Manager at the Schmidt + Clemens Group, world-market leader in special steel solutions for the Petrochemical Industry. Ansgar is always eager to learn about new technologies that increase efficiency, reduce emissions and find new ways to create a future-proof raw material industry.

Schmidt + Clemens has made significant strides in reducing its carbon footprint in its own production. What is on the agenda to strengthen these efforts in the future? Which technologies are applied?

At the moment there is a massive project under way to connect and digitize the spun casting production with the goal to lower the scrap rates and thus saving energy and reducing emissions. Also it is planned to use more renewable energy by installing more Photovoltaics in our plants in all our plants in Germany, Malaysia, Spain and Czech Republic.
Already all our melting furnaces run on electricity instead of natural gas and the transition from heating with natural gas to using electricity (which can be step by step sourced renewably) is on its way in more and more of our production steps. Also we have started to explore the feasibility to work with startups to „regain cold" from our Liquid Gas tanks when the liquid gas is brought to room temperature from -186 °C.

As a world-market leader for heat-resistant tube systems for the Petrochemical Industry, what role do you see Schmidt + Clemens playing in the global transition towards renewable energy and a circular economy?

Our heat-resistant tube systems have a very high quality and longevity which allows our customers to use them as long as possible which saves resources. Our customers use oil and natural gas as normal feedstock still but in their specific industry these valuable resources are not used for energy production but for raw materials production. These raw materials are needed predominantly for the polymer industry which is a vital industry for e.g. light-weight solutions in Battery Electric Vehicles, producing Photovoltaic panels and in the wind power industry. But we also see that more and more oils from polymer pyrolysis (one of many polymer recycling technologies) are being applied as feedstocks instead of naphta from oil. Therefore our material specialists make sure that our special steel solutions work perfectly in these new conditions to ensure the transition to more circularity. Also we look at opportunities for our solutions to work in processes where e-fuels are produced from renewable resources.

How does Schmidt + Clemens plan to collaborate with other stakeholders in the industry to accelerate the adoption of sustainable technologies, particularly in sectors like energy, petrochemicals, and heavy industries?

Our preferred way of collaboration is through partnerships where every stakeholder provides the expertise where it is best at. Of course we see great potential in collaborating with established technology providers. But we also see rising potential to work with Climate Tech startups as they often have completely new and fresh ideas and work on disruptive technologies where established players tend to be too slow to adapt them.

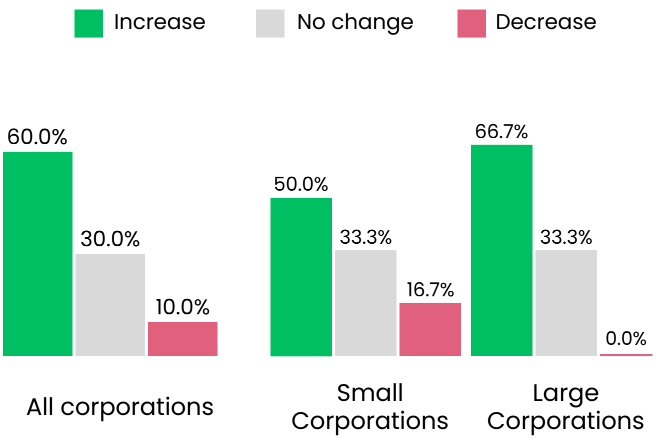
What key challenges do you anticipate in the future, and how do you plan to leverage collaboration with Climate Tech startups to address them?

Key challenges will be speed to adapt new technologies to reduce emissions. Often industries like the ones of our customers are very conservative as they deal with processes which are very delicate and can be dangerous if not handled with the corresponding experience. So we need to find ways to speed up the transition without compromising on safety and regulations.

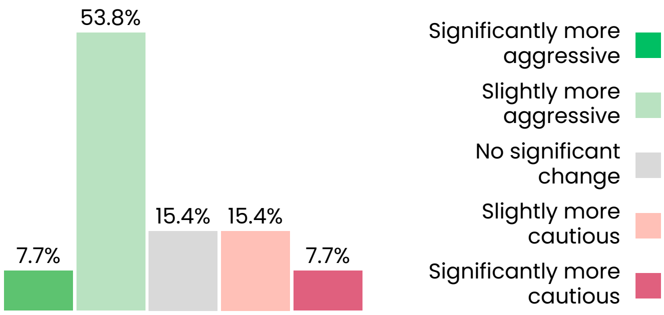
When collaborating with Climate Tech Startups they have to bring in the speed and „out-of-the-box" thinking and we have to balance it with our sense for safety, quality and our decades of experience.

M&A activity
from NZI's SOCT 2024 survey

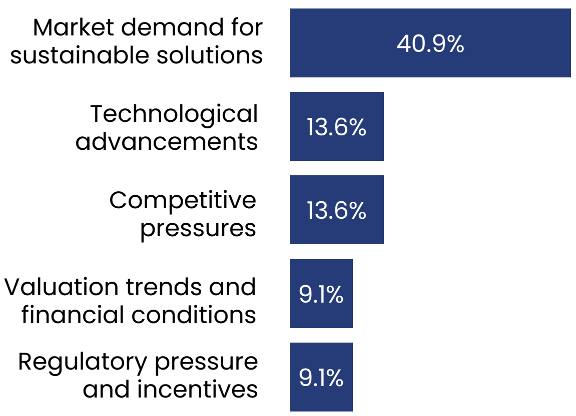
How did your corporation's Climate Tech M&A activity in 2024 compare to previous years?



How do corporations expect their Climate Tech M&A strategy to evolve over the next 2–3 years based on their 2024 experience?



What factors influenced your corporation's Climate Tech M&A decisions in 2024?



Corporations see increased Climate Tech M&A activity in 2024

Most corporations in the SOCT 24 survey reported increased M&A activity in 2024 compared to previous years, with 22.7% more corporations noting an increase than those citing a decrease. This trend was especially pronounced among large corporations, which outpaced medium and small corporations in M&A growth.

Competitive pressure influences large corporations' M&A decisions

While market demand for sustainable solutions drove M&A decisions across all corporations, competitive pressure was a factor for large corporations but was not cited by medium or small corporations. Regulatory changes, particularly in the automotive and transport sectors, were cited by 50% of respondents.

Optimism for future M&A strategies

Small corporations were more optimistic about future M&A, with 50% expecting a more aggressive strategy in the next 2–3 years, compared to 20% of large corporations. Among collaboration types, 57% of corporations with equity investments and 62.5% with supplier/customer relationships expressed optimism. These partnerships offer faster returns and stronger ties compared to joint ventures or R&D collaborations.

04 Ecosystem supporters

The Climate Tech innovation ecosystem thrives on collaboration among diverse stakeholders. Key supporters—accelerators, incubators, governments, NGOs, research institutions, and specialized services like recruitment and PR/marketing—play a vital role in advancing the sector. This section provides an overview of the most active and impactful ecosystem supporters of the year.

Profile of 2024 ecosystem supporters

Overview

Startup programs and support initiatives focused their efforts on assisting over 2,000 organizations in 2024, with a primary emphasis on transforming the electricity sector.

While accelerator and incubator activity is expected to decline by nearly 20% year-over-year in 2024, governmental entities involved in grant support to Climate Tech organizations are projected to grow by more than 24% compared to 2023.

180+

governamental entities provided grant support to Climate Tech organizations in 2024.

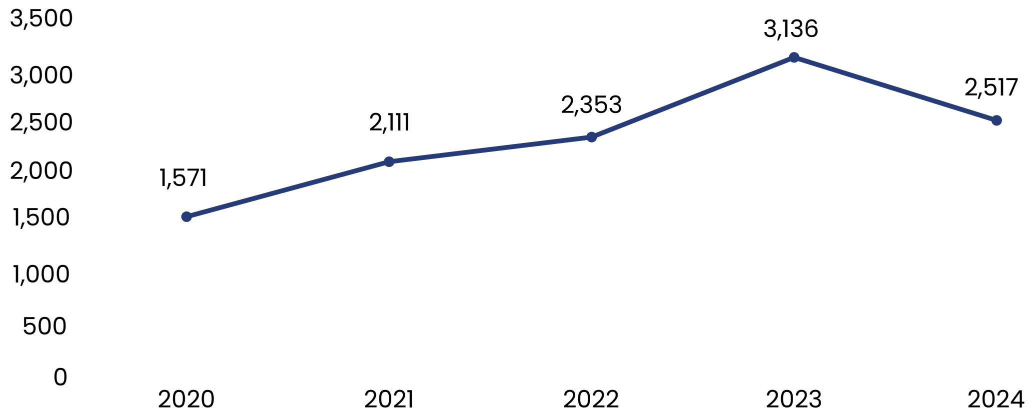
2,000+

organizations benefited from support programs provided by accelerators, incubators, research insitutes, and universities in 2024.

electricity

was the sector receiving the highest support from startup support initiatives and programs.

Yearly global accelerator and incubator activity

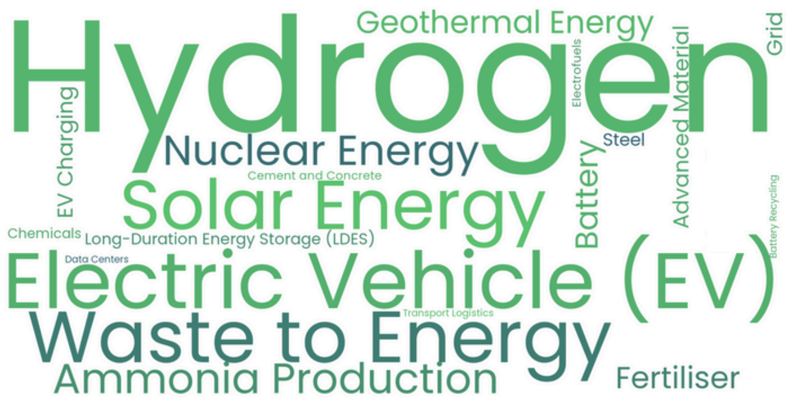


Source: Net Zero Insights

The figures represent total activity in private Climate Tech ventures, including equity, debt, grants, and other instruments, excluding exits and post-exit financing. 2024 data is based on linear projections through year-end.

Most notable grant providers

2024 most funded solutions by governments



2024 most notable grant providers

U.S Dept of Energy
308 Grants
United States

UK Research & Innovation
77 Grants
United Kingdom

European Commission
67 Grants
Belgium

ARENA
56 Grants
Australia

National Science Foundation
52 Grants
United States

European Innovation Council
32 Grants
Belgium

Thought Leader Interview

Prime Coalition ("Prime") is a nonprofit organization on a mission to unlock catalytic capital and change the future of climate finance. We envision a world in which humanity thrives, without the threat of climate change.

Keri Browder
Director of Impact Accountability and Director of Project Frame

As Prime Coalition's Director of Impact Accountability and Director of Project Frame, Keri Browder creates opportunities for investors to share tools and best practices that catalyze, deepen, and accelerate diverse solutions that massively reduce GHG emissions. An environmentalist at heart, Keri has dedicated her career to tackling tough problems. Before Prime, she led the expansion of global programs, including Ocean Conservancy's Urban Ocean, Tetra Tech's USAID WASH-FIN, and supported initiatives to protect voting rights in the US and expand community-based policing reform in Colombia. Her experiences emboldened her call to action against climate change and belief that partnership, investment, policy, and impact evaluation can work in concert to meet the challenge. She holds a graduate certificate in Social Finance from Oxford University as well as undergraduate and graduate degrees in political science from the College of Charleston and Universidad de Los Andes.

What unique advantages does Prime Coalition, as a climate-focused nonprofit, offer over traditional venture capital in supporting early-stage Climate Tech startups?

Prime Coalition takes a unique nonprofit approach to supporting early-stage Climate Tech startups, using catalytic capital to target high-impact companies often deemed too risky for traditional investors. This strategy enables us to back transformative climate solutions while tying every investment to meaningful impact outcomes. We also align investments with impact-driven incentives, linking team compensation to climate milestones to ensure a mission-first focus. Our expertise in regulatory frameworks and fund design allows us to address critical capital gaps, exemplified by Trellis Climate, our new program for first-of-a-kind project finance. With a portfolio of 39 innovative companies, Prime is reshaping climate finance by bridging gaps and redefining how capital drives systemic impact.

How do you attract philanthropic partners and align their goals with the high-risk, high-reward nature of Climate Tech investments?

Prime attracts philanthropic partners by emphasizing impact as our core principle, aligning with mission-driven donors. We build trust and long-term collaboration by understanding partners' goals, sharing our vision transparently, and setting clear expectations.
Since our inception, we've mobilized over $315 million from 360+ philanthropic partners, including 24 donor-advised fund sponsors and 78 private foundations. Remarkably, 25% are first-time climate investors, and 47% are first-time catalytic investors, showcasing our success in driving new commitments to impactful solutions. In addition to deploying catalytic capital, Prime empowers nonprofits and philanthropic asset owners to navigate early-stage Climate Tech investing, simplifying complexities to help them embrace its unique risks and rewards.

How does Prime Coalition identify the most impactful early-stage ventures for investment using catalytic capital in a rapidly evolving field?

We are researching priority areas for systemic investing, focusing on geographies and industries where Prime's catalytic capital can have the greatest climate impact.
Our three core investment criteria remain:

- Impact potential: solutions with large-scale climate outcomes.
- Bridge to somewhere: solutions with paths to follow-on capital.
- Additionality: solutions that need catalytic capital to scale.

This expanded focus enables us to tackle broader climate outcomes, including adaptation and resilience, alongside emissions reduction. Growing partner interest underscores Prime's potential in areas like climate litigation finance and nature-based solutions. As our portfolio expands, new catalytic funds enhance our philanthropic ecosystem, amplifying impact.

Looking ahead, how does Prime Coalition plan to accelerate the deployment of catalytic capital for high-impact climate solutions?

To meet our 2033 goals, we aim to accelerate catalytic capital for underserved enterprises and strengthen nonprofit infrastructure to amplify impact. We're exploring scalable models and reassessing our additionality criteria to adapt to the evolving climate finance landscape.
Beyond catalytic investing, we aim to shift the broader climate finance field. Our CRANE tool, co-created with Rho Impact, with 5,300+ users in 89 countries, helps assess emissions reductions of early-stage solutions. Project Frame brings together 900+ professionals managing $670B+ in VC and PE assets to develop greenhouse gas impact standards. As a pioneer in catalytic capital, we share insights through resources like the Catalytic Capital Resources Library, which has reached 1,500+ users.

Thought Leader Interview

EIT InnoEnergy operates at the centre of the energy transition and is the leading innovation engine in sustainable energy. It brings the technology, business model innovation and skills required to accelerate the green deal, progress towards Europe's decarbonisation and re-industrialisation goals, whilst also improving energy security.

InnoEnergy backs innovations across a range of areas. These include energy storage, transport and mobility, renewables, energy efficiency, hard to abate industries, smart grids and sustainable buildings and cities.

InnoEnergy has a portfolio of more than 200 companies, which are estimated to generate €110 billion in revenue and save 2.1G tonnes of CO_2e accumulatively by 2030. Collectively, these companies have raised more than €25 billion in investment to date.

Szymon Kwiatkowski
Investment Manager

Szymon is an Investment Manager at EIT InnoEnergy. EIT InnoEnergy is one of the world's largest early-stage investors in the field of sustainable energy and backs Climate Tech innovations across a wide range of areas, including energy storage, transport and mobility, renewables, circular economy and sustainable cities and buildings.

As part of the investment team, Szymon is responsible for InnoEnergy´s investments in startups and scaleups in the CEE region, focusing on innovative technologies that helps to achieve Europe's decarbonization goals. Szymon has been involved with venture capital and private equity investments for more than last 13 years and is a CFA charterholder.

InnoEnergy focuses on investing in sustainable energy innovations. How do you evaluate the long-term impact and scalability of projects working on breakthrough technologies in clean energy?

Every two years, we measure the impact of emissions avoided at the portfolio level and publish a comprehensive impact report. However, because we typically invest at a very early stage, we do not precisely quantify the impact of every new investment. Instead, we assess the general potential of the problem being addressed by the company and the size of the target market, considering both its dollar value and its potential for emission avoidance. Hence, we are not afraid to include in our strategy such ambitious and long-term projects such as nuclear fusion and hyperloop technologies.

In recent months, InnoEnergy has supported a wide range of Climate Tech startups. What key trends or sectors are you prioritizing in the clean energy and sustainability space right now?

Currently, we are prioritizing several key trends and sectors in the clean energy and sustainability space, and the mentioned companies are good examples of this. In battery technologies, we are actively seeking innovations that explore untapped areas and create new value in the sector. This includes advancements in energy density, safety, and cost-effectiveness, not just to improve these by a few percent, but to solve actual problems in the industry. That's exactly what Wattalps is doing for heavy-duty construction equipment and special machinery. In bioeconomy, we seek new ways to valorize biomass beyond energy recovery by developing technologies to convert it into high-value products and materials. The same philosophy guided us to support not only Nordic SeaFarm but also EcoBean, which valorizes spent coffee grounds into highly valued specialty and sustainable chemicals. On top of these, we strongly believe in the need for investments in strategic and critical materials, including sustainable mining innovations, recycling, and solutions that limit their use.

How does InnoEnergy work with its portfolio companies to ensure their technologies can integrate into existing energy infrastructure and address real-world sustainability challenges?

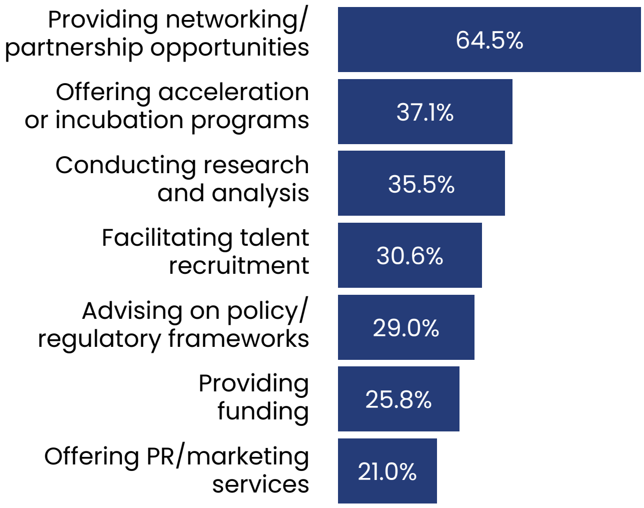
In the past nearly 14 years of operation, InnoEnergy has created a unique ecosystem of startups, partners, investors, research centers, and more, which can be efficiently leveraged by our portfolio companies. Together with our thematic teams, business developers, and investment teams, we involve key stakeholders from the very beginning. This approach allows for pilots, PoCs, and strategic alliances to emerge. Our investments aim to support early-stage startups, de-risk them through our ecosystem to accelerate time to market, and drive growth. We also co-create major industrial initiatives to deploy impactful technologies in high-potential niches. This approach has led to the creation of companies like FertigHy, which provides low-carbon fertilizers, and GravitHy, which accelerates the decarbonization of the steel industry.

As the energy transition accelerates, what role will InnoEnergy play in advancing technologies like ocean turbines and quasi-solid-state sodium batteries?

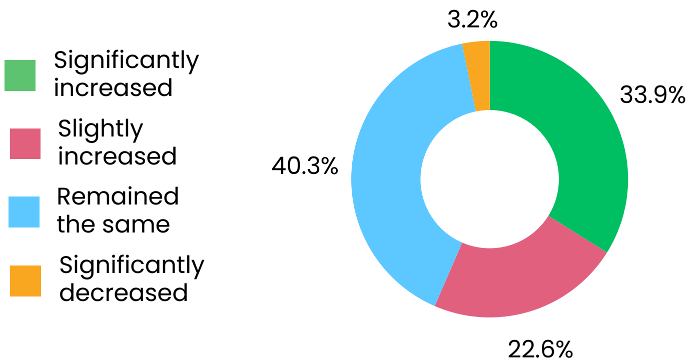
We have invested in a few alternative ocean energy technologies, such as Minesto, SeaQurrent, and CorPower Ocean. The latter recently secured EUR 32 million in Series B1 funding, led by NordicNinja VC, SEB Greentech, and us. We continue to seek opportunities, as the ocean holds vast energy potential if harnessed sustainably. Sodium-ion batteries, made from abundant, recyclable materials, align with our investment strategy. This led us to invest in Altris, and we're eager to find more innovations in this field.

Role and impact
from NZI's SOCT 2024 survey

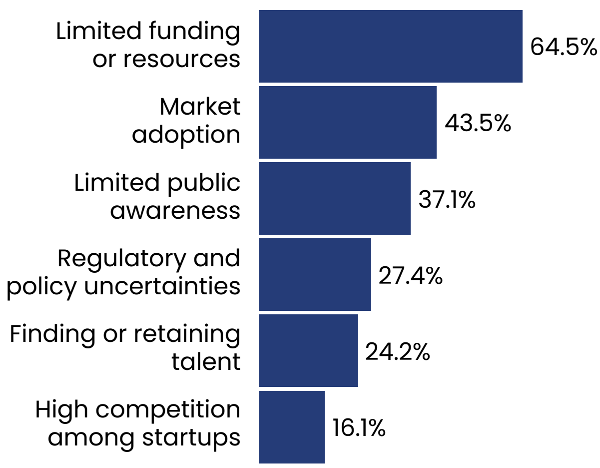
What is the primary way your organization supports the Climate Tech ecosystem?



How has your organization's involvement in Climate Tech evolved in 2024 compared to previous years?



What are the main challenges encountered by your organisation in supporting Climate Tech companies?



Networking and partnership opportunities were the most common ways ecosystem supporters collaborated with Climate Tech companies, cited by 79% of Startup Accelerators.

In North America, over 50% of ecosystem supporters facilitated talent recruitment and workforce development. In Europe, nearly 50% focused on policy and regulatory advisory as their primary support mechanism.

Ecosystem supporters report increased involvement in Climate Tech

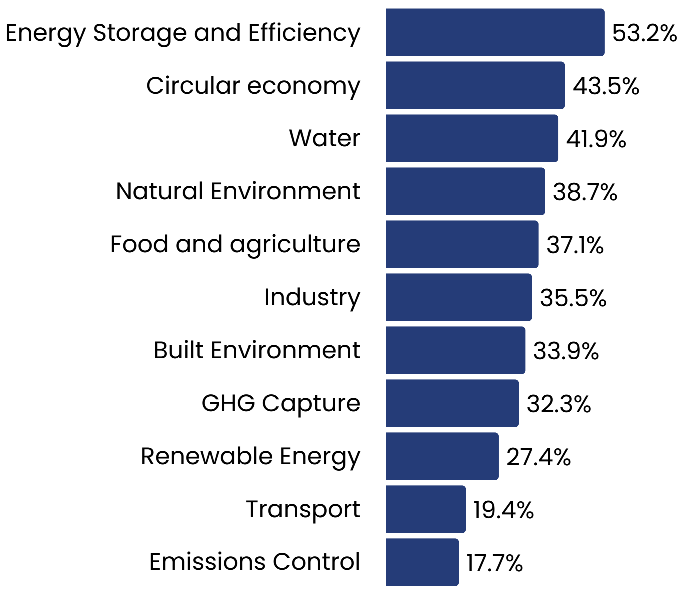
The majority of interviewed ecosystem supporters reported greater involvement in Climate Tech in 2024 compared to previous years, with only 3.2% noting any decline.

Resource constraints the biggest hurdle

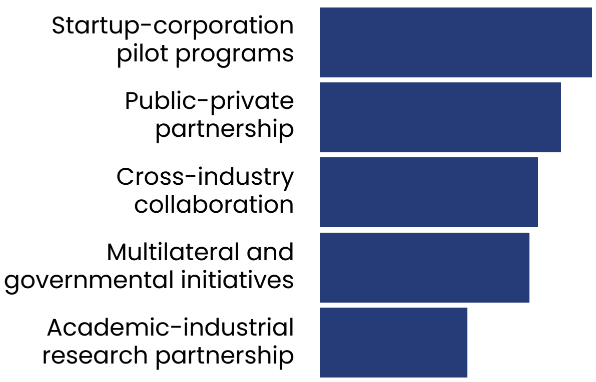
Funding and resource limitations were identified as the top challenge by two-thirds of respondents, with over 90% of North American ecosystem supporters highlighting this issue.

Role and impact
from NZI's SOCT 2024 survey

Which sectors are most in need of support from ecosystem contributors?



What types of collaborations have been most effective in advancing solutions?



Emerging sectors Identified as needing the most support

Energy Storage was the most frequently cited sector needing support, largely due to its capital-intensive infrastructure requirements. Technologies like hydrogen and biofuels also face significant challenges due to high costs and associated risks.

Other emerging sectors identified as needing support included water, circular economy, food and agriculture, and the natural/built environment. Over one-third of respondents highlighted GHG Capture, Natural Environment, Built Environment, and Food-Agriculture as underfunded sectors requiring greater attention.

With funding heavily skewed toward mitigation-focused sectors like Transport and Renewable Energy, these emerging sectors will increasingly rely on government and ecosystem contributors for essential support.

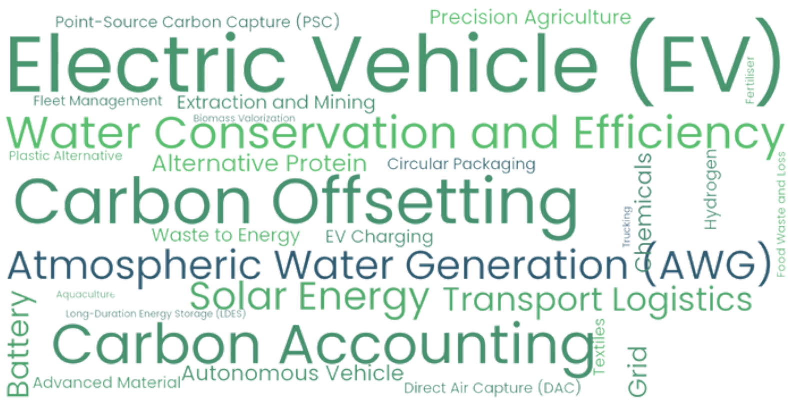
Partnerships with corporations and governments deemed most effective

Over half of ecosystem contributors identified startup-corporation pilot programs and public-private partnerships as the most effective ways to collaborate with Climate Tech startups.

These partnerships are valued for their ability to provide long-term capital and support for early-stage development and growth, offering startups the resources needed to scale successfully.

Most notable accelerators

2024 most funded solutions by Accelerators and Incubators



2024 most notable Accelerators and Incubators

Plug And Play	96 Deals	United States
MassChallenge	75 Deals	United States
EIT Food	72 Deals	Belgium
Third Derivative	59 Deals	United States
Cleantech Open	58 Deals	United States
Remove	48 Deals	Netherlands

Thought Leader Interview

Greentown Labs is a 501(c)(3) nonprofit accelerating climatetech innovation and commercialization by empowering entrepreneurs and enabling collaboration.

As the largest climatetech startup incubator in North America—with locations in Somerville, Mass. and Houston, Texas—Greentown convenes the climatetech ecosystem to provide entrepreneurs the community, connections, and resources they need to thrive.

Sam Crum
Director of Membership

As Greentown Labs' Director of Membership, Sam and his team source and guide new startups into the incubator - connecting entrepreneurs to key support resources and a community of their peers.

He oversees Greentown's strategy and team for new member recruitment and the incubator's application process. .

Since joining the Greentown team in 2019, Sam has worked directly with thousands of climate entrepreneurs, including 200+ startups who joined Greentown Labs' incubator program

What emerging trends in Climate Tech are you excited about, and how is Greentown Labs supporting startups in these areas?

2024 was a reality check for Climate Tech, marking a return to steadier growth after the funding bubble of 2021-2022. While there were exciting breakthroughs and growing interest in climate-focused careers, challenges arose, including cautious investors, stalled companies, rising temperatures, and shifts in US climate policy with potential global effects.

Despite these challenges, key Climate Tech trends are advancing:

- Methane mitigation: High-impact methane solutions, like those from Capwell and M2X Energy, are driving better methane management.
- Mineral processing: New methods are unlocking minerals from low-grade ores, with startups like Hertha Metals innovating lower-emission steelmaking.
- Geologic hydrogen: This emerging zero-emission fuel source is advancing with companies like Eden GeoPower.
- Hydrogen logistics: Transportation and storage advancements, led by alumni like Verne, are essential to scaling hydrogen's use.

How does Greentown Labs balance risk and innovation in early-stage investments, and what role do corporate or governmental partnerships play?

Startups are risky, especially at early stages. Greentown helps de-risk technology, prepare for market challenges, and set up startups for long-term success.
While Greentown doesn't invest, we provide resources and connections to help startups navigate their journey efficiently. Our rigorous application process focuses on promising solutions, allowing for calculated risks.
Corporate partners are essential for scaling, but Greentown ensures startups avoid overly customized solutions that limit growth. Our 80+ corporate partners open doors for pilots, procurement, and investments.

Government offers crucial funding and support, though its pace can be challenging. Founders should cast a wide net early on to keep funding and growth options open.

What key challenges do your portfolio companies face when scaling in Climate Tech, and how does Greentown Labs support them beyond initial investment?

Team, team, team. The challenges around accessing capital and customers are real and areas Greentown supports too, but one theme that will overlay all of these obstacles and will make or break a company's ability to succeed is the people behind the startup.

Greentown supports founders in hiring and team culture through our EIR program and monthly CEO roundtables for peer learning. Our online job board, career fairs, and university partnerships help startups find top talent.

Finally, our incubator locations in Boston, Massachusetts and Houston, Texas strategically position our members nearby top global talent—engineers, chemists, and business leaders who are increasingly eager to deploy their expertise on the greatest challenge of our generation.

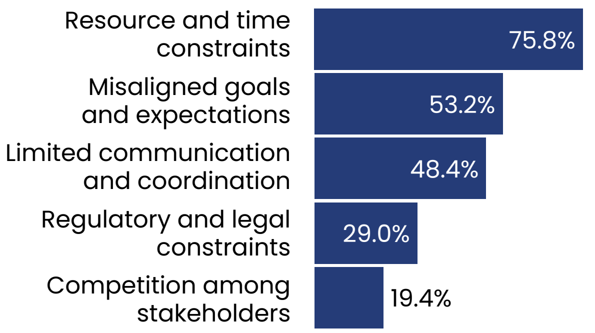
How do you measure the success of your portfolio companies, and can you share examples of significant scale-ups that you have previously supported?

For Greentown, success is measured by our startups' growth, deployment, and impact on emissions reduction and resilience. We track metrics like project count, revenue, and technology stage, but prioritize companies with high impact potential by 2050. Startups estimate CO_2e reductions their solution could achieve and report milestone progress via surveys.
Notable alumni include Form Energy, Sublime Systems, Fervo Energy, Via Separations, ZwitterCo, Cemvita Factory, and American Battery Technology Company.

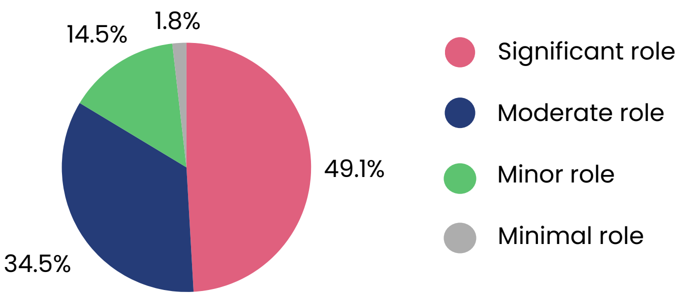
Collaboration and ecosystem dynamics

from NZI's SOCT 2024 survey

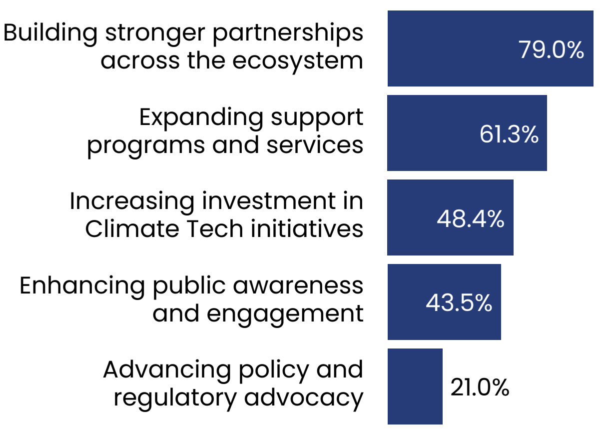
What barriers do you encounter when trying to foster collaboration within the Climate Tech ecosystem?



What role does your organization play in shaping the direction and success of the Climate Tech sector?



What are your organization's strategic goals in supporting the Climate Tech ecosystem over the next 1-3 years?



Resource and time constraints were the most commonly cited barriers to fostering collaboration across categories and geographies. Regulatory and legal constraints were also significant, affecting 50% of organizations operating for 1–3 years, 33% of those operating for less than 1 year, and 25% of those with 10+ years of experience.

Among startup contributors, accelerators, incubators, and recruitment agencies had the highest percentage of respondents viewing their role as major or significant in Climate Tech success. This sentiment was shared by 66% of contributors operating for 7–10 years, 62.5% of those active for 4–6 years, and 56% of those operating for 1–3 years.

Europe-based contributors prioritized "advancing policy and regulatory advocacy" (24%) more than North America-based contributors (11.1%). Conversely, 78% of North America-based contributors cited "expanding support programs and services" as a goal, compared to 44% in Europe. Both regions showed strong alignment on "building stronger partnerships," with 78% in North America and 72% in Europe sharing this objective.

Building stronger partnerships across the ecosystem emerged as the most common strategic goal among ecosystem supporters. This aligns with the importance respondents placed on partnerships with governments and corporate pilot programs in earlier survey responses.

Thought Leader Interview

Fondazione Bruno Kessler (FBK), based in Trento, is a multidisciplinary research institution specializing in technology, innovation, humanities, and social sciences. Established in 2007, FBK continues the legacy of the Istituto Trentino di Cultura, founded in 1962 by Bruno Kessler. With over 620 staff members, 150 doctoral students, and numerous visiting researchers, FBK is a hub of innovation and collaboration, hosting experts from across Europe, the , China, India, and beyond.

FBK operates across two campuses: one in Povo, focused on technology and innovation, and the other in Trento's city center, dedicated to humanities and social sciences. Together, they span 40,000 square meters, including 5,000 square meters of laboratories. The institution comprises 12 research centers, 7 laboratories, and 2 libraries, fostering an ecosystem of businesses, joint labs, and spin-offs.

Francesco Matteucci
Senior Innovation Advisor

Francesco, a C-level innovation expert with 20 years of experience, has made significant contributions to the field of energy and environmental sustainability. His expertise spans from materials science research to managing corporate R&D in renewable energy technologies.

As a co-founder of several start-ups and joint labs, a former European Innovation Council Programme Manager, and Senior Innovation Advisor at Fondazione Bruno Kessler, Francesco tries to bridge the gap between research and real-world application, driving innovation forward.

With your extensive background in renewable energy production, storage, and materials science, how do you see the role of advanced materials evolving in Climate Tech and what do you consider to be the biggest challenges?

I believe that advanced materials play a crucial role in the entire Climate Technology value chain, but it's often an underestimated part. We see many discussions about critical raw materials and circularity, but Europe still lacks alternatives to reduce its dependency on non-EU sources for many materials. The two paths to address this challenge are: first, focusing on the circular economy approach, and second, developing completely new advanced materials. The European Commission is promoting efforts on both fronts, such as co-funded partnerships on critical raw materials and advanced materials as well as on policy with different actions such as the EU Circular Economy act and the Safe and sustainable by design framework. The key challenges here are the development of new materials using AI and computational tools, as well as scaling up manufacturing facilities for advanced materials, where Europe is lagging behind. These are areas that require urgent investment to create a sustainable circular value chain and technological innovations that can enhance the whole renewable energy value and supply chain.

What do you see as the biggest challenges and opportunities in public-private partnerships within Climate Tech, and how can stakeholders improve collaboration to accelerate innovation?

The biggest challenge is creating true win-win collaborations. Financial partners often focus on numbers, overlooking technical or market considerations, while scientists may ignore business and scaling concerns. The solution is bringing together businesses, scientists, and the public sector, ensuring each party's interests are addressed. We need skilled intermediary managers fluent in both business and science to navigate these partnerships.

The public sector can also play a key role with more proactive public procurement, which could drive innovation in Climate Tech. Effective collaboration will require transparency, flexibility, and a focus on long-term sustainability and trust.

What advice would you give to emerging Climate Tech startups in 2024, particularly on scaling their impact?

My advice to emerging Climate Tech startups is to balance the team carefully. If the team is made up only of scientists, they may overlook market needs. If it's market-driven without technical knowledge, innovation may suffer. A strong startup team should blend scientific expertise with market-driven leadership. I believe in venture building—incubating companies with the right teams, especially in leadership roles. Startups should recruit a CEO and CFO early to ensure both technical and business perspectives. Fundraising strategy is crucial; startups must know when and how to secure funding at each growth stage. Additionally, exploring creative compensation models like equity-for-work can help attract business talent.

What promising regional or EU-funded initiatives in Europe could advance climate resilience and sustainability?

Emilia-Romagna faces significant climate resilience challenges, as highlighted by recent floods. To improve resilience, we need long-term investments addressing these risks. Optimizing the electric grid with AI is a promising solution to enhance energy system resilience. AI and machine learning can also help design future climate scenarios and identify the most viable solutions. There is increasing support for greater synergy between European, national, and regional projects, combining public and private efforts. Public-private initiatives are key to advancing local agendas and creating a model for Europe to tackle these complex challenges.

Most notable banks

2024 most funded solutions by Banks

2024 most notable Banks

Bpifrance	32 Deals	France
EIB	27 Deals	Luxembourg
Bnp Paribas	15 Deals	France
HSBC	13 Deals	United Kingdom
SMFG	13 Deals	Japan
Mizuho	12 Deals	Japan

Thought Leader Interview

LLUT University (Lappeenranta-Lahti University of Technology LUT est. 1969) in Finland is a forerunner in clean energy, water and air with the expertise of 9,000 students and faculty members in technology, business and social sciences.

With a clear focus on solving global climate problems through scientific research, academic education and societal impact LUT University is ranked as the 10th best university in the world in UN's Sustainable Development Goal (SDG) #13 - Climate Action.

Pia Erkinheimo
Member of the board

"As for the future, your task is not to foresee it, but to enable it." - Antoine de Saint Exupéry

Pia Erkinheimo is a board member at LUT University and a seasoned angel and growth company investor in ClimateTech, being currently active in FiBAN - one of the largest angel investor network in Europe. She just recently resigned from the Finnish Climate Fund where she headed the dealflow and co-investor relations.

Her background is from companies like Capgemini, Nokia and Nordea. The European Commission has benefitted from her S&T&I experience for +15 years as innovation expert and jury member, and she has advised the G7 Countries in digital technologies, advanced robotics, and data economy. Her other board responsibilities cover e.g. Industry Commons Foundation in Sweden and Atlan – a venture studio for Nature-Tech. Pia is a co-founder of Women in Tech Finland, a network of companies standing for diversity, equity, and inclusion.

How is Finland using renewable energy and emerging technologies to achieve carbon neutrality by 2035, and which innovations are are being prioritized in this journey?

Finland wants to be first country in the world in reaching carbon neutrality by 2035. In 2023 94% of the electricity produced in Finland was emission-free, and the share of the renewable energy sources in the Finnish energy production was 52%.

The secret sauce of the Finnish energy production palette is the diversity of its production mix – both in electricity and heat production. It is critical to balance supply and demand, especially as more intermittent renewable energies like wind and solar are brought online, new innovations are needed.

The focus is currently in smart grid and energy storage solutions as the Nordic innovation ecosystem contains Europe's first pan-European power exchange Nord Pool and one of the best maintained grids (Fingrid). Also sustainable aviation fuel (SAF) production is well on its way, and SMRs (small modular reactors) are being developed, with the aim being in production late this decade.

What role should academia and research organisations play in advancing Climate Technology, and how can we strengthen academic involvement in driving Climate Tech innovation?

We are looking innovation ecosystem (Penta Helix) by adding civil society and media&culture in the "Triple helix" model, in order to broaden the range of perspectives, since the innovation ecosystems need to address complex societal challenges with more inclusive, sustainable and holistic solutions. Research organisations are driving the change in creation of a society that respects planetary boundaries. It is said, that in order to maintain somewhat similar kind of societal order in the decades to come, only 50% of the needed climate innovations have seen the daylight by now.

What do you see as the key challenges in the technology transfer process, and what specific obstacles must be overcome to move Climate Tech innovations from the lab to the market?

We need "language skills" at the delicate interface between academia and business, people who speak both science and economics. The laws of scaling up science-based technology companies are not of interest to the professoriate, and tech transfer professionals in turn need to be "learning machines" to understand enough about the technology to be able to determine its level of maturity (TRL) and at what point the market would be ready for the next radical innovation. Interpersonal skills and timing are everything, as this is people business, and not something that AI could do. A brilliant technology transfer pre/seed investor is a rare breed and a special skill in the venture capital world.

Given the critical role of partnerships among diverse stakeholders in driving impactful innovation, how is LUT University fostering collaboration within the innovation ecosystem to enhance this impact?

About a decade ago, LUT University made a strategic choice to focus on clean energy, water and air, which are life-creating resources for which we at LUT are looking for new solutions. This strategy has started to bear fruit: top scientists have joined our faculty, the level or research has risen and international companies are keen to collaborate. A virtuous circle is emerging. The research community values collaboration with industry, while supporting each other in the fierce academic competition. Also, there is a growing and genuine desire in business to take climate and biodiversity into account, rather than just relying on regulation or coercion. The university's ability to work with industry has been rehearsed in recent years and now this expertise is being put to good use.

Market intelligence for the Net Zero Economy

This report, produced by Net Zero Insights, leverages a proprietary database of millions of data points on business, technology, and financial information related to Climate Tech companies, deals, and investors worldwide.

Trusted by top investors, corporates, and advisors, the Net0 Platform helps identify new technologies and monitor emerging trends and opportunities.

Get in Touch

Report scope and definitions

The data used for this report either refer to global venture funding and deal activity in private Climate Tech ventures, covering equity, debt, grants, and other financial instruments, or liquidity events or exits such as M&A, IPOs, and SPAC while excluding post-exit financing.

Sourced from public announcements, our extensive network of partner accelerators and investors, conferences as well as self-reported data, this information undergoes rigorous daily curation on the Net0 Platform by our dedicated analysts.

By meticulously reviewing hundreds of innovations daily, we take pride in developing an ever-expanding methodology for granular and consistent classification of climate solutions. You can find our definition of Climate Tech here.

Conceptually aligned with the framework of the EU taxonomy for sustainable activities, Net Zero Insights refers to "Climate Tech" as encompassing any technology or innovation contributing to at least one of the six specified environmental objectives.

Our analysts classify each organisation based on their alignment with these environmental objectives, encompassing climate change adaptation, climate change mitigation, sustainable use and protection of water and marine resources, protection and restoration of biodiversity & ecosystem; pollution prevention and control; and the transition to a circular economy.

Centred on identifying the most pressing thematic trends in Climate Tech, Net Zero Insights developed a high-level framework to classify each innovation according to the main climate change challenge areas its technology or innovation aims to address. The 10 challenge areas encompass Energy, Transport, industry, Food and agriculture, Circular economy, Built environment, Emissions control, GHG capture, Water, and Natural environment.
Our analysts classify each organisation based on their alignment with these challenge areas,

It's important to note that a single organisation may be categorised under multiple environmental objectives and also make significant contributions to more than one challenge area.

For a thorough analysis, Net Zero Insights uses a unique framework to discern whether a company provides breakthrough or adoption innovation and categorize the solution type as digital or physical. For consistency in the analysis, our analysts manually categorised all deals exceeding $70 million announced in the past five years.

Breakthrough innovation entails extensive research and development, characterized by long-term efforts and innovations in the proof-of-concept phase. Conversely, adoption concentrates on implementing and deploying established solutions, with a focus on improving metrics like cost or efficiency.

The differentiation between the physical and digital aspects of an innovation aims to determine if a business' core offering predominantly revolves around a digital or physical solution.

To compare investor types and better understand their investment activity in Climate Tech, we have created seven groups that cluster investors based on their profile and funding activity.

- Venture investors: venture capital firms, angels, angel groups and family offices
- Growth equity investors: private equity firms, asset managers and investment companies
- Banks: investment banks, commercial banks and merchant banks
- Corporations: corporations, corporate venture capital, insurance companies and holding companies
- Governments: governments and sovereign wealth funds
- Startup support and programs: accelerators/ incubators, competition/challenges, academic / research institutions and university
- Others: infrastructure, investment companies, non-profit organizations, real estate, hedge fund, lender / debt provider, wealth management firm, advisory firm, asset manager, funds of funds, limited partner.

Capital stage definitions

Pre-seed and seed: These are the earliest funding stages, focused on turning an idea into a viable product. Pre-seed funds idea validation and early development, while seed supports building the MVP, acquiring initial customers, and achieving product-market fit.

Early stage: This phase primarily includes Series A and Series B rounds, focusing on scaling operations, expanding the customer base, and building revenue. The company aims to refine its product and establish a strong market presence.

Later stage: Later stage funding (Series C and beyond) supports scaling established businesses, market expansion, and preparing for IPO or acquisition. Companies at this stage typically have proven models, steady revenue, and clear growth strategies.

Research Methodology

This report leverages data from the Net0 Platform to analyze global funding and deal activity for Climate Tech startups, scale-ups, and SMEs.
The analysis covers equity, debt, grants, and other forms of investments. Exits are classified as acquisitions, mergers, IPOs, and SPACs, with post-exit financing rounds being excluded from the analysis.

Given the continuous expansion of the Net0 Platform's scope and the enrichment of its data, this quarterly analysis offers broader coverage, particularly in later-stage funding rounds and emerging markets. Due to these methodological enhancements, the data in this report is not directly comparable to that of previous reports. Consequently, new historical series data has been calculated to analyze market insights effectively.

Our data is meticulously sourced from a variety of channels, including public announcements, our extensive network of partner accelerators and investors, conferences and self-reported data.
This information undergoes rigorous daily curation on the Net0 Platform by our dedicated analysts, ensuring accuracy and consistency of the data in alignment with the scope and definitions outlined in this report.

When simultaneously comparing funding amounts and deal counts, the deal count refers to the total number of deals with disclosed round amounts. This distinction is crucial for maintaining clarity and precision in our analysis.

The data referenced in the present report is accurate as of 30/11/2024. It is essential to acknowledge that typical reporting delays may lead to potentially incomplete information. Nevertheless, this analysis offers valuable insights into the funding landscape of the Climate Tech sector, offering stakeholders a reliable overview of current trends and activities.
In some analysis data was projected by the end of the year by using a linear projection method.

The comprehensive, rigorously curated data provided by the Net0 platform is enriched with insights from Climate Tech stakeholders, gathered through the State of Climate Tech 2024 survey, interviews, and focus groups with industry experts.

By leveraging a diverse blend of data points and a mixed-method research approach—integrating both quantitative and qualitative analyses—this report aims to provide an in-depth understanding of the evolving dynamics within the Climate Tech investment landscape.

Selection of startups to watch:
Startups featured in the startups to watch section of this report were chosen based on a combination of criteria, including:

- Developing solutions that are highly representative of the challenge area under analysis.
- Securing one of the largest early stage deals by amount in 2024.
- Ranking prominently in total funding raised to date.

This approach ensures a balanced representation of innovative and impactful startups within the Climate Tech landscape.

Selection of notable investors and ecosystem supporters featured:
The report features most notable investors and ecosystem supporters based on their activities in the Climate Tech space during the analyzed period. It also ensures a clear representation of the respective investor or ecosystem supporter categories under review and number of deals made in Climate Tech.

Surveys research methodology

The methodology used for this survey analysis was designed to ensure the collection of reliable, valid, and actionable insights from a diverse range of Climate Tech stakeholders. This includes investors, innovators, ecosystem supporters, and corporations, to better understand trends, behaviors, and challenges in the climate tech ecosystem. The survey design, data collection, and analysis were guided by rigorous scientific principles to ensure the robustness of the findings.

1. Survey design
The survey was developed with the goal of collecting both qualitative and quantitative data to gain insights into the various perspectives and roles of stakeholders in the Climate Tech sector. The questions were designed to capture key information on the stakeholders' experiences, priorities, challenges, and future expectations related to climate tech solutions. The survey consisted of a combination of multiple-choice questions, Likert scale ratings, and some open-ended questions, allowing for both structured and unstructured data collection.
The survey design was informed by established survey methodology best practices, including clarity of questions, use of neutral wording to avoid bias, and a logical flow from general to specific questions. The questions were carefully reviewed and pretested with a small sample of stakeholders to ensure clarity and relevance, following the guidelines set by Dillman et al. (2014) on improving survey design and reducing nonresponse bias.

2. Survey Sample
The survey was administered to a targeted sample of 262 respondents, representing four key groups within the climate tech ecosystem: innovators, investors, ecosystem supporters, and corporations.

- Innovators (79 respondents): These individuals are involved in the development and implementation of climate solutions, and they represent the core of the innovation pipeline in the Climate Tech sector.
- Investors (99 respondents): This group includes venture capitalists, private equity firms, angel investors, and other financial stakeholders who provide funding to support the growth and scalability of climate tech solutions.
- Corporations (22 respondents): These organizations contribute to the procurement, integration, and operationalization of climate solutions within their business practices. They do not engage in direct investment but are key players in adopting and scaling technologies.
- Ecosystem supporters (62 respondents): This category includes government agencies, non-governmental organizations, research institutions, and other entities that support and foster the growth of the climate tech ecosystem.

The proportion of each group in the final sample reflects their respective significance in the field.

3. Data Collection
Data was collected through an online survey administered via a secure platform and anonymity was ensured. Invitations to participate were sent to a curated list of stakeholders across various regions and sectors, selected through purposive sampling. This approach was chosen to ensure the inclusion of key actors with expertise or involvement in the climate tech space. The survey was open for responses over a 2 months period including September and October 2024.

4. Data Analysis
- Once the data collection phase was completed, the responses were analyzed using a combination of descriptive and inferential statistical methods to identify trends, correlations, and patterns within the data. The analysis followed a rigorous framework to ensure the reliability and validity of the findings:
- Quantitative data analysis: descriptive statistics were used to summarize demographic information and responses to closed-ended questions. Measures of central tendency (mean, median) and dispersion (standard deviation) were calculated for key variables, providing insights into the general trends within each stakeholder group. Chi-square tests were employed to analyze the relationships between variables (e.g. region).

- Qualitative data analysis: open-ended responses were coded using thematic analysis, a widely used qualitative research method (Braun & Clarke, 2006). The responses were categorized into themes and subthemes to identify recurring patterns and insights.
- Comparative analysis: differences between the four stakeholder groups (innovators, investors, corporations, and ecosystem supporters) were examined to identify group-specific patterns in responses.

5. Reliability and Validity
The reliability of the survey was assessed using Cronbach's alpha, a measure of internal consistency, which was calculated for the Likert scale items to ensure they reliably measure the same construct. A Cronbach's alpha value greater than 0.7 was considered acceptable for the scales used in the survey (Tavakol & Dennick, 2011).

6. Limitations
While the survey sample provides valuable insights, certain limitations exist. The purposive sampling method, while appropriate for targeting key stakeholders, may limit the generalizability of the findings to the broader Climate Tech ecosystem. Additionally, the self-reported nature of survey data can introduce response biases, such as social desirability bias or recall bias, which may influence the results. These limitations are mitigated by the application of robust statistical methods and the transparency of the research design.

References:
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2), 77-101.
- Dillman, D. A., Smyth, J. D., & Christian, L. M. (2014). Internet, phone, mail, and mixed-mode surveys: The tailored design method (4th ed.). Wiley.
- Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. International Journal of Medical Education, 2, 53-55.

Focus groups research methodology

The three key topics of the focus groups were selected from the key insights collected in the survey, and aimed to analyze the perspectives of stakeholders within the Climate Tech sector on:

- Technological trends in Climate Tech.
- The role of corporations in advancing Climate Tech.
- Public-private partnerships in advancing Climate Tech innovation.

1. Sampling and participant recruitment
The participants in this study were selected from a pool of Climate Tech industry leaders, with specific attention to the diversity of their roles and expertise in the sector. Each focus group comprised from a minimum of 4 and a maximum of 6 participants. Participants were recruited using purposive and snowball sampling techniques, ensuring representation from diverse regions and backgrounds, but focusing on individuals with in-depth expertise and experience relevant to each topic.

2. Focus group questions

Technological trends in Climate Tech

- Why do you believe Energy Storage, CCUS, and Hydrogen have emerged as key climate tech technologies that garner consensus from all stakeholders? What makes them the most relevant for addressing and advancing solutions to climate challenges?
- What do you see as the biggest barriers to widespread adoption of these technologies in today's market? How can these be addressed?
- What are the primary challenges in scaling and integrating these emerging technologies across industries? Are there specific sectors that are more receptive to adoption than others?
- How can we attract more industry-specific investors to support the development of these technologies? What role do traditional investors and governments play in this ecosystem?
- Looking ahead, what actionable steps should be taken to accelerate the deployment of these technologies?

The role of corporations in advancing Climate Tech

- What do you see as the most important role that corporations play in the development and deployment of climate tech solutions? Are there particular sectors where corporate leadership is making a greater impact?
- What are the primary challenges corporations face when adopting or integrating climate technologies into their operations? How can these challenges be addressed to accelerate corporate adoption of sustainable practices and technologies?

- How can corporations leverage their financial resources and industry expertise to support Climate Tech innovation?
- How important are partnerships between corporations, startups, and other climate tech stakeholders in driving innovation? What are the best practices for fostering successful collaborations in this space?
- What are the main barriers to increased corporate investment in Climate Tech? How can we incentivize more corporations to allocate resources to climate technology development?

Public-Private Partnerships in Advancing Climate Tech Innovation

- Why do you believe Public-Private Partnerships (PPPs) are becoming more central to the Climate Tech ecosystem? What makes them uniquely positioned to drive innovation and address climate challenges?
- What do you see as the most significant challenges PPPs face in scaling climate technologies across industries? Are there particular sectors or technologies that face greater hurdles in achieving widespread adoption?
- How can governments better support PPPs in advancing climate technologies? What policies, incentives, or regulatory frameworks would most effectively enhance these partnerships?
- What role does the private sector play in driving PPP success, particularly in terms of funding, innovation, and expertise? How can we encourage greater private sector involvement in climate tech partnerships?
- What financial mechanisms or funding models have been most successful in PPPs for Climate Tech? How can we attract more private investors to participate in these partnerships, especially for early-stage technologies?
- Looking forward, what actionable steps can both the public and private sectors take to strengthen PPPs in the climate tech space? How can we ensure these partnerships lead to rapid commercialization and deployment of innovative solutions?
- What emerging trends in climate tech do you see as particularly suited to be advanced through PPPs? How can we leverage these trends to create more impactful collaborations between the public and private sectors?

3. Conducting the focus groups

Each focus group session was designed to foster open dialogue, allowing participants to express their views and engage in discussion around the key topics. The sessions were moderated by Net Zero Insights' Head of Market Insights with expertise in Climate Tech and qualitative research methods, adhering to the best practices outlined by Krueger and Casey (2015).

The following procedures were implemented during the focus group sessions:

- Pre-session preparation: participants were provided with a briefing document outlining the key topics and focus group objectives, along with any relevant background materials.
- Facilitation: focus groups were conducted virtually to accommodate participants across different geographies, using video conferencing tools. Sessions were recorded (with participant consent) to ensure accurate data capture.
- Duration: each focus group session lasted approximately 60 minutes to allow for comprehensive discussion.
- Data collection: notes were taken during the sessions to capture key themes and insights, and recordings were transcribed for detailed analysis.

4. Data analysis

Data from the focus groups were analyzed using thematic analysis, following the steps outlined by Braun and Clarke (2006). This included:

- Familiarization with the data through transcription and note-taking.
- Coding the data into meaningful segments.
- Identifying recurring themes and sub-themes that reflect the participants' perspectives.
- Synthesizing the themes to provide insights into each key topic.

Triangulation: To enhance the validity of the findings, triangulation was employed by comparing survey results with insights derived from the focus groups. This cross-validation method strengthens the robustness of the study's conclusions (Creswell & Plano Clark, 2017).

5. Ethical Considerations

The study followed ethical guidelines to ensure participant confidentiality and no personal identifying information was shared. Participants were also asked for explicit consent before the focus group sessions were recorded.

.
6. Limitations
While the study provides valuable insights, there are limitations that must be acknowledged:

- Sampling bias: The purposive sampling method may have led to a non-random sample, which limits the generalizability of the findings.
- Focus group dynamics: The results are influenced by the dynamics of group discussions, where dominant voices may overshadow others, potentially affecting the diversity of perspectives.

References:

- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. Qualitative Research in Psychology, 3(2), 77-101.
- Creswell, J. W., & Plano Clark, V. L. (2017). Designing and Conducting Mixed Methods Research (3rd ed.). SAGE Publications.
- Pearce, J. M., Campbell, T. D., & Parkinson, D. (2019). Technological
- Krueger, R. A., & Casey, M. A. (2015). Focus Groups: A Practical Guide for Applied Research. Sage Publications

Authors

Sofia Esteves

Head of Market Insights

Sofia combines a background in science and engineering with nearly 15 years of experience in innovation and technology transfer, at academia and industry. In her role as the Head of Market Insights at Net Zero Insights, she is focused on enhancing the impact created by Climate Tech innovation stakeholders.

Mayuri PK

Market Research Analyst

Mayuri has a strong background with over 6 years of experience in startup ecosystem analysis and market research. As a Market Research Analyst at Net Zero Insights, she focuses on analyzing trends in Climate Tech investments and providing strategic insights for global stakeholders. Her expertise spans quantitative research, data science, and innovation ecosystems.

Piyush Singh

Market Research Analyst

Piyush has over 4 years of combined experience in business and policy consulting across multiple sectors, including energy. His role as a Market Research Analyst at Net Zero Insights involves extracting actionable insights on Climate Tech investments.

Ashwini Bhawsar

Startup Ecosystem Manager

Ashwini has over 3 years of experience in the startup ecosystem and is involved in marketing operations and communications at Net Zero Insights. She also focuses on helping Climate Tech startups achieve credible visibility on the Net0 platform while providing resources to accelerate their growth.

Collaborators

Carolina Bentley
Proofreader

Emet Zeitz
Proofreader

Federico Cristoforoni
Editor

Haya Khoury
Visual designer

Natalia Bogatykh
Data Engineer

Pooneh Amini Naeini
Technical Analyst

Karin Killander
Proofreader and PR manager

Contributors

Adi Vagman
Co Founder & Managing Partner, E44 Ventures

Alp Katalan
Business Development Manager, Desolenator

Ansgar Niehoff
Strategic Technology & Investment Manager, Schmidt + Clemens Group

Clea Kolster
Partner & Head of Science, Lowercarbon Capital

Francesco Matteucci
Senior Innovation Advisor, Fondazione Bruno Kessler (FBK)

Frederico Gonçalves
Partner & Managing Director, EDP Ventures

Ibon Iribar
Investment Advisor, CEMEX Ventures

Jan Lozek
Founder, Future Energy Ventures

Keri Browder
Director of Impact Accountability, Prime Coalition

Marcin Lewenstein
Innovation Officer, EIT InnoEnergy

Mark Frayman
Managing Partner, Orion Ventures

Nadav Steinmetz
Co-Founder & Managing Partner, Climate First

Olivia Page
Co-Founder & CEO, Mykor

Pia Erkinheimo
Board member, LUT University

Rosario Cannata
Executive Manager, EDP

Sam Crum
Director of Membership, Greentown Labs

Szymon Kwiatkowski
Investment Manager, EIT InnoEnergy

Tulika Raj
Co-founder, SunGreenH2